【書類名】　意見書
【整理番号】　GNP-20887
【提出日】　令和 8年 8月 3日
【あて先】　特許庁審査官殿
【事件の表示】
　【出願番号】　特願2023-534208
【特許出願人】
　【識別番号】　519302132
　【氏名又は名称】　シャタック　ラボ，インコーポレイテッド
【代理人】
　【識別番号】　110001519
　【氏名又は名称】　弁理士法人太陽国際特許事務所
　【代表者】　中島　崇晴
　【電話番号】　（省略）
　【連絡先】　担当は西山　崇
【発送番号】　086227
【意見の内容】
１．拒絶理由の概要
　令和８年３月３日発送（令和８年２月２４日起案）の拒絶理由通知書には、以下の拒絶理由が通知されています。

理由１（発明の単一性：特許法第３７条）

理由２（明確性：特許法第３６条第６項第２号）
・請求項　１～４

理由３（実施可能要件：特許法第３６条第４項第１号）及び理由４（サポート要件：特許法第３６条第６項第１号）
・請求項　１～４

２．補正の説明
　同日付で提出した手続補正書により特許請求の範囲に対する以下の補正を行いました。
（１）請求項１において、医薬がＴｒｉｍ７調節剤を含むことを規定しました。本補正は、明細書００３１等に支持されております。
　また、明確化のため、「前記抗がん処置による抗腫瘍応答、又はＴＲＩＭ７の過剰発現及び／又は活性化が、ＴＲＩＭ７の過剰発現及び／又は活性化について前記対象から得られた生物学的試料を評価する工程を用いて評価されており」との記載から、「又はＴＲＩＭ７の過剰発現及び／又は活性化が」との語を削除しました。
　加えて、明確化のため、「ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を用いる抗がん処置」との語を、「ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する抗がん処置」に修正しました。
　「プロテアソーム阻害剤」についての記載は削除しました。
（２）請求項２においても、医薬がＴｒｉｍ７調節剤を含むことを規定しました。
　また、「ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を用いる抗がん処置」との語を、「ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する抗がん処置」に修正しました。
　加えて、「プロテアソーム阻害剤」についての記載は削除しました。
　その他軽微記載整備を行いました。
（３）旧請求項３に記載の内容の一部を削除し、新たに請求項４、５に規定しました。

（４）旧請求項４に記載の内容を、新請求項６～１５に分けて規定しました。
（５）上記補正に伴い、従属関係の整備等の記載整備を行いました。
以上の補正は、特許法第１７条の２第３項及び第４項の規定に違反するものではありません。

３．本願請求項に係る発明の説明
本願請求項１に係る発明は以下の通りです。
「がんの処置を必要とする対象においてがんを処置するための、Ｔｒｉｍ７調節剤を含む医薬であって、
（ａ）前記対象において、ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する抗がん処置が実施されており（has been administered）、
（ｂ）ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する前記抗がん処置による抗腫瘍応答が、ＴＲＩＭ７の過剰発現及び／又は活性化について前記対象から得られた生物学的試料を評価する工程を用いて評価されており（has been evaluated）、ここで、
（Ｉ）ＴＲＩＭ７の過剰発現及び／又は活性化が観察される場合、Ｔｒｉｍ７調節剤が投与され（to be administered）、
（ＩＩ）ＴＲＩＭ７の過剰発現及び／又は活性化について前記対象から得られた第２の生物学的試料を用いて評価する工程を用いて、ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する前記抗がん処置による抗腫瘍応答、又はＴＲＩＭ７の過剰発現及び／又は活性化が再評価され（to be re-evaluated）、かつ
（ＩＩＩ）ＴＲＩＭ７の過剰発現及び／又は活性化が観察されない場合、Ｔｒｉｍ７調節剤の投与が取りやめられる（to be withdrawn）、
医薬。」

本願請求項２に係る発明は以下の通りです。
「がんの処置を必要とする対象においてがんを処置するための、Ｔｒｉｍ７調節剤を含む医薬であって、
（ａ）前記対象において、ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する抗がん処置が実施されており（has been administered）、
（ｂ）ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する前記抗がん処置による抗腫瘍応答が、前記対象における腫瘍サイズの縮小をモニタリングすることによって評価されており（has been evaluated）、ここで、
（ｉ）腫瘍サイズ縮小の欠如が観察される場合、Ｔｒｉｍ７調節剤が前記対象に投与され（to be administered）、
（ｉｉ）前記対象における前記腫瘍縮小をモニタリングすることにより、ＰＤ-１、ＰＤ-Ｌ１及び／又はＰＤ-Ｌ２の機能及び／又は活性を阻害する能力を有する抗がん処置による抗腫瘍応答が再評価され（to be re-evaluated）、
（ｉｉｉ）腫瘍サイズの縮小が観察される場合、Ｔｒｉｍ７調節剤の投与が取りやめられる（to be withdrawn）、
医薬。」

４．本願が特許を受けることができるものであることの説明
４-１．単一性について
拒絶理由通知書において単一性要件違反のご指摘をうけ、旧請求項１～４に係る発明が審査対象とされました。補正後の特許請求の範囲において、旧請求項５以降は削除されており、本拒絶理由は解消しております。

４-２．明確性について
（１）拒絶理由通知書において、旧請求項１～４に係る「医薬」の活性成分が特定されておらず、発明が明確でないとのご指摘を受けました。補正後の請求項１及び２では、医薬がＴｒｉｍ７調節剤を含むことが規定されており、請求項１、２及び従属項において拒絶理由は解消しているものと思料致します。

（２）拒絶理由通知書において、旧請求項１～４に「（Ｉ）ＴＲＩＭ７の過剰発現及び／又は活性化が観察される場合、Ｔｒｉｍ７調節剤及び／又はプロテアソーム阻害剤が投与され」、「（ＩＩＩ）ＴＲＩＭ７の過剰発現及び／又は活性化が観察されない場合、Ｔｒｉｍ７調節剤の投与が取りやめられる（to be withdrawn）」と記載されているが、プロテアソーム阻害剤のみを投与する場合に、Ｔｒｉｍ７調節剤の投与が取りやめられることの意味が不明である、とのご指摘を受けました。補正後の請求項１及び２では、プロテアソーム阻害剤の投与に係る記載が削除されており、ご指摘の拒絶理由は解消しているものと思料致します。

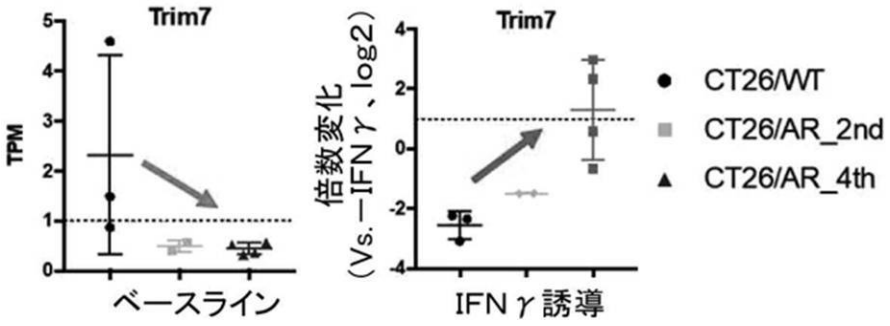
（３）拒絶理由通知書において、旧請求項１～４に係る発明は、「Ｔｒｉｍ７調節剤」、「プロテアソーム阻害剤」、「ＭＳＫ１阻害剤」という性質により特定された化合物を有効成分とする医薬（組成物）に関する発明であるが、一般に、所望の性質を特定することのみで、その性質を有する物質自体を把握することは困難であるから、該記載により特定される化合物が何であるのか不明であるとのご指摘を受けました。
　補正後の本願各請求項において、プロテアソーム阻害剤及びＭＳＫ１阻害剤に係る記載は削除されております。また、Ｔｒｉｍ７調節剤について、以下に詳細にご説明する通り、その範囲は当業者に十分に明確に把握されるものであって、ご指摘の拒絶理由は解消しているものと思料致します。

４-３．実施可能要件及びサポート要件について
（１）
（ａ）拒絶理由通知書において、旧請求項１～４に係る発明は、「Ｔｒｉｍ７調節剤」、「プロテアソーム阻害剤」、「ＭＳＫ１阻害剤」という性質により特定された化合物を有効成分とする医薬（組成物）に関する発明であるところ、本願実施例には、親ＣＴ２６腫瘍細胞と抗ＰＤ-１耐性ＣＴ２６腫瘍細胞と比較して、ＴＲＩＭ７が差次的に発現していることが試験されているだけであり、あらゆる「Ｔｒｉｍ７調節剤」、「プロテアソーム阻害剤」、「ＭＳＫ１阻害剤」が、がん治療剤として機能することについてまで具体的に示されていないし、そもそもＴＲＩＭ７発現を阻害または増加させる化合物を用いて、実際にがん治療を行い得ることや抗ＰＤ-１耐性がんに対して治療効果を発揮できることを具体的に示すものではないことから、実施可能要件及びサポート要件が満たされないことのご指摘を受けました。

　「プロテアソーム阻害剤」、「ＭＳＫ１阻害剤」との語は特許請求の範囲から削除されており、本願発明の医薬にはＴｒｉｍ７調節剤が含まれることが規定されています。

　拒絶理由通知書でも認められているように、本願実施例５及び６には、「ＴＲＩＭ７発現が親ＣＴ２６細胞と比較して抗ＰＤ-１耐性ＣＴ２６腫瘍細胞において減少していたこと、ＩＦＮγ刺激条件下ではＴＲＩＭ７発現が抗ＰＤ-１耐性ＣＴ２６腫瘍細胞において増加したこと（図５Ｃ参照）」が示されています。このように、実施例５では、ＩＦＮγの存在下又は非存在下で、抗ＰＤ-１耐性細胞におけるＴｒｉｍ７などの遺伝子の逆説的調節不全が見出されています。明細書段落０２３１に「これは、とりわけ、ＩＦＮγに対する細胞応答が抗ＰＤ-１及び他の免疫治療薬に対する感受性に関与することが広く知られているので、驚くべきことであった。」と記載されるように、抗ＰＤ-１耐性細胞では親細胞に比べてＴｒｉｍ７遺伝子が抑制されていることや、ＩＦＮγに応答して通常は（

親細胞では）抑制されるＴｒｉｍ７遺伝子が、抗ＰＤ-１耐性細胞では逆に過剰発現されるという逆説的な挙動は驚くべきものでした（明細書段落００３８の図５Ａ～Ｄの説明、図５Ｄ、０２４１等）。以下に図５Ｃを再掲します。



また、本願実施例６では、野生型腫瘍及び耐性腫瘍を有するマウスでのトランスクリプトーム解析に基づき、抗ＰＤ-１療法への耐性に関与するドライバー遺伝子には「ＴＲＩＭファミリータンパク質（特に、Ｔｒｉｍ７）」が含まれることが特定されています。

さらに、本願実施例９では、Ｃｃｌ５（ＲＡＮＴＥＳ）、Ｃｘｃｌ１０（ＩＰ-１０）、及びＩｆｎｂ１といったＴｒｉｍ７に制御される遺伝子も逆説的な調節不全を示すことが明らかにされています。実施例９では、「これらの結果は、とりわけ、抗ＰＤ-１に対する獲得耐性が、Ｃｃｌ５（ＲＡＮＴＥＳ）、Ｃｘｃｌ１０（ＩＰ-１０）及びＩｆｎｂ１などのＴｒｉｍ７調節遺伝子の逆説的調節に関連することを示している。したがって、これら３つの結果は、抗ＰＤ-１耐性腫瘍がＴｒｉｍ７調節剤、例えばＴｒｉｍ７阻害剤、によって処置しうることを示唆する。」と結論付けています（明細書段落０２５３）。

本願に示される状況依存的なＴｒｉｍ７の上方制御又は下方制御に鑑み、当業者はＴｒｉｍ７阻害剤を含むＴｒｉｍ７調節剤の有用性を理解するものと考えられます。したがって、ＴＲＩＭ７の過剰発現及び／又は活性化を伴うがんや抗ＰＤ-１耐性がん等に対するＴｒｉｍ７調節剤の有用性は本願明細書に支持されており、本願発明は実施可能要件及びサポート要件を満たしているものと思料致します。

（ｂ）また、拒絶理由通知書において、旧請求項３に係る発明は、様々な「ｓｉＲＮＡ」、「ｓｈＲＮＡ」、「ｍｉＲＮＡ」、「アンチセンスＲＮＡ」、「Ｔｒｉｍ７タンパク質を阻害するｇＲＮＡ」、様々な「小分子」、「抗体」、「ペプチド」、「ペプチド模倣体」等を活性成分とする発明であるが、該化合物はＴｒｉｍ７に直接作用するものだけでなく、間接的に作用するような化合物まで包含しており、それらの化合物については本願発明の治療剤として使用できる程度に記載されていないこと、Ｔｒｉｍ７に直接作用する化合物であることを特定したとしても、Ｔｒｉｍ７の機能を阻害または増強できる「小分子」や「ペプチド」、「ペプチド模倣体」等が知られていたわけでもないことから、実施可能要件及びサポート要件が満たされないことのご指摘を受けました。

補正後の請求項３では「前記Ｔｒｉｍ７調節剤がＴｒｉｍ７阻害剤である」ことが規定されており、また、請求項４にはＴｒｉｍ７調節剤が「Ｔｒｉｍ７タンパク質の産生を阻害する」ものであることが規定されています。請求項５には、Ｔｒｉｍ７調節剤が「Ｔｒｉｍ７の活性を阻害する」ことが規定されております。

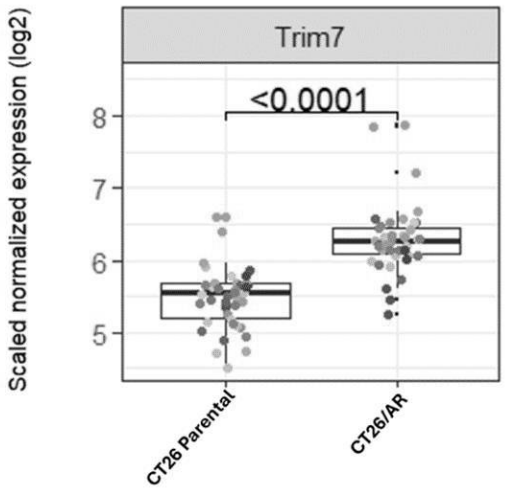
また、本願明細書では、例えば、「実施形態では、Ｔｒｉｍ７阻害剤は、ＭＡＡＶＧＰＲＴＧＰＧＴＧＡＥＡＬＡＬＡＡＥＬ（配列番号１）、ＡＡＴＲＡＰＰＦＰＬＰＣＰ（配列番号２）、ＨＧＳＱＡＡＡＡＲＡＡＡＡＲＣＧ（配列番号３）及びＮＶＳＬＫＴＦＶＬ

ＫＧＭＬＫＫＦＫＥＤＬＲＧＥＬＥＫＥＥＫＶ（配列番号４）から選択されるタンパク質セグメント又はその近傍においてＴｒｉｍ７タンパク質に結合する小分子又はペプチド阻害剤である。」（段落０１７１）といった指針が含まれています。
加えて、入手可能な最新の構造情報（例えば、Sosa et al., Crystal structure and mutational analysis of the human TRIM7 B30.2 domain provide insights into the molecular basis of its binding to glycogenin-1, J Biol Chem 296: 100772-100772 (2021); Ru et al., C-terminal glutamine acts as a C-degron targeted by E3 ubiquitin ligase TRIM7, Proc Natl Acad Sci U S A 119: e2203218119-e2203218119 (2022); Luptak et al., TRIM7 Restricts Coxsackievirus and Norovirus Infection by Detecting the C-Terminal Glutamine Generated by 3C Protease Processing, Viruses 14(8): 1610 (2022)など）や、Ｅ３ユビキチンリガーゼ活性測定、及び他のＥ３ユビキチンリガーゼ阻害剤の開発実績に基づき、当業者であれば困難を伴うことなく小分子又はペプチド阻害剤をスクリーニングし、開発できるものと思料致します。加えて、当業者は本技術分野の通常の知識に基づき、抗体や核酸医薬（ｓｉＲＮＡ、アンチセンスＲＮＡなど）を設計及び開発できるものと思料致します。
したがって、補正後の請求項３～５についても、ご指摘の拒絶理由は当てはまらないものと思料致します。

（２）拒絶理由通知書において、以下の理由から、旧請求項１～４の実施可能要件及びサポート要件が満たされない旨のご指摘を受けています。
「本願請求項１-４に係る発明は、抗ＰＤ-１抗体等を用いた抗がん療法を受けた患者を対象とし、ＴＲＩＭ７の過剰発現に基づいてＴＲＩＭ７調節剤またはプロテアソーム阻害剤を投与することを特徴とするがんを処置するための医薬の発明である。
ここで、一般に、物質名、化学構造だけからその用途を予測することは困難であるから、出願時の技術常識及び出願当初の明細書に記載された作用の説明等からでは、含有成分がその医薬用途として機能することが推認できない場合には、明細書に有効量、投与方法、製剤化方法が記載されている場合であっても、それだけでは当業者は当該医薬が実際にその用途として使用できるか否かを知ることはできないので、明細書に特定の薬理試験の結果である薬理データ又はそれと同視すべき程度の記載をしてその用途を裏付ける必要がある。また、特定化合物の疾患モデル細胞に対する影響を、差次的遺伝子発現解析手法を用いてマーカー遺伝子を同定したからといって、該遺伝子が疾患の原因遺伝子であるとまではいえないし、疾患の原因遺伝子であるか不明な一部マーカー遺伝子の発現のみで治療効果を予測することは難しいということが技術常識である。
しかし、本願実施例５、６等には、ＴＲＩＭ７発現が親ＣＴ２６細胞と比較して抗ＰＤ-１耐性ＣＴ２６腫瘍細胞において減少していたこと、ＩＦＮγ刺激条件下ではＴＲＩＭ７発現が抗ＰＤ-１耐性ＣＴ２６腫瘍細胞において増加したこと（図５Ｃ参照）が示されているに過ぎず、それら両細胞間におけるＴＲＩＭ７の発現差異が治療効果を示すものではない。そして、本願の実施例５、６等における実験結果が、抗ＰＤ-１耐性の結果なのか、抗ＰＤ-１耐性から感受性への変化を意味しているのかも不明であり、該結果からＴｒｉｍ７阻害剤が治療剤として機能するのか、Ｔｒｉｍ７増強剤が治療剤として機能するのかも不明であり、プロテアソーム阻害剤が薬理活性を有すると認められる実験がなされているものでもないから、Ｔｒｉｍ７調節剤またはプロテアソーム阻害剤が、当該疾患への医薬として使用できることを推認できる程度に発明の詳細な説明が記載されているということはできない。」

上述の通り、本願の実施例５及び実施例６では、ＴＲＩＭ７の逆説的な調節不全を明らかにしています。加えて、出願人は、ＣＴ２６親細胞（チェックポイント阻害剤応答性モデル）と比較して、ＣＴ２６チェックポイント阻害剤獲得耐性（ＣＴ２６／ＡＲ）モデルにおいてＴｒｉｍ７が有意に上方制御されていることをさらに裏付ける出願後の補足データを以下に提示致します。以下の図Ａにおいて、横軸（Scaled normalized expression（

log2))はＴｒｉｍ７の正規化した発現レベルを表し、この発現レベルをＣＴ２６親細胞とＣＴ２６／ＡＲで比較しています。

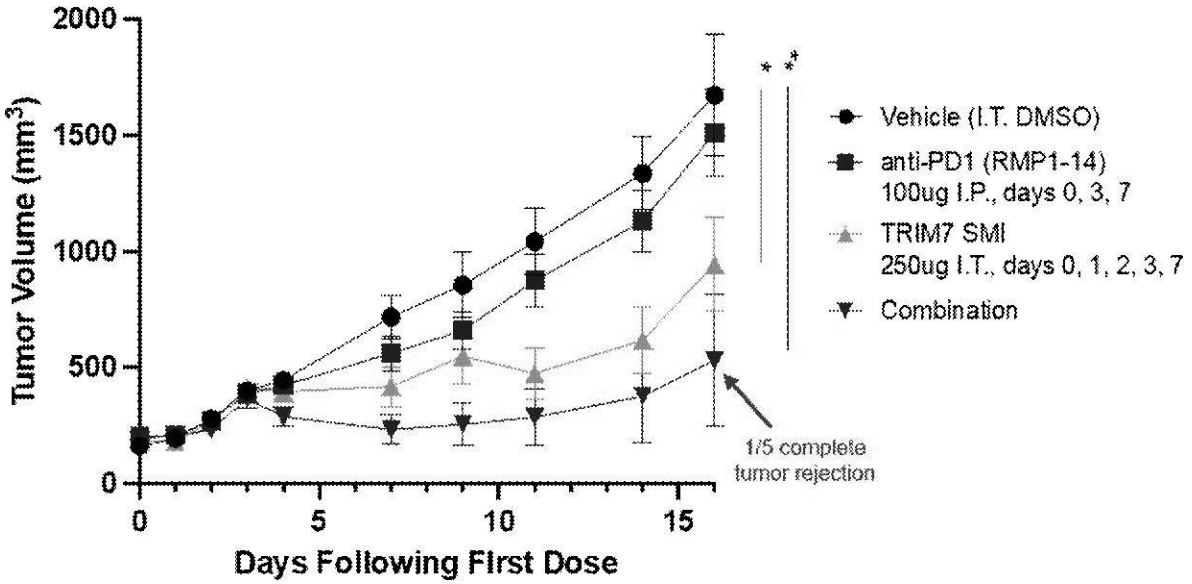


図Ａ：ＴＲＩＭ７は、ＣＴ２６親細胞（チェックポイント阻害剤応答性モデル）と比べて、ＣＴ２６チェックポイント阻害剤獲得耐性（ＣＴ２６／ＡＲ）モデルにおいて有意に情報制御されている。

また、上述のように、実施例９には「抗ＰＤ-１耐性腫瘍がＴｒｉｍ７調節剤、例えばＴｒｉｍ７阻害剤、によって処置しうる」ことが記載されています。この点について、以下にこれを支持する出願後の追加データを示します。下記図Ｂは、抗ＰＤ-１抗体耐性ＣＴ２６細胞同種移植片を有するマウス（ＣＴ２６／ＡＲモデル）にＴｒｉｍ７小分子阻害剤（「ＴＲＩＭ７　ＳＭＩ」）を投与して得られたデータを示しています。具体的には、実施例１に記載されているように、ＢＡＬＢ／Ｃマウスの後側腹部に５００，０００個の抗ＰＤ-１抗体耐性ＣＴ２６細胞を接種しました。平均腫瘍体積が８０～１００ｍｍ^{3}に達した時点（０日目）で、マウスを以下の４つの処置群に分けました。

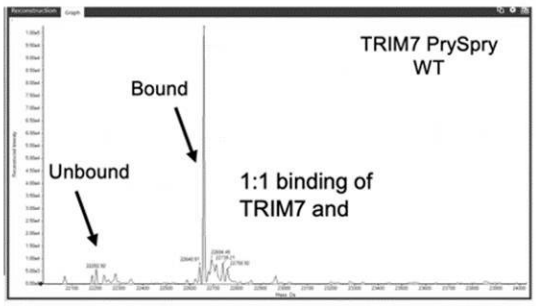
群１：ビヒクルのみ。
群２：抗ＰＤ-１抗体（クローンＲＭＰ１-１４；BioXcell社）を０、３、６日目に投与。
群３：ＴＲＩＭ７　ＳＭＩを０、１、２、３、７日目に投与。
群４：抗ＰＤ-１抗体とＴＲＩＭ７　ＳＭＩを併用。

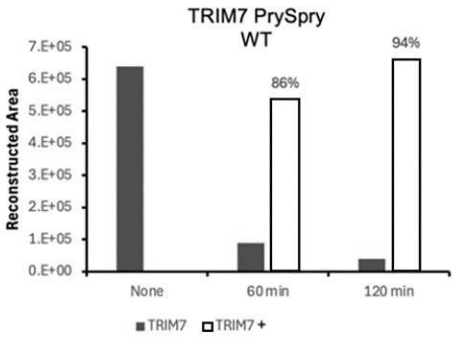
以下の図Ｂにおいて、横軸（Days Following First Dose）は初回投与後の日数を表し、縦軸（Tumor Volume（mm^3））は腫瘍体積（ｍｍ^3）を表します。図Ｂに示すように、ＴＲＩＭ７　ＳＭＩによる処置により腫瘍サイズは有意に縮小しました（ｐ≦０．０５）。一方、抗ＰＤ-１抗体単独の処置では、予想通り、腫瘍サイズに有意な縮小は観察されませんでした。興味深いことに、抗ＰＤ-１抗体とＴＲＩＭ７　ＳＭＩを併用すると、ＴＲＩＭ７　ＳＭＩ単独療法と比較してさらに腫瘍サイズが縮小し（ビヒクルのみの処置マウスに対してｐ≦０．０１）、５分の１のマウスで腫瘍の完全な消失（完全拒絶）が観察されました。

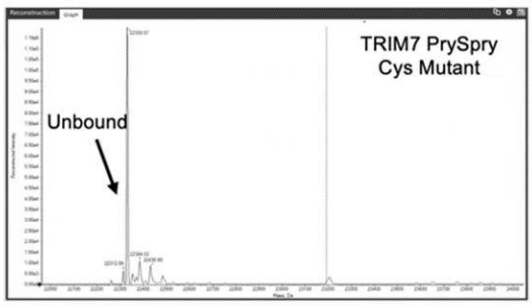


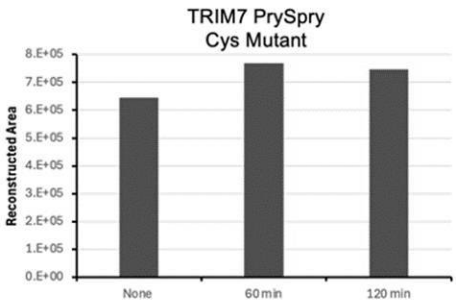
図Ｂ：ＴＲＩＭ７阻害はＣＴ２６／ＡＲ腫瘍の成長を制御し、それらを抗ＰＤ-１阻害に対して再感受性化する。

さらに以下の図では、ＴＲＩＭ７に共有結合するＴＲＩＭ７　ＳＭＩが実際にＴＲＩＭ７の阻害剤であることの証拠が示されています。図Ｃでは、質量分析を用いて、ＴＲＩＭ７　ＰｒｙＳｐｒｙドメイン（Ｂ３０．２ドメインとしても知られ、脊椎動物のＴＲＩＭファミリータンパク質のＣ末端に主に見られるタンパク質相互作用モジュール）に対するＴＲＩＭ７　ＳＭＩの結合を、システインを除去した変異体ＴＲＩＭ７　ＰｒｙＳｐｒｙドメインへの結合と比較しています。図Ｃ上段が野生型ＴＲＩＭ７　ＰｒｙＳｐｒｙドメインとの結合試験、下段が変異体ＴＲＩＭ７　ＰｒｙＳｐｒｙドメインとの結合試験結果を表しています。図Ｃに示すように、野生型ＴＲＩＭ７　ＰｒｙＳｐｒｙドメインでは特異的結合と１：１の複合体形成が観察されました。一方、変異体ＴＲＩＭ７　ＰｒｙＳｐｒｙドメインではこれらの結合及び複合体形成は観察されず、結合の特異性が証明されています。

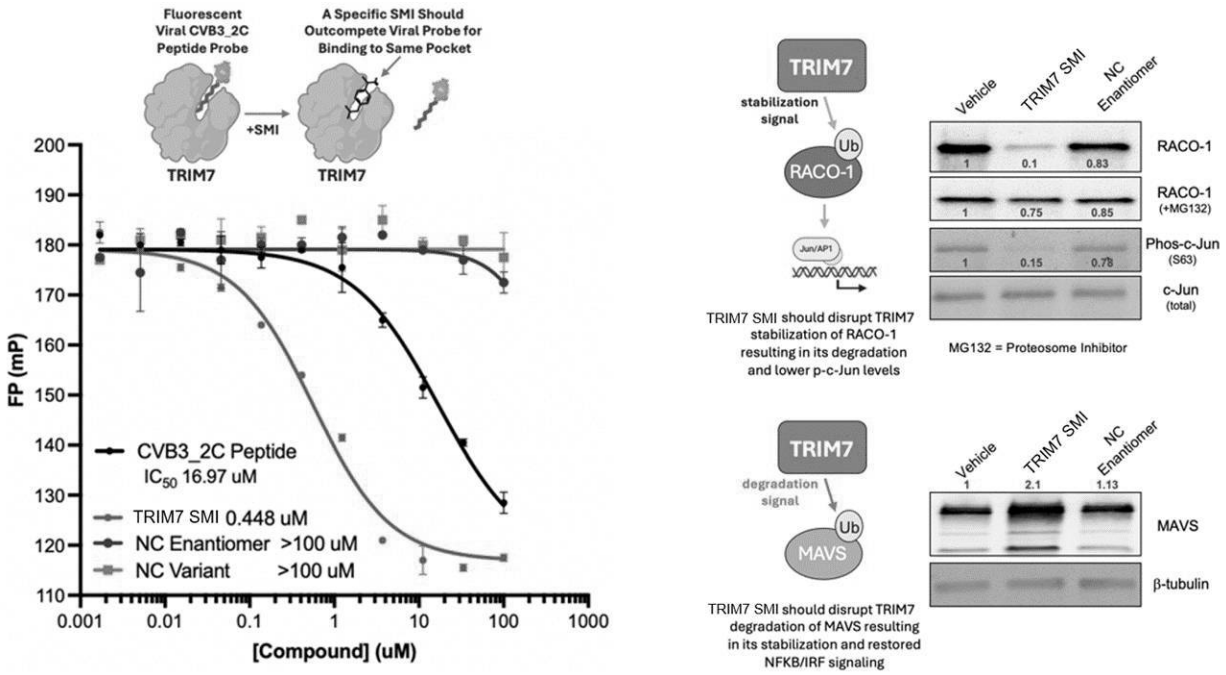








図Ｃ：質量分析によりＴＲＩＭ７　ＳＭＩのＴＲＩＭ７　ＰｒｙＳｐｒｙドメインへの１：１結合が示され、システイン変異を通して共有結合の特異性が確認された。



図Ｄ：左図は、コクサッキーウイルスＢ３（ＣＶＢ３）ペプチドプローブと競合した蛍光偏光アッセイを用いてさらに確認された、ＴＲＩＭ７　ＳＭＩの結合特異性及び活性を示す。ＴＲＩＭ７　ＳＭＩのエナンチオマー（NC Enantiomer）や陰性対照バリアント(NC Variant)では競合活性は観察されなかった。右図は、Ｈ１２９９肺がん細胞においてＴＲＩＭ７　ＳＭＩがＴＲＩＭ７によるＲＡＣＯ-１の安定化及びＭＡＶＳの分解を逆転させたことを示す。

なお、最近の文献ではＴＲＩＭ７に結合するペプチドが同定されています。そのようなペプチドの一つとして、コクサッキーウイルスＢ３（ＣＶＢ３）ペプチドがあります（例えば、参照文献１：Luptak et al., TRIM7 Restricts Coxsackievirus and Norovirus Infection by Detecting the C-Terminal Glutamine Generated by 3C Protease Processing, Viruses 14(8): 1610 (2022)）。ＴＲＩＭ７タンパク質への結合に対するＣＶＢ３ペプチドとＴＲＩＭ７　ＳＭＩとの競合を評価するために、蛍光偏光アッセイを実施しました。図Ｄ（左図）に示すように、ＴＲＩＭ７　ＳＭＩはＣＶＢ３ペプチドプローブ（CVB3_2C Peptide）に競合して打ち勝っています。一方、ＴＲＩＭ７　ＳＭＩのエナンチオマー（NC Enantiomer）や陰性対照バリアント（NC Variant）は競合しなかったことから、結合の特異性と活性が確認されています。

ＴＲＩＭ７の生物学的活性に関し、本願明細書段落０１５８には、「実施形態では、Ｔｒｉｍ７経路活性化は、ＡＰ１コアクチベーターＲＡＣＯ-１のユビキチン化及び安定化並びに／あるいはＡＰ１媒介性遺伝子発現の増加を用いてアッセイされ得る。ＡＰ１媒介遺伝子発現シグネチャは当技術分野で周知である。」と記載されています。そのため、ＴＲＩＭ７　ＳＭＩがこのＲＡＣＯ-１の安定化及びＡＰ１活性化の増加を逆転させるか否かを試験しました。

Ｈ１２９９肺がん細胞を、ビヒクル、ＴＲＩＭ７　ＳＭＩ、又は陰性対照バリアントで処理し、ＲＡＣＯ-１レベル、ｃ-Ｊｕｎリン酸化レベル（ＡＰ-１活性化の直接的かつ

十分に確立された生化学的指標）、及びミトコンドリア抗ウイルスシグナル伝達タンパク質（ＭＡＶＳ）のＫ４８結合ユビキチン化を評価しました（なお、本願明細書では、「実施形態では、評価することは、インターフェロン遺伝子刺激因子（ＳＴＩＮＧ）及び／又はＡＰ-１共活性化因子ＲＡＣＯ-１のタンパク質ユビキチン化及び／又はＫ４８結合ユビキチン化をアッセイすることによって行われる。」との記載があります（段落０１７５））。
図Ｄ（右図）に示すように、ＴＲＩＭ７　ＳＭＩによる細胞の処理は、ＲＡＣＯ-１レベルの低下、ｃ-Ｊｕｎリン酸化レベルの低下、及びＭＡＶＳレベルの上昇をもたらしました。これらは陰性対照では観察されていません。このことは、ＡＰ１コアクチベーターＲＡＣＯ-１の安定化、ＡＰ-１活性化、及びＭＡＶＳの分解の逆転を示唆しています。したがって、ＴＲＩＭ７　ＳＭＩはＴＲＩＭ７の機能阻害剤です。

本願の出願当初の明細書には、上述のように、ＴＲＩＭ７の過剰発現及び／又は活性化を伴うがんや抗ＰＤ-１耐性がん等に対するＴｒｉｍ７調節剤の有用性が十分に示されております。また、Ｔｒｉｍ７小分子阻害剤により抗腫瘍効果が示された上記の薬理データは、いずれの本願当初明細書の記載を裏付けるものです。したがって、補正後の本願請求項１～４について、ご指摘の拒絶理由は当てはまらないものと思料致します。

（３）拒絶理由通知書において、旧請求項１～４に係る発明は、抗ＰＤ-１抗体の他に、抗ＰＤ-Ｌ１抗体や抗ＰＤ-Ｌ２抗体等の阻害剤を用いた場合の獲得耐性に関して、ＴＲＩＭ７の過剰発現に基づいてＴＲＩＭ７調節剤またはプロテアソーム阻害剤を投与することを特徴とするがんを処置するための医薬の発明であるが、本願実施例には、抗ＰＤ-１抗体を用いて得られた抗ＰＤ-１耐性ＣＴ２６腫瘍細胞における、発現遺伝子の変動について調べているに過ぎず、抗ＰＤ-Ｌ１抗体や抗ＰＤ-Ｌ２抗体を用いて得られる細胞の発現遺伝子の変化については何ら具体的に調べられておらず、抗ＰＤ-１耐性腫瘍細胞におけるＴＲＩＭ７の関与が、抗ＰＤ-Ｌ１耐性や抗ＰＤ-Ｌ２耐性腫瘍においても同様に関与していることを明らかにしておらず、実施可能要件及びサポート要件が満たされないことのご指摘を受けました。

しかしながら、ＰＤ-Ｌ１及びＰＤ-Ｌ２の制御メカニズムがインターフェロン応答を介して相互に関連していることは当業者によく知られています（例えば、参照文献２：Garcia-Diaz et al., Interferon Receptor Signaling Pathways Regulating PD-L1 and PD-L2 Expression, Cell Rep. 19(6):1189-1201 (2017)）。同様に、ＰＤ-１及びＰＤ-Ｌ１遮断に対する耐性メカニズムも、インターフェロン応答を介して関連していることがよく知られています（例えば、参照文献３：Nowicki et al., Mechanisms of Resistance to PD-1 and PD-L1 blockade, Cancer J. (1):47-53 (2018)、特に５頁）。
また、本出願の実施例９では、「抗ＰＤ-１に対する獲得耐性が、Ｃｃｌ５（ＲＡＮＴＥＳ）、Ｃｘｃｌ１０（ＩＰ-１０）及びＩｆｎｂ１などのＴｒｉｍ７調節遺伝子の逆説的調節に関連する」ことを実証しています。少なくともこれらの科学的考察に基づき、当業者は、抗ＰＤ-Ｌ１耐性腫瘍及び抗ＰＤ-Ｌ２耐性腫瘍においてもＴＲＩＭ７が同様に関与していることを十分に理解するものと思料致します。したがいまして、補正後の本願請求項１～４においてご指摘の拒絶理由は当てはまらないものと思料致します。

５．結論
以上ご説明した通り、本出願は、拒絶理由を有するものではありません。従って、再度ご審査のうえ、特許査定を賜りますようお願い申し上げます。

以上

（参照文献１：Luptak et al., TRIM7 Restricts Coxsackievirus and Norovirus Infection by Detecting the C-Terminal Glutamine Generated by 3C Protease Processing, V

iruses 14(8): 1610 (2022))

viruses

MDPI

Article

TRIM7 Restricts Coxsackievirus and Norovirus Infection by Detecting the C-Terminal Glutamine Generated by 3C Protease Processing

Jakub Luptak [1], Donna L. Mallery [1], Aminu S. Jahun [2], Anna Albecka [1], Dean Clift [1], Osaid Ather [1], Greg Slodkowicz [3], Ian Goodfellow [2] and Leo C. James [1,*]

[1] MRC Laboratory of Molecular Biology, Francis Crick Avenue, Cambridge CB2 0QH, UK; jluptak@mrc-lmb.cam.ac.uk (J.L.); dlm@mrc-lmb.cam.ac.uk (D.L.M.); aam@mrc-lmb.cam.ac.uk (A.A.); dclift@mrc-lmb.cam.ac.uk (D.C.); oa260@cam.ac.uk (O.A.)
[2] Division of Virology, Department of Pathology, University of Cambridge, Addenbrooke's Hospital, Cambridge CB2 0QQ, UK; asj40@cam.ac.uk (A.S.J.); ig299@cam.ac.uk (I.G.)
[3] The Francis Crick Institute, London NW1 1AT, UK; grzegorz.slodkowicz@crick.ac.uk
* Correspondence: lcj@mrc-lmb.cam.ac.uk

Abstract: TRIM7 catalyzes the ubiquitination of multiple substrates with unrelated biological functions. This cross-reactivity is at odds with the specificity usually displayed by enzymes, including ubiquitin ligases. Here we show that TRIM7's extreme substrate promiscuity is due to a highly unusual binding mechanism, in which the PRYSPRY domain captures any ligand with a C-terminal helix that terminates in a hydrophobic residue followed by a glutamine. Many of the non-structural proteins found in RNA viruses contain C-terminal glutamines as a result of polyprotein cleavage by 3C protease. This viral processing strategy generates novel substrates for TRIM7 and explains its ability to inhibit Coxsackie virus and norovirus replication. In addition to viral proteins, cellular proteins such as glycogenin have evolved C-termini that make them a TRIM7 substrate. The 'helix-ΦQ' degron motif recognized by TRIM7 is reminiscent of the N-end degron system and is found in ~1% of cellular proteins. These features, together with TRIM7's restricted tissue expression and lack of immune regulation, suggest that viral restriction may not be its physiological function.

Keywords: TRIM7; coxsackievirus; norovirus; 3C protease; 3Cpro; Mpro; SARS-CoV-2; restriction; degradation; ISG

check for updates

Citation: Luptak, J.; Mallery, D.L.; Jahun, A.S.; Albecka, A.; Clift, D.; Ather, O.; Slodkowicz, G.; Goodfellow, I.; James, L.C. TRIM7 Restricts Coxsackievirus and Norovirus Infection by Detecting the C-Terminal Glutamine Generated by 3C Protease Processing. *Viruses* **2022**, *14*, 1610. https://doi.org/10.3390/v14081610

Academic Editor: Shan-Lu Liu

Received: 28 June 2022
Accepted: 20 July 2022
Published: 23 July 2022

Publisher's Note: MDPI stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Copyright: © 2022 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

TRIM proteins comprise the largest family of E3 ligases in mammals. They are characterized by a tripartite motif of RING, B-Box and coiled-coil and may encode additional domains, of which the PRYSPRY domain is the most common. Many TRIM proteins have proposed roles in infection and immunity (for recent review, see [1]) where they typically inhibit viral replication, although some are reported to promote replication [2,3]. A commonly reported mechanism is one in which the TRIM binds a viral protein and causes it to be degraded via ubiquitination. Targeting is usually mediated by the C-terminal PRYSPRY domain, whilst ubiquitination is catalyzed by the N-terminal RING domain, though TRIMs have been reported to degrade their target even when they lack a RING [4]. Typical targets for degradation include the viral capsid or nucleocapsid: TRIM5 [5], TRIM11 [6], and TRIM34 [7] are reported to target the capsid of HIV and TRIM14 [4], TRIM22 [8] and TRIM41 [9] the nucleocapsid of Influenza.

Antiviral TRIMs have also been shown to stimulate immune signaling, providing a second mechanism of viral inhibition. This can be indirect, by promoting the activity of pattern-recognition receptors (PRRs). For instance, TRIM65 is reported to K63-polyubiquitinate MDA5 and promotes its signaling [10]. RIPLET, technically not a TRIM but containing

RING, coiled-coil and PRYSPRY domains, modifies RIG-I with K63-chains and stimulates its activity [11,12], while TRIM56 is implicated in DNA sensing by promoting the activity of cGAS [13]. Other TRIMs promote signaling directly as primary PRRs. TRIM5 [14] and TRIM21 [15] both stimulate innate immune signaling upon detection of incoming virions. This activity occurs in parallel with their direct-acting restriction mechanisms, in which targeted viral capsids are degraded. The restriction and signaling activities of both TRIM5 and TRIM21 are inextricably linked—synthesis of K63-polyubiquitination leads to proteasomal recruitment, followed by concomitant liberation of free K63-chains and protein degradation [16,17].

Exactly how many members of the TRIM family are involved in antiviral immunity, what proportion directly restrict or promote immune signaling and which are capable of both is unknown. In a transient over-expression screen of 36 human and 19 mouse TRIMs, over one-third were found to possess antiretroviral activity [18]. In a follow-up study, 16 out of 43 over-expressed human TRIMs were shown to induce NF-κB or AP-1 signaling [19]. Tellingly, there was a close correspondence between restriction and signaling; of 17 TRIMs that restricted MLV, 12 were also immune stimulatory. A more recent over-expression screen of 118 RING E3 ligases (including over 50 TRIMs) identified TRIM7 as a potent inhibitor of picornavirus Coxsackievirus B3 (CVB3) [20]. Restriction required PRYSPRY binding to viral protein 2BC, followed by its RING-dependent ubiquitination and degradation. TRIM7 was also previously identified in a genome-wide overexpression screen as an inhibitor of the picorna-like virus calicivirus murine norovirus (MNV), an activity similarly found to be PRYSPRY-dependent [21]. However, these reports of antiviral activity are at odds with other TRIM7 studies. TRIM7 has been proposed as a pro-viral factor in Zika virus infection, where it was shown to K63-polyubiquitinate the viral envelope protein [3]. TRIM7 has also been reported as an agonist of TLR4-mediated signaling in macrophages [22], a tumour suppressor through degradation of Src [23], an oncogene through the K63-ubiquitination and stabilization of AP-1 co-activator RACO-1 [24] and an interactor and regulator of glycogenin—the latter activity giving rise to TRIM7s original designation as 'glycogenin-interacting protein' (GNIP) [25]. There is little that connects these disparate cellular roles.

In this study, we sought to determine how TRIM7 is capable of interacting with such diverse substrates and the mechanism by which it targets viral proteins for degradation. We show that the PRYSPRY domain of TRIM7 possesses a unique binding site that promiscuously interacts with proteins containing a C-terminus ending in a helix-ΦQ motif. The reason why TRIM7 has been reported to bind glycogenin-1 (GYG1), RACO-1 and CVB3 2BC is because they all possess this structural signature. We also show that it is this promiscuous helix-ΦQ motif that allows TRIM7 to restrict infection by both norovirus and Coxsackievirus. These viruses possess a viral 3C-protease that cleaves their polyproteins to generate viral proteins ending in glutamine. A TRIM7 point mutant that no longer binds this glutamine motif loses all viral restriction. 3C-like proteases are common to positive-strand RNA viruses from the Picornaviridae, Caliciviridae and Coronaviridae families, and all generate viral proteins with C-terminal glutamines that are potential substrates for TRIM7. However, we show that despite possessing multiple proteins ending in glutamine, SARS-CoV-2 is not restricted by TRIM7. Moreover, the tissue expression of TRIM7 and its gene regulation is not consistent with an antiviral TRIM, such as TRIM5 or TRIM21, suggesting that its described function as a restriction factor may be an artefact of overexpression.

2. Material and Methods

2.1. Proteins

Trim7 PRYSPRY constructs and all GYG constructs with an N terminal hexahistidine tag were expressed in *E. coli* (C41) and purified using Nickel affinity chromatography and Size Exclusion Chromatography (SEC). Briefly, cells were grown in 2XTY (supplemented with 0.5% glucose, 2 mM $MgSO_4$ and appropriate antibiotics) at 37 °C for 2–3 h

(OD600 around 0.6–1), after which they were induced with 1 mM IPTG and incubated at 18 °C overnight. Cells were pelleted with a Sorvall SLC-6000 compatible centrifuge at 4500× *g* for 25 min and the pellet frozen until processed. The pellet was resuspended in lysis buffer (50 mM Tris pH 8, 1 M NaCl, 10% *v*/*v* BugBuster (Merck, Gillingham, UK), 10 mM imidazole, 2 mM DTT and 1 × complete protease inhibitors (Roche, Basel, Switzerland) and sonicated for 15 min total time (10 s on/20 s off) at 70% amplitude. The soluble fraction was recovered by centrifugation at 40,000× *g* in a JLA25.50 rotor and put through a gravity flow column with 5 mL of NiNTA Agarose (Qiagen). The bound fraction was washed in Buffer B (300 mM NaCl, 50 mM Tris pH 8, 10 mM imidazole and 1 mM DTT) and eluted with Buffer E (300 mM NaCl, 50 mM Tris pH 8, 400 mM imidazole and 1 mM DTT). Fractions containing the protein were pooled, filtered and separated by SEC using HiLoad 26/600 Superdex 75 pg column (Cytiva, Marlborough, MA, USA) in 150 mM NaCl, 50 mM Tris pH 8 and 1 mM DTT. Full-length GYG proteins were separated using an equivalent 200 pg column. The appropriate fractions were pooled and concentrated to 10–15 mg/mL.

MBP-Trim7-CCPS was expressed and purified as described above but with the following adjustments; after elution from NiNTA agarose, the fractions were pooled and loaded onto an MBP-Trap (5 mL) column (300 mM NaCl, 50 mM Tris pH 8, 1 mM DTT) washed with the same buffer and then eluted with buffer supplemented with 10 mM Maltose. The peak fractions were then separated by SEC using HiLoad 26/600 Superdex 200 pg column in 300 mM NaCl, 50 mM Tris pH 8, 10% *w*/*v* glycerol and 1 mM DTT. The appropriate fractions were pooled and concentrated to 20 mg/mL.

TRIM7-RING and TRIM7-RING-Box were expressed in *E. coli* C41 cells as GST–TEV fusion protein. Cleared cell lysates were prepared by sonication in 50 mM Tris at pH 8 (pH 9 for RING-Box), 150 mM NaCl, 2 mM DTT with the addition of 20% (*v*/*v*) BugBuster and complete protease inhibitors, followed by centrifugation 16,000× *g* for 30 min. Lysates were loaded onto GST beads and washed with lysis buffer, then cleaved with TEV protease overnight at 4 °C. Cleaved proteins were concentrated and separated using a HiLoad 26/600 Superdex 75 size exclusion column. The peak fractions were pooled, concentrated and frozen in aliquots at −80 °C. Ube1, Ube2N, Ube2V2, Ube2W were produced as previously described [26].

2.2. Constructs and Cloning

TRIM7-PRYSPRY constructs (both mouse and human) were ordered as synthetic genes encoding the residues 342–511 of TRIM7 and cloned into a double-digested pOPTH (introduces uncleavable MAHHHHHHM sequence at the N terminus) expression vector (KpnI, NdeI). MBP-hTRIM7-CC-PS was cloned by amplifying the hTrim7-PRYSPRY sequence and Gibson assembly with a synthetic gene fragment containing the CC sequence (residues 165–341) into a double-digested pOPTM (introduces hisMBP-TEVsite at the N terminus) expression vector (KpnI, NdeI). TRIM7-RING (residues 1–123) and TRIM7-RING-BOX (residues 1–168) constructs were cloned by PCR amplification from codon optimized (*E. coli*) synthetic DNA and Gibson assembly into double-digested (NdeI, HindIII) pOPTG expression vector (introduces GST-TEVsite at the N terminus).

The pEXN-2XHA plasmid backbone was used in comparison of signaling of different TRIM proteins. pEXN-2XHA-TRIM7 was generated by inserting a synthetic gene encoding full-length codon optimized TRIM7 sequence (GeneArt) into a double-digested pEXN-2XHA vector (EcoRI, XhoI). pEXN-2XHA-TRIM5 was used as previously described [27]. pEXN-2XHA-TRIM21 was cloned from constructs published previously [27]—sequence encoding full-length human TRIM21 was PCR amplified and inserted into double-digested pEXN-2XHA vector (EcoRI, XhoI).

pSMPP-2xHA-TRIM7 constructs were cloned from the pEXN-2XHA-TRIM7 plasmid. Relevant fragments (FL, ΔPRYSPRY(1–341), ΔRING(128–511)) were PCR amplified and cloned into BamHI linearized pSMPPv1 using Gibson assembly. Untagged pSMPP-TRIM7 constructs (FL, ΔPRYSPRY(1–341), ΔRING(128–511)) were ordered as synthetic gene fragments (GeneArt) designed for Gibson assembly into double-digested (NotI, BamHI)

pSMPP vector. N terminally mCherry tagged Trim7 constructs were generated from pSMPP-2X11A-Trim7 by PCR amplification and Gibson assembly into double-digested (XhoI, KpnI) pmCherry-C1 plasmid vector.

The pmEGFP-GYG1 plasmid was generated by restriction cloning of a synthesized gene fragment encoding the hsGYG1-GN1 sequence into the pmEGFP-C1 plasmid vector (BsrG1, EcoR1). For bacterial expression of hGYG1-GN1, the plasmid pOPTH-hGYG1-GN1 was generated by Q5 PCR amplification of the sequence from pmEGFP-GYG1 and Gibson assembled into a double-digested pOPTH (introduces uncleavable MAHHHHHH sequence at the N terminus) plasmid vector (KpnI, NdeI). The rest of the GYG1 protein constructs were ordered as synthetic genes and cloned into double-digested pOPTH (KpnI, HindIII). Plasmids encoding rbGYG1 and mGYG1 were ordered as synthetic genes encoding the full-length proteins and cloned into the pOPTH plasmid vector using restriction cloning. Truncated constructs rbGYG1VR-CIV, rbGYG1CIV, rbGYG11-246, rbGYG11-299 were generated from pOPTH-rbGYG1 by PCR amplification and restriction cloning into the same pOPTH vector.

All site-directed mutagenesis was performed with a Q5 PCR kit apart from mutagenesis of pSMPP-TRIM7-R385A, which was cloned by Gibson mutagenesis following manufacturer's instructions (NEB).

2.3. Cell Culture

HEK293T, BV-2 and HeLa-CD300lf cells were grown in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% FBS, 2 mM L-glutamine, 100 U/mL penicillin and 100 mg/mL streptomycin (GIBCO) at 37 °C with 5% CO_2. U2OS cells were grown in McCoy-5a medium supplemented with 10% FBS, 1XGlutaMAX, 100 U/mL penicillin and 100 mg/mL streptomycin.

2.4. NF-κB and AP-1 Signalling Experiments

HEK293T cells were seeded at 1×10^5 cells per well in 24 well plates 24 h before transfection. Each well was transfected with the indicated amounts of pEXN-2XHA-TRIM5/7/21 plasmids, together with 10 ng pGL4.32[luc2P/NF-κB-RE/Hygro] (NF-κB response element-dependent firefly luciferase, Cat. no 9PIE849) or pGL4.44 [luc2P/AP-1-RE/Hygro] (AP-1 response element-dependent firefly luciferase, Cat. no 9PIE411) and 5 ng pRL-TK (Cat. no E2241) using the Dual Luciferase Reporter Assay (Promega, Madison, WI, USA). Twenty-four hours post-transfection, cells were lysed in Passive Lysis Buffer, and sequential firefly and Renilla luminescence measured (BMG PHERAstar plate reader), according to manufacturer's instructions (Promega). Firefly luciferase luminescence was normalized to Renilla luciferase luminescence, and these values normalized to those of empty vector-transfected cells.

2.5. qPCR Signalling Experiments

HEK293T cells were seeded at 2.5×10^5 cells per well in a 6-well plate. The next day the cells were transfected with 250 ng of either pEXN-2X11A-TRIM7 or empty pEXN-11A vector, and the cells harvested after 8 h. RNA was extracted using the RNeasy mini kit (Qiagen, Hilden, Germany) and cDNA produced from 1 µg RNA using oligo dT and Superscript RT. Gene expression was monitored by TaqMan Gene Expression Assays (Applied Biosystems, Waltham, MA, USA) on a StepOnePlus Real-Time PCR System (Life Technologies, Carlsbad, CA, USA). Gene expression assays were from Applied Biosystems: ACTB (Hs01060665_g1), CXCL10 (Hs01124251_g1). Relative expression was quantified using the $2-\Delta\Delta Ct$ method.

2.6. Stable Cell Line Generation

Stable cell lines were generated by stable lentiviral transduction of recipient cell lines. To generate the relevant lentiviruses, the untagged pSMPP-TRIM7 constructs were co-transfected into HEK293T cells with pMDG2 (a gift from D. Trono [EPFL; Addgene, plasmid 12259]) and pCRV-GagPol (a gift from S. Neil, Kings College London) plasmids

using FuGENE6 (Promega) reagent and incubated for 3 days. Supernatants were harvested by filtration and used for transduction of HeLa cells expressing the mouse CD300lf [28] as previously described [20]. Briefly, 2.5 × 10^5 cells were infected with the appropriate lentivirus in a 6-well plate in DMEM supplemented with 3% FBS, 20 mM HEPES and 4 µg/mL Polybrene, and spinoculated at 1000× *g* and 25 °C for 45 min. The cells were then incubated at 37 °C for 48 h. Antibiotic selection was carried out with 2 µg puromycin until all untransduced control cells were dead. For SARS-CoV-2 infection experiments, HeLa cells expressing TRIM7 constructs and mouse CD300lf were further transduced with lentiviruses encoding the ACE2 receptor [29] and selected with 10 µg/mL of Blasticidin.

2.7. CVB3 Experiments

For production of Coxsackievirus, plasmid encoding for strain pH3 with an eGFP (eGFP-CVB3) (kind gift from Professor Charles Rice) was transfected into HEK293T cells using PEI. After 48 h, stock was harvested by 3 cycles of freeze-thawing and centrifugation to remove the debris. Collected stock (P0) was used to generate working stock (P1) by infecting fresh plate for HEK293T cells and harvesting as above. Virus titer was assessed by TCID50 using HeLa cells. For infection experiments, Hela-CD300lf stably expressing different TRIM7 constructs were seeded onto 24-well plates 24 h prior to infection. Cells were infected at MOI of 0.2, and infection levels determined by measuring eGFP fluorescence after 12 h on the incuCyte.

2.8. MNV1 Experiments

HeLa-CD300lf-TRIM7 cells suspended in DMEM supplemented with 10% FBS were incubated with the virus on an end-to-end rotor at 37 °C for 1 h. The cells were then washed twice in DMEM to remove the unbound virus and then incubated at 37 °C for the duration of the infection. The plate (both cells and supernatants) was frozen at −80 °C for 24 h post-infection, and TCID50 assays were subsequently carried out in BV2 cells as previously described [30].

2.9. SARS-CoV-2 Experiments

HeLa cells expressing versions of TRIM7, CD300lf and ACE2 were seeded into 96-well plates at a density of 1.5 × 10^4 day prior to infection. For infections, we used isolate of omicron variant BA.1 (EPI_ISL_7886688), which was a gift from Ravi Gupta (University of Cambridge). Media containing 2% FBS and virus at MOI = 0.2 PFU/cell was added to wells and incubated for 18 h. Cells were freeze-thawed and then lysed with buffer containing 0.25% Triton X-100, 50 mM KCl, 100 mM Tris–HCl pH 7.4, glycerol 40% and RNAsecure (1/100) for 5 min. Infectious virus was then inactivated at 95 °C for 5 min, and samples processed for RT-qPCR as described previously [31]. We used Luna Universal Probe One-Step kit following manufacturer's recommendations. Primer/probe for genomic viral RNA were CDC-N1. Primer probes for 18S control were described previously [32]. SARS-CoV-2_N_Positive control RNA was used as standard for the viral genomic N reactions. For 18S standard, DNA was synthesized and kindly gifted by Jordan Clarks and James Stewart (University of Liver-pool). Final concentrations of 500 nM for each primer and 125 nM for the probe were used. RT–qPCRs were run on ABI StepOnePlus PCR System with the following program: 55 °C for 10 min, 95 °C for 1 min and then 40 cycles of 95 °C denaturation for 10 s and 60 °C extension for 30 s. RNA copy numbers were obtained from standards, and then genomic copies of N normalized to 1010 copies of 18S.

2.10. Co-Localisation and Degradation Experiment

U2OS cells were seeded into 8-well µ-slide (ibidi, polymer) for co-localization or into standard tissue culture 24-well plates for bulk fluorescence quantification and protein quantification. For the former, 2–3 × 10^4 cells were seeded, and 5 × 10^5 cells were seeded in the latter. The next day, cells were co-transfected by a 1:1 mixture of the relevant plasmids using FuGENE6 and placed in the incuCyte (Sartorius, Göttingen, Germany) for

live imaging. After 18–20 h of transfections, the cells were harvested, and the cell pellets processed as described below. To quantify the degradation of the EGFP-GYG1 using bulk fluorescence by incuCyte, the total integrated intensity in each condition was divided by the green fluorescence area to account for different transfection efficiencies. Co-localization experiment was imaged using live-cell imagining with the LUMASCOPE LS720 (Etaluma, Carlsbad, CA, USA). Cells expressing low levels of each plasmid were selected for analysis to avoid overexpression artefacts and aggregated proteins. Images were processed in ImageJ, and data plotted in PRISM (GraphPad Software, San Diego, CA, USA).

2.11. Protein Quantification Using Capillary Based Western Blot (Simple Western)

Cells were lysed in 40 µL RIPA buffer supplemented with 1 × complete protease inhibitors (Roche, Basel, Switzerland), and the soluble protein recovered by centrifugation (40 min at 21 k × g); 30 µL of the lysate was diluted with 90 µL 0.1 × Sample Buffer 2 and further processed according to manufacturer's instructions. Jess instrument was used to analyze the samples (Protein Simple). Actin was used as the loading control (MAB8929, 1:1000) and detected with anti-mouse HRP. mCh-Trim7 was detected with rabbit polyclonal anti-mCherry (ab167453, 1:200), and EGFP-GYG1 was detected with rabbit polyclonal anti-GFP (NB600-303, 1:1000), both detected with anti-rabbit NIR secondary antibody. Protein levels were normalized to actin and to the fraction of cells transfected (determined with the incuCyte as the fraction of cells with detectable green fluorescence).

2.12. In Vitro Ubiquitination Experiment

In vitro ubiquitination reactions were carried out in 1× ubiquitination buffer (50 mM Tris·HCl, pH 7.4, 2.5 mM $MgCl_2$, 0.5 mM DTT) with the addition of 2 mM ATP, 0.5 µM His-E1, 1 µM Ube2W, Ube2N/Ube2V2, 8 µg ubiquitin (Ub) and 400 ng TRIM7-RING. Reaction mixtures were incubated at 37 °C for 1–4 h, quenched by the addition of LDS sample buffer and boiling at 95 °C for 5 min. Samples were resolved by LDS-PAGE and detected by immunoblot with TRIM7 or anti-Ub-HRP (Santa Cruz, sc8017-HRP P4D1, 1:1000) antibodies.

2.13. Ubiquitin Discharge Assay

To prevent auto-ubiquitination, lysine 92 of Ube2N was replaced with arginine [33]. Ube2NK92R was loaded with ubiquitin by mixing 40 µM of the E2 with 1 µM Ube1, 0.37 mM Ub and 3 mM ATP in 50 mM HEPES pH 7.5, 150 mM NaCl, 20 mM $MgCl_2$ and incubating the reaction at 37 °C for 30 min. The reaction was transferred to ice and used immediately. To observe E3 mediated discharge of ubiquitin, 2 µM ubiquitin-loaded E2 was mixed with 1.5 µM TRIM7-RING in 50 mM HEPES pH 7.5, 150 mM NaCl, 20 mM $MgCl_2$, 50 mM L-lysine, 2.5 µM Ube2V2. Samples were taken at the time points indicated in the text and mixed immediately with LDS sample buffer at 4 °C. The samples were boiled for exactly 20 s, resolved by LDS-PAGE and observed by immunoblot using anti-Ube2N (Bio-Rad, Hercules, CA, USA, AHP974, 1:1000) and detected by near-infrared detection (Odyssey, LI-COR, Lincoln, NE, USA).

2.14. Western Blotting

Stable cell lines were confirmed for overexpression of TRIM7 by Western blotting. Cells were lysed in RIPA buffer supplemented with 1 × complete protease inhibitors (Roche). Lysates were separated on a precast SDS-PAGE gel (NuPAGE, 4–12%), transferred onto nitrocellulose membranes and blotted for GAPDH (Ambion, AM4300; 1:20,000) and TRIM7 (generated in house, αTrim7; 1:250)

2.15. Antibody Generation and Animal Handling

C57BL/6 wild-type mice were obtained from Jackson Laboratories and bred in LMB ARES facility under SPF conditions. Thirteen-week-old mice were used for the immunization experiment, which was conducted in accordance with the moderate severity limit

protocol 7 and Home Office Animals (Scientific Procedures) Act (1986). All animal work was licensed under the UK Animals (Scientific Procedures) Act, 1986 and approved by the Medical Research Council Animal Welfare and Ethical Review Body. Mice (n – 12) were immunized subcutaneously (s.c) with 100 µg TRIM7-Ring-Box protein in PBS mixed with complete Freund's adjuvant, 200 µl of emulsion per animal. Mice received 3 rounds of s.c boosting with 50 µg protein mixed with incomplete Freund's adjuvant. Boosting was done on days 21, 42 and 79. Tail bleeds for ELISA analyses were collected on days 31, 58. Mice were sacrificed on day 93, and intracardiac blood was collected.

2.16. ELISA

A 96-well plate (Nunc) was coated overnight with 1 µg/mL of TRIM7-RING-BOX. The plate was blocked with 2% Marvell in PBS, 0.05% Tween 20 (MPBST) and incubated with diluted mouse sera. Bound antibodies were detected with goat anti-mouse IgG-GRP (Jackson Immunoresearch, 115-035-071).

2.17. Peptides

The peptides DNIKKKLDTYLQ and NLGLSMLLQ were ordered from St John's Innovation Centre, Cambridge, UK. Peptides DNIKRKLDTYLQ, TIEALFQ, AIEALFQ, TIEALFA, TIEALAQ, TIEALRQ, TIEALVQ, TIEALFE, TIEALEQ, LQ, LLQ, Ac-LLQ, AAAAAAA, AAAAALQ were synthesized by DesignerBioscience (Cambridge, UK), dissolved in DMSO to 50–200 mM and stored at 80 °C. Peptides HDDFGLQ, LEALEFQ were synthesized by GenScript (Oxford, UK) and dissolved in DMSO to 50 mM and stored at −80 °C.

2.18. Isothermal Titration Calorimetry (ITC)

Titrations were performed using the MicroCal iTC200 and MicroCal Auto-iTC200 instruments (Malvern Panalytical, Malvern, UK). Binding assays were carried out in 150 mM NaCl, 50 mM Tris pH 8, 1 mM DTT and 0–1% *v*/*v* DMSO. Generally, 13–20 injections were carried out with a DP of 6 µcal/s, 750 RPM, 20 25 °C, 150 180 s spacing and 5 6 s injection times of 2.5–3 µL. Titrations were performed with protein and peptide in cell and syringe interchangeably. Generally, TRIM7 in the cell was kept at 20 µM with peptides diluted to 400 µM and up to 2 mM for dipeptides. See Table S2 for details. Control titrations of titrant into buffer were carried out where appropriate. Data were analyzed using MicroCal PEAQ-ITC analysis software, using one site model to fit the data. For low c-value experiments (such as the LQ titration), the N was fixed to 1. For experiments with MBP-T7CCPS, the buffer was 165 mM NaCl, 50 mM Tris pH 8, 1 mM DTT, 0.5% *w*/*v* glycerol and 0–0.8% *v*/*v* DMSO. For experiments with rbGYG, mGYG and mTrim7-PRYPSRY, the protein samples were dialysed against 50 mM phosphate buffer, pH 6.5 and 1 mM DTT. ITC experiments were conducted on MicroCal ITC 200 at 15 °C or an AutoITC and analysed using a standard one-state model within MicroCal instrument software.

2.19. Differential Scanning Fluorimetry

Thermal stabilization assays were performed using the NanoTemper Prometheus NT48 instrument (Nanotemper Technologies, München, Germany). All experiments were performed in 150 mM NaCl, 50 mM Tris pH 8, 1 mM DTT and 1% *v*/*v* DMSO unless otherwise noted. Trim7 PRYSPRY was used at 10 µM and peptides at 90 µM. Samples were heated at 2 °C/min from 15 to 95 °C. Data were analyzed using the NanoTemper Prometheus Control Software, and the first derivative of the melt curves used to define the Tm for the apo and complexed samples. Independent experiments used the same peptide and protein stocks.

2.20. Crystallography

TRIM7 PRYSPRY protein was dialyzed against 20 mM HEPES, 50 mM NaCl, 1 mM DTT, pH 8.0) and concentrated to 15 mg/mL; 4 mM peptide was added and the complex crystallized in 0.2 M KCSN, 15% PEG 4000, 0.1 M NaOAc, pH 5.5 for GYG1 (N112-

DNIKRKLDTYLQ-COOH, 100 mM in DMSO), for RACO-1 (NH2-NLGLSMLLQ-COOH, 50 mM in DMSO) in 1.5 M $NaPO_4H_2$ and 1 M Sodium Citrate pH 5.5. For 2BC, Trim7 PRYSPRY at 13.15 mg/mL in 150 mM NaCl, 50 mM Tris pH 8 and 1 mM DTT was mixed 100:1 with the 2BC peptide (NH2-TIEALFQ-COOH, 100 mM in DMSO) and crystallized in 1 M Sodium Potassium Tartrate, 0.1 M HEPES pH 6.8. For the NS6 peptide (NH2-LEALEFQ-COOH, 50 mM in DMSO), the protein was mixed 50:1 and co-crystallized in 1 M Na/K Phosphate pH 5. Crystals were cryoprotected with reservoir solution containing 20–30% *v*/*v* glycerol. Data for GYG1 was collected on an in-house FR-E SuperBright high-brilliance rotating anode linked to an automated crystal mounting system (ACTOR; Rigaku, Tokyo, Japan) and processed with CCP4i [34] package as follows: data were indexed in iMOSFILM, point group was determined in POINTLESS and scaled in SCALA. Molecular replacement was performed with PHASER [35] using the TRIM21 PRYSPRY model 2IWG. After the first iteration, further structures were solved using the Trim7 model from the GYG structure. The MR model was refined with COOT [36] and REF-MAC5 [37]. Data for RACO-1 was collected at the ESRF on beamline ID23 and processed as for GYG1. Data for 2BC and NS6 were collected at the DLS (Didcot, UK) on beamline i24 and i04, respectively, processed using the FASTDP [38] and Xia2dials [39] pipeline and processed as described.

2.21. Phylogenetic Analysis

Coding sequences for TRIM7 and GYG1 were obtained from Ensembl v. 106 [40]. Multiple sequence alignments were constructed with Muscle v. 3.8.31 with default arguments [41]. Phylogenetic trees were reconstructed using raxml v. 8.2.12 (with "-f a -m PROTCATLGX -# 100" arguments) [42].

3. Results

3.1. TRIM7 Binds a Linear Epitope of GYG1

TRIM7 is a canonical TRIM protein with the same domain architecture of RING, B Box, Coiled-coil and PRYSPRY as the antiviral TRIM proteins TRIM5 and TRIM21 (Figure 1A). To understand how TRIM7 can interact with multiple diverse substrates, we initially focused on its originally reported ligand, rabbit glycogenin-1 (rbGYG1) [25]. Previously, the PRYSPRY domain was reported to be responsible for mediating binding [43], consistent with other TRIMs. Using recombinant proteins, we probed the interaction of TRIM7-PRYSPRY (PRYSPRY) with nested truncations of the rbGYG1 protein using ITC (Figure 1B). PRYSPRY clearly interacted with the full-length rbGYG1 with an affinity of 2.3 μM. This interaction was maintained in different orthologues of TRIM7 and GYG1 (Figure S1A–C). Glycogenin is an ancient enzyme found in early unicellular eukaryotes, but while the catalytic domain is highly conserved, the C-terminal sequence has undergone more recent divergence. Binding to TRIM7 was maintained when the catalytic domain was deleted, suggesting a possible regulatory relationship. The non-catalytic region could be subdivided further into a variable region (VR) that differs between classes and a C-terminal section that is relatively conserved in vertebrate subphylum ('conserved in vertebrates'; CIV). Binding was maintained to a minimal construct comprising the CIV alone (KD = 5.2 ± 0.3 μM) but lost when this region was deleted (Figures 1B and S1D,E). Structural prediction of the CIV region suggested it contains two α-helices. We, therefore, synthesized peptides corresponding to these predicted helices and found that a peptide comprising the 12 carboxy-terminal (C-term) residues (DNIKKKLDTYLQ—rbGYG1322-333) was sufficient to maintain TRIM7 binding (Figure 1B). Co-crystallization of the PRYSPRY with the equivalent peptide from human glycogenin-1 (DNIKRKLDTYLQ—hsGYG1322-333) confirmed binding and the helical nature of the GYG322-333 epitope (Table S1, Figure 1C, left). TRIM7 binding takes place at the same location within the PRYSPRY as used by TRIM21 PRYSPRY to bind its ligand, IgG Fc, further indicating that this is a specific interaction (Figure 1C, right).

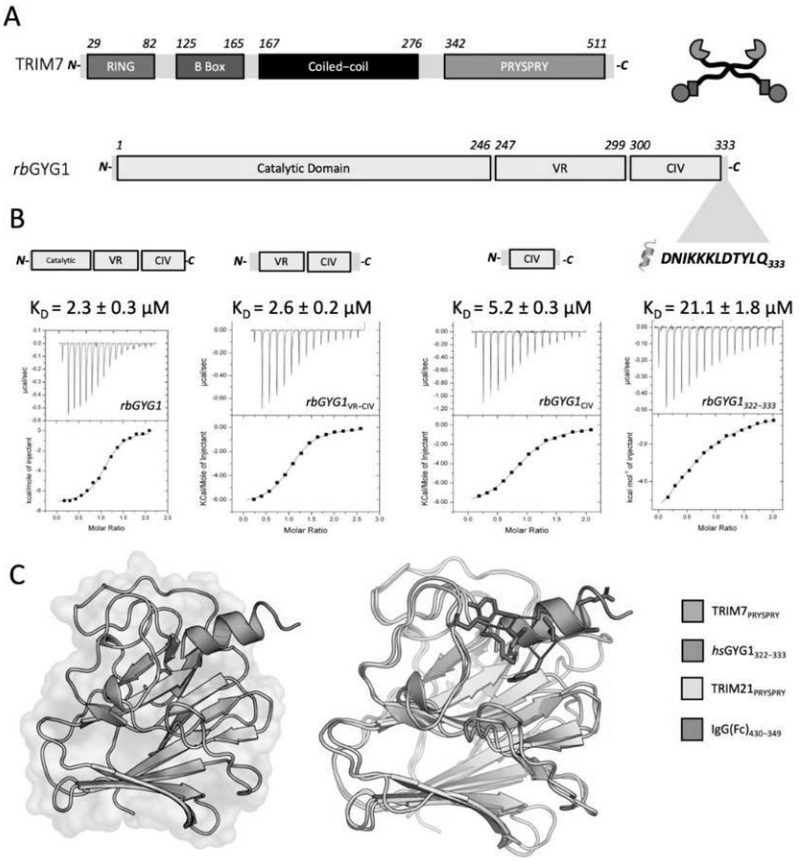


Figure 1. A linear epitope of GYG1 is sufficient to explain TRIM7 binding. (**A**) Sequence and domain organization of TRIM7 (grey) and Glycogenin1 (pale orange). TRIM7 is also depicted as a cartoon (circle—RING, rectangle—B-Box, line—Coiled-coil, cut-out circle—PRYSPRY) used throughout this paper. Highlighted are sequences determining TRIM7 binding. (**B**) ITC binding experiments with sequential truncations of rbGYG1 (shown above each titration) titrated into human hisTRIM7-PRYSPRY. Representative traces and their accompanying fitted KD's are shown. (**C**) Cartoon overview of the crystal structure of hisTRIM7-PRYSPRY (grey) with the GYG1 peptide (orange) on left (PDB accession 7OVX). Comparison with TRIM21:Fc structure (2IWG, PRYSPRY wheat, Fc magenta) on the right.

3.2. Cellular and Viral Substrates of TRIM7 Terminate with a Glutamine

Having identified the precise epitope within GYG1 bound by TRIM7, we searched for a similar sequence in TRIM7's other reported ligands. RACO-1 contains no similar epitope, but we noted that it also has a C-terminus ending in 'LQ'. TRIM7 has also been reported as a restriction factor for the picornavirus CVB3 and the calicivirus MNV. The genome of picornaviruses, and many other positive-stranded RNA viruses such as SARS-CoV-2, encode their major replicase components as a single polypeptide that is subsequently processed into individual proteins by the viral 3C protease (3Cpro). Importantly, 3Cpro cleaves after glutamine (Q), meaning this is the C-terminal residue of many of the viral proteins. CVB3 contains eight proteins ending in Q, while MNV1 contains two (Figure 2A). We hypothesized that this C-terminal Q may be the motif that TRIM7 recognizes in all its reported ligands, explaining its promiscuous binding. To test this, we synthesized peptides

corresponding to the C-termini of RACO-1, CVB3 2BC, which was previously identified as a possible ligand [20], and both MNV NS6 and NS3 (Figures 2B and S1J). Remarkably, in ITC experiments with TRIM7 PRYSPRY, all peptides bound with comparable affinity (Figures 2D and S1F–K and Table S2).

3.3. Crystals Structures of TRIM7 Bound to Substrate Peptides Reveal a Common ΦQ Binding Mechanism

Solving the co-crystal structures of each complex (Figures 2B–D and S2A–D,F) revealed that the different peptides are bound in the same PRYSPRY pocket, with the penultimate two residues responsible for most of the binding interaction. In each case, there is the potential for hydrogen bonds between the terminal glutamine residue and TRIM7 residues N383, R385, G408, Q436 and S499 (Figure 2C,D). An additional hydrogen bond is also formed between the peptidyl oxygen of the penultimate peptide residue (L or F) and the peptidyl nitrogen of TRIM7 residue T384. We tested the importance of these residues in ITC binding experiments (Figure 2D). Mutants N383A, R385A, F426 and Q436A abolished binding to all tested peptides, whilst T384A bound with reduced affinity (Figures 2D and S3). Mutation L423A effectively abolished binding, albeit there was some evidence of a weak binding signal in the ITC traces for the three peptide ligands. These results are consistent with the side-chains of R385 and Q436 participating in binding and only the main chain of T384. It is surprising that the bidentate salt bridge formed with R385 by the carboxylic acid of each peptide seems to play such an important role, given this interaction has no inherent residue specificity. Interestingly, an apo structure of TRIM7 was recently solved in which one of several malonate ions coordinated by the protein mimics both this carboxylic acid:R385 salt-bridge and the hydrogen bond with the main-chain of T384 (Figure S2E). Investigating this further, we found that mutating the C-terminal glutamine in the 2BC peptide to alanine abolished binding (Figure 2E). This suggests that interaction between the glutamine side-chain and Q436 in TRIM7 may be necessary to position the carboxy terminus for binding to R385. As 3Cpro occasionally cleaves after glutamate, we also tested whether this residue is also accommodated by TRIM7. However, a glutamine to glutamate substitution also abolished binding. This suggests that despite binding being driven by a seemingly nonspecific interaction involving a free carboxy termini, there is exquisite sequence specificity. The penultimate residue in 3Cpro cleavage has some variability. For instance, CVB3 proteins end with 'AQ', 'RQ', 'EQ', 'HQ', 'FQ' and 'VQ'. However, binding was only maintained in an 'FQ' peptide (Figure 2E). Previously, residue T323 was suggested as important in TRIM7 restriction of CVB3, as a T323A mutation allowed the virus to escape TRIM7 restriction [20]. However, the T323A mutation had no effect on binding, suggesting that this escape mutant may be altering viral fitness independent of TRIM7 (Figure 2E), or disrupting binding in the context of the full protein. To confirm these results with an independent method, we tested all of our 2BC peptides for TRIM7 interaction using differential scanning fluorimetry. We only observed an increase in PRYSPRY thermostability upon the addition of 2BC peptides ending in 'FQ' (Figure 2F).

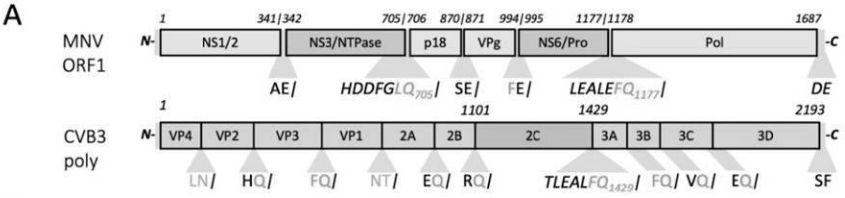


Figure 2. *Cont.*

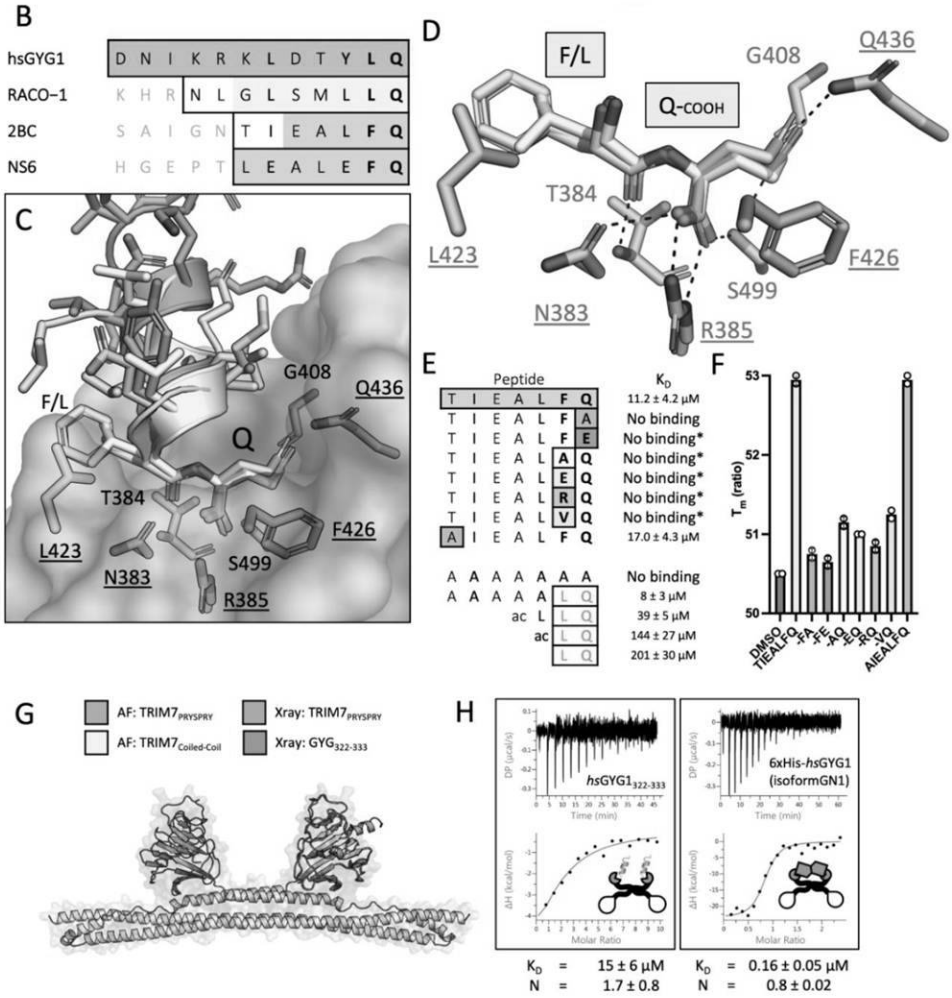


Figure 2. TRIM7 binds diverse substrates using a C-terminal helix-ΦQ motif. (**A**) MNV1 polyprotein (pale blue) and the relevant proteins in a darker shade (NS3 and NS6), CVB3 polyprotein in pale green and the relevant 2C protein in a darker shade. (**B**) Main peptide sequences used in the binding and structural experiments. Outlined are the sequences synthesized, whilst the shaded residues are those resolved in the crystal structures. Color coding is maintained throughout the figure. (**C**) hisTRIM7-PRYSPRY:Peptide complex structures superposed. Shows PRYSPRY as a transparent surface representation with a few key residues in stick. Peptides are color coded as described. PDB accession codes are 7OW2, 7OVX, 8A5L and 8A5M). (**D**) Detail of the PRYSPRY pocket with the recognition motif bound. Several key residues are highlighted. Dashes indicate charged or H-bonding between the peptide and the PRYSPRY residues. Underlined residues are essential for binding (see Figure S3). (**E**) Peptide substitutions based on the TIEALFQ peptide and possible polyprotein processing ends. Binding experiments with the minimal LQ motif, sequence is shown and the derived KD from ITC titrations. No binding* denotes where the peptide was only tested using nanoDSF. (**F**) Thermal denaturation data of hTRIM7-PRYSPRY derived using the Prometheus nanoDSF. Shows the Tm of the protein in the presence of peptide or DMSO. Peptides which bind stabilize the protein.

The results of two independent measurements are shown. (**G**,**H**) Avidity enhancement of binding affinity with hisMBP-TRIM7-CC-PRYSPRY. (**G**) Shows the AlphaFold prediction of the TRIM7 dimer (grey and wheat) aligned with the crystal structure (pale blue and orange). (**H**) The full-length GYG1 protein (isoform GN1) shows clear 1:1 binding between the protein dimers, whilst no difference in affinity is observed when the peptide is the substrate. Representative traces and their accompanying fitted KD's are shown.

3.4. Constrained C Terminal -ΦQ Motif Is Sufficient to Explain Monomeric TRIM7 Binding, but the Interaction Is More Potent When Both TRIM7 and Substrate Are Dimeric

The above data suggest that TRIM7 can interact with proteins ending in either 'LQ' or 'FQ'. To determine whether these residues alone are sufficient for binding, we compared the dipeptide LQ with N-terminal acetylated LQ (ac-LQ) and ac-LLQ. These bound with increasing affinity from 200 to 39 μM. Next, we compared the binding of a simple polyA peptide with one where the last two residues were LQ (Figure 2E). While no binding could be detected for polyA, binding of 8 μM was measured for AAAAALQ—the same as that of the peptides from identified TRIM7 ligands. Taken together, these data indicate that the L/FQ alone is sufficient for TRIM7 binding, but it needs to be presented in a constrained format, as seen with increasing affinity from the dipeptide to the full protein. In the context of TRIM7 as a restriction factor, this means that CVB3 proteins VP3, 3A and 2C and MNV proteins NS3 and NS6 are possible TRIM7 ligands. More generally, our results define TRIM7 ligands as proteins or peptides containing a C-terminal helix-ΦQ motif. However, although there are many human proteins that end with 'LQ' or 'FQ', most of these are unlikely to be TRIM7 substrates as other constraints such as steric accessibility are likely to play a factor. Affinity may also be important; those measured here represent monomer binding, whereas TRIM proteins are dimers. To test how dimerization may alter target affinity, we produced MBP-TRIM7-CC-PRYSPRY (CC-PS) and compared its binding to the hGYG1322-333 peptide, and a hexahistidine-tagged GYG1 construct comprising the full-length human GN1 isoform of GYG1, which most closely matches rabbit GYG1 (Figures 2G,H and S4). In the case of the peptide, the affinity was virtually unchanged (15 ± 6 μM), but the stoichiometry from the analysis ($N = 1.7 \pm 0.8$) indicated two peptides were binding to the dimer, as expected. For the full-length protein, the affinity increased 100-fold (0.16 ± 0.05 μM) compared to the peptide or 10-fold from PRYSPRY-hsGYG1-GN1 interaction ($KD = 2.1 \pm 0.6$), with the stoichiometry indicating binding is 1:1 between the dimers.

3.5. TRIM7 Contains an Active RING E3 Domain Capable of Self-Ubiquitination

TRIM7 has been shown to induce degradation of CVB3 2BC by catalyzing K48-chain ubiquitination, a signal for proteasomal recruitment [20]. This is similar to related proteins TRIM5 and TRIM21, which catalyze the ubiquitination of their substrates via their E3 RING domains, leading to substrate degradation and immune signaling via K48- and K63-linked ubiquitin chain synthesis. K48-chains are a signal for proteasomal recruitment, whereas K63-chains are immune second messengers that activate multiple signaling hubs such as TAK1 and TBK1. TRIM5 and TRIM21 first use the E2 Ube2W to monoubiquitinate their own N-termini. This monoubiquitin then acts as a primer for K63-chain extension by the hetero-dimeric E2 Ube2N/2V2. To investigate whether TRIM7 utilizes this same mechanism, we carried out a series of in vitro ubiquitination reactions using these E2s (Figure 3A,B). Incubation with Ube2W resulted in monoubiquitination of the TRIM7 RING, whilst Ube2N/2V2 did not modify the protein. However, the addition of both Ube2W and Ube2N/2V2 together resulted in the modification of TRIM7 with an extended ubiquitin chain (Figure 3C). To confirm that the TRIM7 RING is not simply the substrate for ubiquitination but is catalyzing the reaction, we performed discharge experiments and measured the kinetics of ubiquitin release from pre-charged Ube2N. The addition of TRIM7 RING substantially accelerated ubiquitin discharge from Ube2N, confirming that it is an active E3 ligase with this E2 enzyme (Figure 3D). Taken together, this shows that TRIM7 behaves similarly to TRIM5 and TRIM21 and can modify itself with K63-polyubiquitin.

To demonstrate that the RING domain is active in the context of the full protein and in cells, we overexpressed different constructs of HA-TRIM7 in 293Ts (Figure 3F). Overexpression of WT and ΔPRYSPRY constructs led to laddering consistent with its ubiquitination. Laddering was abolished upon over-expression of a ΔRING construct.

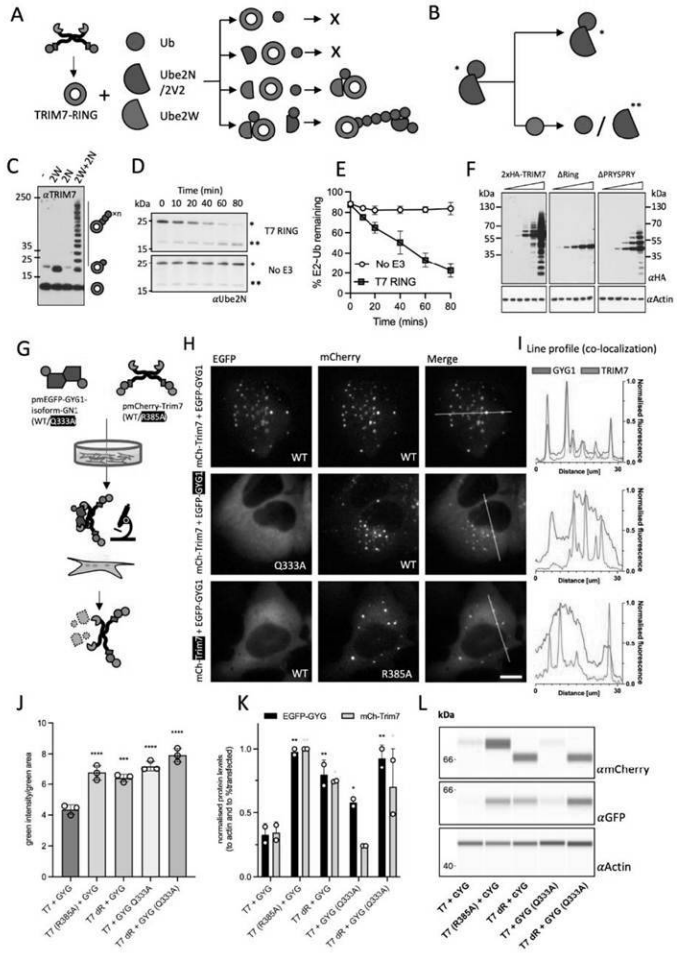


Figure 3. TRIM7 co-localizes with helix-ΦQ containing substrates inside cells and degrades them. (**A**–E) Schematic representation of in vitro ubiquitination experiments. (**A**) TRIM7-RING (grey circle) was mixed with combinations of Ubiquitin (Ub, blue circle), Ube2N/Ube2V2 (purple semi-circle) and Ube2W (green teardrop) as shown and the reactions followed by immunoblotting (**C**) with an anti-TRIM7 antibody. A representative blot from at least three independent experiments is shown. (**B**) E2 discharge experiment: Ube2N~Ub complex was incubated with TRIM7-RING, and the reaction followed over time as indicated in (**D**). Blots were probed with anti-ubiquitin antibody. A single asterisk denotes the charged Ube2N~Ub, whereas a double asterisk denotes the uncharged Ube2N. A representative blot from at least three independent experiments is shown. (**E**) Densitometry quantification of band intensities from 3D was plotted to show the kinetics of E2-Ub discharge. Error bars in all graphs depict the mean +/− SEM. Data represent three independent replicates. (**F**) Western

blots from cells overexpressing indicated constructs of epitope-tagged TRIM7 (probed with anti-HA antibody). Ubiquitin-laddering is lost when the RING domain is deleted. A representative blot from two independent experiments is shown. (**G**) Schematic overview of the experiments presented in (**H**). EGFP-GYG1 is represented as a green-brown shape. TRIM7 is shown as usual but with N-terminal mCherry (magenta circle). Plasmids were co-expressed, and the fluorescence monitored using live imaging and the protein levels quantified using either the fluorescence intensity or using cell lysates. (**H**) Live-cell microscopy of U2OS cells expressing mCherry-TRIM7 and EGFP-GYG1 constructs. Left column shows the EGFP signal (green), middle column shows the mCherry signal (magenta) and the right column shows the false-colored merged image (EGFP—green; mCherry—magenta; merge—white). The scale is the same in all images, and the bar represents 10 µm. Rows represent different conditions: Top is both WT sequences. Middle has a Q333A mutation in the EGFP-GYG1 construct. Bottom has the R385A mutation in the mCh-TRIM7 con-struct. Example images are shown from at least two independent experiments. (**I**) Line profile analysis (ImageJ) of the fluorescent signal. Green trace shows the EGFP signal whilst the magenta trace shows the mCherry signal. The line used in the analysis is shown on the merged signal images. (**J**) Fluorescence-based quantification (using the IncuCyte) of EGFP-GYG1 protein levels, graph shows the total integrated intensity of the EGFP signal divided by the EGFP area from three biological replicates. ANOVA was used for statistical analysis and significant differences from T7 + GYG condition indicated ($p < 0.0005$ (***), $p < 0.0001$ (****)). (**K**,**L**) Protein quantification and blots using cell lysates and capillary-based Western Blot (Jess). Values from two biological replicates were normalized to loading control (actin) and the fraction of cells transfected (see methods). ANOVA was used for statistical analysis and significant differences from T7 + GYG condition indicated ($p < 0.05$ (*), $p < 0.005$ (**)). Black asterisks indicate significance in EGFP-GYG expression and pink asterisks in mCh-TRIM7 expression.

3.6. TRIM7 Binding in Cells Leads to Codegradation with the Substrate

Next, we investigated whether TRIM7 binding of helix-ΦQ substrates leads to degradation in cells. We designed an experiment (Figure 3G) where we co-transfected cells with plasmids encoding mCherry-Trim7 (mCh-T7) constructs (WT and R385A) and EGFP-hsGYG1-GN-1 (EGFP-GYG1) constructs (WT and Q333A). mCh-T7 formed puncta, consistent with the cytoplasmic bodies formed by many TRIM proteins (Figures 3H and S5). When co-expressed, EGFP-GYG1 formed puncta that co-localized with mCh-T7 (Figure 3H,I; top row). Either Q333A substitution in GYG1 or R385A substitution in TRIM7 resulted in a loss of co-localization, suggesting that GYG1 is recruited to TRIM7 puncta through direct binding. To test if TRIM7 binding causes degradation of EGFP-GYG1, we quantified cellular protein levels by fluorescence microscopy (Figure 3J) and immunoblotting (Figure 3K,L). Strikingly, EGFP-GYG1 protein levels were ~3 times higher when binding to TRIM7 was prevented with either GYG1-Q333A or TRIM7-R385 substitutions, or the TRIM7 RING domain was deleted (dR). This data supports the hypothesis that TRIM7 targets helix-ΦQ substrates for degradation by direct binding and RING-dependent degradation. Importantly, we noted that TRIM7 is co-degraded alongside GYG1. This is a hallmark of TRIM ligases, and similar co-degradation has been observed during the function of both TRIM5 and TRIM21 [44–46].

3.7. Restriction of MNV1 and CVB3 Viral Replication Is Dependent on TRIM7 Binding and Substrate Degradation but Not Its Immune Signalling Activity

As TRIM7 is reported to restrict both MNV1 and CVB3 viruses, we sought to understand if the antiviral effect was due to binding and subsequent degradation of 3Cpro-liberated viral proteins with accessible helix-ΦQ C-termini or because of immune activation. We performed a series of experiments where we expressed increasing levels of TRIM7 or known antiviral proteins TRIM5 and TRIM21, which in addition to directly restricting viruses by mediating their degradation, have also been shown to stimulate immune signaling pathways. This resulted in dose-dependent activation of both NF-κB and AP-1 pathways (Figure 4A–C). Strikingly, TRIM7 activated both pathways substantially more strongly than either TRIM5 or TRIM21 and sufficiently to induce transcription of antiviral

cytokines TNF and CXCL10 (Figure 4D). Immune stimulation was dependent upon the RING domain but not the PRYSPRY, consistent with activity being induced by RING-mediated ubiquitination (Figure 4C). Consistent with this, the R385A mutant did not prevent TRIM7 from activating NF-κB (Figure 4E). Next, we generated cell lines stably expressing various TRIM7 constructs (Figures 4F and S6) in HeLa cells expressing the MNV1 receptor CD300lf. These cells were then used in infection experiments with recombinant CVB3-GFP or WT MNV1 virus (Figure 4G). For CVB3, we used a target cell infection assay in which we quantified the expression of a virally-encoded GFP gene using an IncuCyte (Figure 4H). Cells expressing WT TRIM7 were able to reduce CVB3 infection by ~4-fold, but neither ΔRING, ΔPRYSPRY or R385A constructs were capable of restriction. For MNV1, we quantified the production of infectious virus from target cells by TCID50 (Figure 4I). Here the antiviral effect was even more striking, with WT TRIM7 expressing-cells reducing MNV1 infection levels by 3–4 log10. However, it is not possible to say that TRIM7 is more effective against MNV1 than CVB3, given the nature of the different assays. Activity against MNV1 was completely lost in constructs R385A or ΔPRYSPRY, where binding to the substrate was abolished. There was some loss of infection in cells expressing ΔRING, which may indicate that the binding of TRIM7 to viral proteins alone partially disrupts their function. Importantly however, because immune signaling is PRYSPRY and R385A-independent, whereas viral restriction is lost in ΔPRYSPRY and R385A cells, our data suggest that TRIM7 antiviral activity is dependent upon binding and degradation and not activating innate immunity. Many ssRNA viruses produce a polyprotein that is cleaved by a virally-encoded 3Cpro, including SARS-CoV-2. Analysis of SARS-CoV-2 suggests there are 10 viral proteins that possess an 'LQ' or 'FQ' and thus are potential substrates for TRIM7 (Figure S7A). To test whether TRIM7 can restrict SARS-CoV-2, we used the same TRIM7/CD300lf HeLa cells that restrict both MNV1 and CVB3 and stably transduced them with hACE2 to make them permissive for SARS-CoV-2 infection (Figure S7B). We infected these hACE2/TRIM7/CD300lf cells with the omicron variant BA.1 (EPI_ISL_7886688) at an MOI of 0.2 TCID50 (Figure S7C). After 18 h, we measured viral replication by qPCR using probes against genomic viral RNA encoding N protein [31,32]. Unlike with MNV1 and CVB3, we observed no TRIM7 restriction of SARS-CoV-2 replication. This is in contrast to our previously published work with SARS-CoV-2, where we observed loss of infection in the presence of neutralizing antibodies [31,47] or reduced replication upon knockout of the cellular co-factor furin [29]. However, we do not rule out that under certain conditions or in different cell types, over-expressed TRIM7 may be capable of restricting SARS-CoV-2.

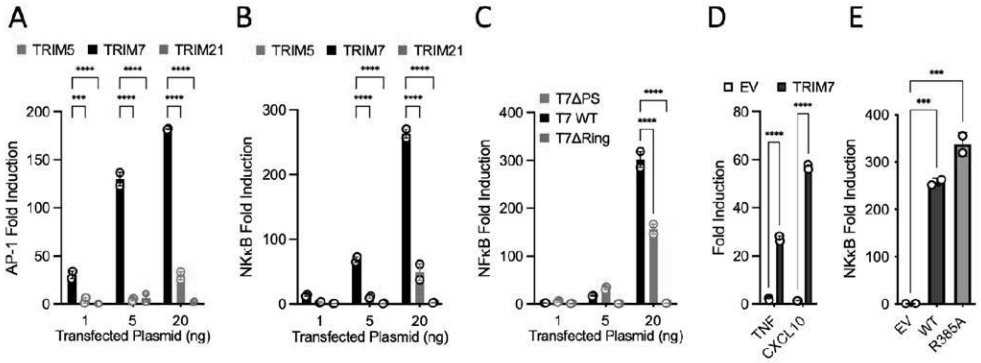


Figure 4. *Cont.*

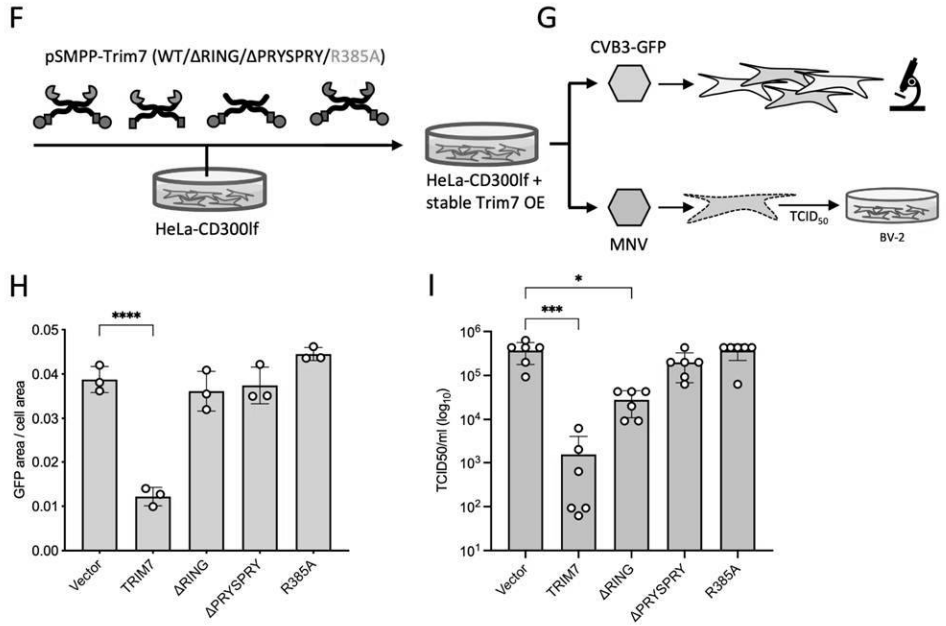


Figure 4. TRIM7 restriction of MNV1 and CVB3 infection is driven by helix-ΦQ binding. (**A**–**C**) Co-transfection of TRIM constructs and reporter plasmids. Overexpression of TRIM7 induces strong signaling by AP-1 and NF-κB that is dependent on the RING domain. (**D**) Overexpression of TRIM7 induces signaling, as measured by qPCR; values are normalized to actin. (E) Point mutation that prevents binding to targets does not impact signaling. For all signaling experiments (**A**–E), representative examples of at least two independent experiments are shown, with values normalized to cells transfected with empty vector. ANOVA was used for statistical analysis and significant differences indicated ($p < 0.0005$ (***), $p < 0.0001$ (****)). (**F**,**G**) Schematic overview of the experiments shown in H and I. Lentivirus generation and stable transfection of HeLa-CD300lf cells (**F**), with the expression confirmed by Western blotting (Figure S6). Cells were infected with recombinant CVB3 virus-producing EGFP in host cells or MNV1 (**G**). (**H**) Quantification of CVB3 infection by measuring the fraction of EGFP-expressing cells. Data shown are from three independent experiments. (**I**) Quantification of MNV1 replication by TCID50 following infection of TRIM7-expressing cells. Virions from lysed cells were titrated onto susceptible BV-2 cells. Data shown are a representative result of three independent experiments. Error bars show the standard deviation. Non-parametric ANOVA was used for statistical analysis and significant differences indicated ($p < 0.05$ (*), $p < 0.0005$ (***), $p < 0.0001$ (****)).

4. Discussion

The data presented here show that TRIM7 targets diverse substrates through a highly unusual binding mechanism in which it recognizes a C-terminal helix-ΦQ motif. This mechanism explains the promiscuous binding reported for TRIM7 against unrelated proteins GYG1, RACO-1, 2C, NS3 and NS6. During preparation of our manuscript similar data were deposited in bioRxiv [48], showing that TRIM7 binds to a ΦQ motif. The helix-ΦQ motif recognized by TRIM7 is reminiscent of the N-end degron system. In the 'N-end rule' system [49], N-recognin proteins hydrogen bond with free α-amino groups at protein N-termini [50]. TRIM7 recognizes the other end of proteins, using an arginine residue (R385) to form a bidentate salt bridge with the carboxylic acid at C-termini. Both systems also share similarities in how substrates can be generated. In the N-end system, proteolytic cleavage of N-terminal methionine exposes amino acids at the termini that differ in their

suitability as substrates for degradation. Likewise, proteolysis can reveal novel substrates for TRIM7 by exposing C-terminal glutamines. We propose that it is the action of 3C proteases that generates viral substrates for TRIM7 and results in the restriction of CVB3 and MNV1 infection. In support of our hypothesis, we show here that when C-terminal glutamine binding is abolished by mutating TRIM7 residue R385, CVB3 and MNV1 are no longer restricted.

As many positive-stranded RNA viruses utilize a 3C or 3C-like mechanism to liberate their proteins from an initial single polypeptide, there is the possibility that TRIM7 could function as a more broadly-acting restriction factor than antiretroviral ligase TRIM5. However, the ends generated by 3C proteases are not strictly conserved. For example, human norovirus proteins do not have identical C-termini as MNV1 proteins (e.g., in Norwalk virus, NS6 ends in EGETALE rather than LEALEFQ, but NS3 maintains a terminal LQ). Moreover, not all viral proteins with a helix-ΦQ motif will be accessible in the full-length form or if the protein is sequestered during viral replication in the cells. Nevertheless, it is surprising that we observed no restriction of live SARS-CoV-2, as the virus possesses 10 proteins ending in 'LQ' or 'FQ'. We purposefully performed our experiments in the same cells where we had observed restriction of both MNV1 and CVB3 so we could be sure there was functional TRIM7 present. However, it is possible that restriction requires a cell line that is more physiologically relevant for SARS-CoV-2 or that expresses even higher levels of TRIM7. During preparation of this manuscript, a pre-print report was submitted that claims TRIM7 can bind to and degrade multiple SARS-CoV-2 proteins, including NSP5 and NSP8 [48]. If similar conditions can be recreated during an infection, then perhaps restriction would be observed. Of course, it may also be that binding does not happen during live virus infection, or there is insufficient degradation to impact viral replication. Alternatively, SARS-CoV-2 may antagonize TRIM7 function.

An important question raised by this study is what the principle physiological function of TRIM7 actually is. Just because TRIM7 possesses antiviral activity does not mean this is its natural cellular function, particularly as we show such activity is dependent upon a highly promiscuous binding mechanism that makes ~1% of cellular proteins potential substrates. Both CVB3 and MNV viral screens were based on TRIM7 over-expression, as were many of the other reports identifying TRIM7 ligands. Ectopic TRIM7 over-expression may have created an opportunity for binding and ubiquitination that does not exist under normal physiological conditions. The link between TRIM7s helix-ΦQ binding mechanism and 3C-protease processing, while highly attractive, may be coincidental. A decreased CVB3 viral yield was reported in mouse myoblasts and cardiomyoctes treated with TRIM7 siRNAs [20], however, this may have been an indirect effect caused by reduced cell growth and metabolism in the depleted cells. TRIM7s link to glycogenin makes this a particularly valid possibility. We strongly suggest that future studies of TRIM7 restriction phenotypes, such as with SARS-CoV-2, avoid overexpression and use a knockout approach to avoid possible artefacts. Even so, it will be important to rule out confounding factors such as altered cell growth, and ultimately viral challenge experiments in TRIM7 knockout animals may be required for a definitive answer.

We have shown here that in addition to its reported degradation activity, over-expressed TRIM7 stimulates signaling pathways, including AP-1 and NF-κB, by synthesizing K63-polyubiquitin. Dual signaling/effector function is a signature of bona fide antiviral TRIMs, TRIM5 and TRIM21. Indeed, TRIM-based overexpression screens have previously identified multiple TRIMs with antiviral restriction and NF-κB activity [19]. However, it is possible that overexpression creates artefactual conditions for signaling. Over-expression can push TRIM proteins into forming cytoplasmic bodies that would not normally assemble under resting conditions [45], and these bodies are a source of ubiquitination and downstream signal transduction.

Phylogeny and tissue expression offer a useful perspective into TRIM7 function. TRIMs have undergone a rapid expansion in mammals: there are over 70 TRIMs in humans, whereas only 20 in worms and 10 in flies [51]. Recently expanded TRIMs comprise a

sub-group that evolves at a faster rate and contains a PRYSPRY domain [52]. Positive selection has been identified in many of these proteins, and while this includes TRIMs that have no known antiviral role, such a signature can be indicative of pathogen evolutionary pressure [53]. TRIM7 contains a single positively selected site in its coiled-coil, in contrast to the 21 sites found in antiretroviral TRIM5 [53]. TRIM7 is also older than antiviral TRIMs such as TRIM5 and TRIM21, which are only found in mammals. The evolutionary age of TRIM7 is a reasonable match to the appearance of the helix-ΦQ motif in glycogenin. Glycogenin arose in early eukaryotes, but the final ~30 residues did not become fixed until jawed vertebrates and the helix-ΦQ motif only in amniotes (~319 Mya; Figure S8 and [54]). TRIM7 is present in birds and reptiles, suggesting it is slightly younger (~261 Mya; Figure S8 and [54]). Tissue expression is another useful predictor of function, and in this respect TRIM7 and glycogenin are also a good match. TRIM7 is expressed principally in skeletal muscle and in the brain, both regions where glycogenin is also highly expressed (Figure S9). TRIM7 expression is highly tissue-specific, unlike antivirals TRIM5 and TRIM21, which are expressed almost everywhere. Moreover, TRIM7 tissue expression does not match that of the MNV1 receptor CD300lf, the CVB3 receptor CXADR or the SARS-CoV-2 receptor ACE2 (Figure S9), which would be strange if it had evolved to restrict these viruses. Finally, in contrast to Trim5 and Trim21, the Trim7 gene does not possess either an interferon-stimulated response element (ISRE) or gamma interferon-activated site (GAS), and its expression is not upregulated by interferon [55]. Taken together, neither phylogeny, tissue expression nor gene regulation are particularly supportive of TRIM7 being a broad-acting antiviral restriction factor. It is possible that none of the currently identified TRIM7 interacting proteins are the natural substrate—indeed, the real TRIM7 substrate may not be a protein at all but a metabolite. The malonate observed to bind apo TRIM7 makes the same core salt bridge as the glutamate side chain and carboxylic acid in the protein ligands. However, this moiety is not sufficient for full binding affinity, meaning that any natural metabolite would likely need to make additional interactions. Typically, an enzyme's function is revealed with its substrate, but TRIM7s unusual binding mechanism obviates such an approach. Future work will need to focus on TRIM7 in the tissues where it is naturally found, using knockout-out approaches and whole organism models.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/v14081610/s1, Figure S1: ITC traces of GYG1 binding to the TRIM7 PRYSPRY domain. Traces for hsGYG1:hTRIM7 binding (A), rbGYG1:mTRIM7 binding (B), mGYG1:mTRIM7 binding (C), rGYG11-246:hTRIM7 binding (D), rGYG11-299:hTRIM7 binding (E), GYG1339-350:hTRIM7 binding (F), RACO1229-237:hTRIM7 binding (G), 2BC1434-1440:hTRIM7 binding (H), NS61172-1178:hTRIM7 binding (I), NS3700-706:hTRIM7 binding (J), NS3700-706:hTRIM7-R385A binding (K), Ac-LLQ:hTRIM7 binding (L), Ac-LQ:hTRIM7 binding (M), AAAAAAA:hTRIM7 binding (N), AAAAALQ:hTRIM7 binding (O), LQ:hTRIM7 binding (P). Fitted values and more experimental information is shown in Table S2. In all cases, representative traces and their accompanying fitted KD's are shown. Figure S2: Crystal structures of TRIM7:peptide complexes. (A) TRIM7 PRYSPRY in surface representation, with the GYG1 peptide in stick (orange). Additionally, a sequence of residues from the PRYSPRY is shown of all four aligned structures in stick (relevant colours for each peptide complex). Below, an example 2FO-FC map is shown for the structure. (B–D) Structures of (B) TRIM7:RACO1 (yellow), (C) TRIM7:2BC (green) and (D) TRIM7:NS6 (blue) with example densities shown below. (E) Previously published apo structure with bound malate (turquoise) in the pocket (6UMB). (F) TRIM7:NS3 (accession 8A8X) overview shows TRIM7 (grey cartoon) in complex with NS3 (lilac, stick). Figure S3: ITC binding traces. Main: Peptides GYG1, RACO-1, 2BC and NS6 titrated against indicated mutant hisTRIM7-PRYSPRY. Right middle: PRYSPRY against 2BC peptide mutant T323A and the Q1440A mutation, which abolishes binding. Bottom right: ITC binding analysis of key PRYSPRY residues (Alanine substitutions). NT = not tested. * potential weak binding with KD estimated >100 μM. Fitted values and more experimental information is shown in Table S2. In all cases, representative traces and their accompanying fitted KD's are shown. Figure S4: Isoforms of GYG1 used in the study. Differences between rabbit GYG1 (top), human GYG1 (middle) and the GN1 isoform of human GYG1 (bottom). Figure S5: GYG1 co-localizes to TRIM7 cytoplasmic

bodies. Live-cell microscopy of U2OS cells expressing mCherry-TRIM7 and EGFP-GYG1 constructs. Left column shows the EGFP signal (green), middle column shows the mCherry signal (magenta) and the right column shows the false-colored merged image (EGFP—green; mCherry—magenta; merge—white). The scale is the same in all images, and the bar represents 10 µm. Rows represent different conditions: Top is both WT sequences. Middle has a Q333A mutation in the EGFP-GYG1 construct. Bottom has the R385A mutation in the mCh-TRIM7 construct. Figure S6: Stable cell line validation. Western blot from cells transduced with TRIM7 expressing lentivirus. Probed with a mouse anti-TRIM7 (1:250) antibody generated in-house. Figure S7: SARS-CoV-2 infection is not restricted by TRIM7. (A) SARS-CoV-2 proteins and their C-terminus dipeptide sequence. (B) Cells over-expressing TRIM7 constructs described in Figure 4 were further transduced with lentivirus en-coding human ACE2 to make the cells permissive for SARS-CoV-2 infection. Cells were infected, and viral load quantified using RT-qPCR. (C) Viral genome copies normalized to 18S. No significant differences were detected. Figure S8: Phylogenetic trees for GYG1 and TRIM7. Maximum-likelihood phylogenetic tree for GYG1 and TRIM7 using sequences from Ensembl v 106 [40]. Branch lengths correspond to the mean number of amino-acid substitutions per site. Figure S9: Tissue expression of TRIM7 and associated proteins. Transcripts per million (TPM) in selected tissues of TRIM proteins TRIM5, TRIM7, and TRIM21 together with GYG1 (glycogenin) and cell-surface receptors for different vi-ruses: CD300LF (MNV1), CXADR (CVB3) and ACE2 (SARS-CoV-2). Data taken from GTEx v8 [56]. Asterisks denote tissues with high TRIM7 and GYG1 expression. Table S1: Data collection and refinement statistics. TRIM7 PRYSPRY complexes with various ligands. Statistics for both data integration and model refinement are given. Table S2: ITC data. Concentration for proteins and peptides is given as monomer. * indicates where N was fixed in analysis. ** Concentration of GYG1 (and MBP-T7-CC-PS) dimer was used in the analysis. NB = no binding.

Author Contributions: Study was conceived by L.C.J. Manuscript was written by L.C.J. and J.L. and reviewed by all authors. Experiments and analysis were performed by all authors. Specifically, structural and biophysical data was collected predominately by J.L., with contributions from D.L.M., O.A. and L.C.J. Cell degradation and co-localization experiments were performed by D.C. and J.L. Signaling and ubiquitination experiments were performed by D.L.M. CVB3 infections were carried out by A.A. and MNV infections by A.S.J. Phylogenetic and evolutionary analysis was performed by G.S. and L.C.J. Funding was acquired by L.C.J. and I.G. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the MRC (UK; U105181010), a Wellcome Trust Investigator Award (200594/Z/16/Z) and a Wellcome Trust Collaborator Award (214344/A/18/Z). I.G. is a Wellcome senior fellow and supported by the Wellcome Trust (207498/Z/17/Z).

Institutional Review Board Statement: The animal study was conducted in accordance with moderate severity protocol 19b6 approved by the Animal Welfare and Ethical Review Body of the LMB MRC (date of approval 09 June 2015).

Informed Consent Statement: Not applicable.

Data Availability Statement: Data used in the study is contained within the article and the supplementary material. Crystallographic data were deposited to the PDB. Any additional information available on request.

Acknowledgments: We thank past and present members of the James lab, including David Jacques for help with data collection and the Biophysics Facility at MRC-LMB.

Conflicts of Interest: The authors declare that they have no competing interests.

References

1. Koepke, L.; Gack, M.U.; Sparrer, K.M. The antiviral activities of TRIM proteins. *Curr. Opin. Microbiol.* **2021**, *59*, 50–57. [CrossRef] [PubMed]
2. Bharaj, P.; Atkins, C.; Luthra, P.; Giraldo, M.I.; Dawes, B.E.; Miorin, L.; Johnson, J.R.; Krogan, N.J.; Basler, C.F.; Freiberg, A.N.; et al. The host E3-ubiquitin ligase TRIM6 ubiquitinates the Ebola virus VP35 protein and promotes virus replication. *J. Virol.* **2017**, *91*, e00833-17. [CrossRef] [PubMed]
3. Giraldo, M.I.; Xia, H.; Aguilera-Aguirre, L.; Hage, A.; van Tol, S.; Shan, C.; Xie, X.; Sturdevant, G.L.; Robertson, S.J.; McNally, K.L.; et al. Envelope protein ubiquitination drives entry and pathogenesis of Zika virus. *Nature* **2020**, *585*, 414–419. [CrossRef] [PubMed]

4. Wu, X.; Wang, J.; Wang, S.; Wu, F.; Chen, Z.; Li, C.; Cheng, G.; Qin, F.X. Inhibition of Influenza A virus replication by TRIM14 via its multifaceted protein-protein interaction with NP. *Front. Microbiol.* **2019**, *10*, 344. [CrossRef] [PubMed]
5. Stremlau, M.; Owens, C.M.; Perron, M.J.; Kiessling, M.; Autissier, P.; Sodroski, J. The cytoplasmic body component TRIM5α restricts HIV-1 infection in Old World monkeys. *Nature* **2004**, *427*, 848–853. [CrossRef]
6. Yuan, T.; Yao, W.; Tokunaga, K.; Yang, R.; Sun, B. An HIV-1 capsid binding protein TRIM11 accelerates viral uncoating. *Retrovirology* **2016**, *13*, 72. [CrossRef]
7. Ohainle, M.; Kim, K.; Komurlu Keceli, S.; Felton, A.; Campbell, E.; Luban, J.; Emerman, M. TRIM34 restricts HIV-1 and SIV capsids in a TRIM5α-dependent manner. *PLoS Pathog.* **2020**, *16*, e1008507. [CrossRef]
8. Di Pietro, A.; Kajaste-Rudnitski, A.; Oteiza, A.; Nicora, L.; Towers, G.J.; Mechti, N.; Vicenzi, E. TRIM22 inhibits Influenza A virus infection by targeting the viral nucleoprotein for degradation. *J. Virol.* **2013**, *87*, 4523–4533. [CrossRef]
9. Patil, G.; Xu, L.; Wu, Y.; Song, K.; Hao, W.; Hua, F.; Wang, L.; Li, S. TRIM41-mediated ubiquitination of nucleoprotein limits vesicular stomatitis virus infection. *Viruses* **2020**, *12*, 131. [CrossRef]
10. Lang, X.; Tang, T.; Jin, T.; Ding, C.; Zhou, R.; Jiang, W. TRIM65-catalized ubiquitination is essential for MDA5-mediated antiviral innate immunity. *J. Exp. Med.* **2017**, *214*, 459–473. [CrossRef]
11. Oshiumi, H.; Matsumoto, M.; Hatakeyama, S.; Seya, T. Riplet/RNF135, a RING finger protein, ubiquitinates RIG-I to promote interferon-beta induction during the early phase of viral infection. *J. Biol. Chem.* **2009**, *284*, 807–817. [CrossRef] [PubMed]
12. Cadena, C.; Ahmad, S.; Xavier, A.; Willemsen, J.; Park, S.; Park, J.W.; Oh, S.W.; Fujita, T.; Hou, F.; Binder, M.; et al. Ubiquitin-dependent and -independent roles of E3 ligase RIPLET in innate immunity. *Cell* **2019**, *177*, 1187–1200.e16. [CrossRef] [PubMed]
13. Seo, G.J.; Kim, C.; Shin, W.J.; Sklan, E.H.; Eoh, H.; Jung, J.U. TRIM56-mediated monoubiquitination of cGAS for cytosolic DNA sensing. *Nat. Commun.* **2018**, *9*, 613. [CrossRef] [PubMed]
14. Pertel, T.; Hausmann, S.; Morger, D.; Zuger, S.; Guerra, J.; Lascano, J.; Reinhard, C.; Santoni, F.A.; Uchil, P.D.; Chatel, L.; et al. TRIM5 is an innate immune sensor for the retrovirus capsid lattice. *Nature* **2011**, *472*, 361–365. [CrossRef]
15. McEwan, W.A.; Tam, J.C.; Watkinson, R.E.; Bidgood, S.R.; Mallery, D.L.; James, L.C. Intracellular antibody-bound pathogens stimulate immune signaling via the Fc receptor TRIM21. *Nat. Immunol.* **2013**, *14*, 327–336. [CrossRef]
16. Dickson, C.; Fletcher, A.J.; Vaysburd, M.; Yang, J.C.; Mallery, D.L.; Zeng, J.; Johnson, C.M.; McLaughlin, S.H.; Skehel, M.; Maslen, S.; et al. Intracellular antibody signalling is regulated by phosphorylation of the Fc receptor TRIM21. *eLife* **2018**, *7*, e32660. [CrossRef]
17. Fletcher, A.J.; Mallery, D.L.; Watkinson, R.E.; Dickson, C.F.; James, L.C. Sequential ubiquitination and deubiquitination enzymes synchronize the dual sensor and effector functions of TRIM21. *Proc. Natl. Acad. Sci. USA* **2015**, *112*, 10014–10019. [CrossRef]
18. Uchil, P.D.; Quinlan, B.D.; Chan, W.T.; Luna, J.M.; Mothes, W. TRIM E3 ligases interfere with early and late stages of the retroviral life cycle. *PLoS Pathog.* **2008**, *4*, e16. [CrossRef]
19. Uchil, P.D.; Hinz, A.; Siegel, S.; Coenen-Stass, A.; Pertel, T.; Luban, J.; Mothes, W. TRIM protein-mediated regulation of inflammatory and innate immune signaling and its association with antiretroviral activity. *J. Virol.* **2013**, *87*, 257–272. [CrossRef]
20. Fan, W.; Mar, K.B.; Sari, L.; Gaszek, I.K.; Cheng, Q.; Evers, B.M.; Shelton, J.M.; Wight-Carter, M.; Siegwart, D.J.; Lin, M.M.; et al. TRIM7 inhibits enterovirus replication and promotes emergence of a viral variant with increased pathogenicity. *Cell* **2021**, *184*, 3410–3425.e17. [CrossRef]
21. Orchard, R.C.; Sullender, M.E.; Dunlap, B.F.; Balce, D.R.; Doench, J.G.; Virgin, H.W. Identification of antinorovirus genes in human cells using genome-wide CRISPR activation screening. *J. Virol.* **2019**, *93*, e01324-18. [CrossRef] [PubMed]
22. Lu, M.; Zhu, X.; Yang, Z.; Zhang, W.; Sun, Z.; Ji, Q.; Chen, X.; Zhu, J.; Wang, C.; Nie, S. E3 ubiquitin ligase tripartite motif 7 positively regulates the TLR4-mediated immune response via its E3 ligase domain in macrophages. *Mol. Immunol.* **2019**, *109*, 126–133. [CrossRef] [PubMed]
23. Zhu, L.; Qin, C.; Li, T.; Ma, X.; Qiu, Y.; Lin, Y.; Ma, D.; Qin, Z.; Sun, C.; Shen, X.; et al. The E3 ubiquitin ligase TRIM7 suppressed hepatocellular carcinoma progression by directly targeting Src protein. *Cell Death Differ.* **2020**, *27*, 1819–1831. [CrossRef] [PubMed]
24. Chakraborty, A.; Diefenbacher, M.E.; Mylona, A.; Kassel, O.; Behrens, A. The E3 ubiquitin ligase Trim7 mediates c-Jun/AP-1 activation by Ras signalling. *Nat. Commun.* **2015**, *6*, 6782. [CrossRef]
25. Skurat, A.V.; Dietrich, A.D.; Zhai, L.; Roach, P.J. GNIP, a novel protein that binds and activates glycogenin, the self-glucosylating initiator of glycogen biosynthesis. *J. Biol. Chem.* **2002**, *277*, 19331–19338. [CrossRef]
26. Fletcher, A.J.; Christensen, D.E.; Nelson, C.; Tan, C.P.; Schaller, T.; Lehner, P.J.; Sundquist, W.I.; Towers, G.J. TRIM5α requires Ube2W to anchor Lys63-linked ubiquitin chains and restrict reverse transcription. *EMBO J.* **2015**, *34*, 2078–2095. [CrossRef]
27. Mallery, D.L.; McEwan, W.A.; Bidgood, S.R.; Towers, G.J.; Johnson, C.M.; James, L.C. Antibodies mediate intracellular immunity through tripartite motif-containing 21 (TRIM21). *Proc. Natl. Acad. Sci. USA* **2010**, *107*, 19985–19990. [CrossRef]
28. Jahun, A.S.; Sorgeloos, F.; Chaudhry, Y.; Arthur, S.E.; Hosmillo, M.; Georgana, I.; Izuagbe, R.; Goodfellow, I.G. Leaked genomic and mitochondrial DNA contribute to the host response to noroviruses in a STING-dependent manner. *bioRxiv* **2021**. [CrossRef]
29. Papa, G.; Mallery, D.L.; Albecka, A.; Welch, L.G.; Cattin-Ortola, J.; Luptak, J.; Paul, D.; McMahon, H.T.; Goodfellow, I.G.; Carter, A.; et al. Furin cleavage of SARS-CoV-2 Spike promotes but is not essential for infection and cell-cell fusion. *PLoS Pathog.* **2021**, *17*, e1009246. [CrossRef]
30. Hwang, S.; Alhatlani, B.; Arias, A.; Caddy, S.L.; Christodoulou, C.; Cunha, J.B.; Emmott, E.; Gonzalez-Hernandez, M.; Kolawole, A.; Lu, J.; et al. Murine norovirus: Propagation, quantification, and genetic manipulation. *Curr. Protoc. Microbiol.* **2014**, *33*, 15K. 2.1–15K. 2.61. [CrossRef]

31. Albecka, A.; Clift, D.; Vaysburd, M.; Rhinesmith, T.; Caddy, S.L.; Favara, D.M.; Baxendale, H.E.; James, L.C. A functional assay for serum detection of antibodies against SARS-CoV-2 nucleoprotein. *EMBO J.* **2021**, *40*, e108588. [CrossRef] [PubMed]
32. Ashraf, N.; Zino, S.; Macintyre, A.; Kingsmore, D.; Payne, A.P.; George, W.D.; Shiels, P.G. Altered sirtuin expression is associated with node-positive breast cancer. *Br. J. Cancer* **2006**, *95*, 1056–1061. [CrossRef] [PubMed]
33. McKenna, S.; Spyracopoulos, L.; Moraes, T.; Pastushok, L.; Ptak, C.; Xiao, W.; Ellison, M.J. Non-covalent interaction between ubiquitin and the human DNA repair protein Mms2 is required for Ubc13-mediated polyubiquitination. *J. Biol. Chem.* **2001**, *276*, 40120–40126. [CrossRef]
34. Collaborative Computational Project Number 4. The CCP4 suite: Programs for protein crystallography. *Acta Crystallogr. D Biol. Crystallogr.* **1994**, *50*, 760–763. [CrossRef]
35. McCoy, A.J.; Grosse-Kunstleve, R.W.; Adams, P.D.; Winn, M.D.; Storoni, L.C.; Read, R.J. Phaser crystallographic software. *J. Appl. Crystallogr.* **2007**, *40*, 658–674. [CrossRef] [PubMed]
36. Emsley, P.; Lohkamp, B.; Scott, W.G.; Cowtan, K. Features and development of Coot. *Acta Crystallogr. D Biol. Crystallogr.* **2010**, *66*, 486–501. [CrossRef] [PubMed]
37. Murshudov, G.N.; Skubak, P.; Lebedev, A.A.; Pannu, N.S.; Steiner, R.A.; Nicholls, R.A.; Winn, M.D.; Long, F.; Vagin, A.A. REFMAC5 for the refinement of macromolecular crystal structures. *Acta Crystallogr. D Biol. Crystallogr.* **2011**, *67*, 355–367. [CrossRef]
38. Winter, G.; McAuley, K.E. Automated data collection for macromolecular crystallography. *Methods* **2011**, *55*, 81–93. [CrossRef]
39. Winter, G.; Waterman, D.G.; Parkhurst, J.M.; Brewster, A.S.; Gildea, R.J.; Gerstel, M.; Fuentes-Montero, L.; Vollmar, M.; Michels-Clark, T.; Young, I.D.; et al. DIALS: Implementation and evaluation of a new integration package. *Acta Crystallogr. D Struct. Biol.* **2018**, *74*, 85–97. [CrossRef]
40. Howe, K.L.; Achuthan, P.; Allen, J.; Allen, J.; Alvarez-Jarreta, J.; Amode, M.R.; Armean, I.M.; Azov, A.G.; Bennett, R.; Bhai, J.; et al. Ensembl 2021. *Nucleic Acids Res.* **2021**, *49*, D884–D891. [CrossRef]
41. Edgar, R.C. MUSCLE: Multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res.* **2004**, *32*, 1792–1797. [CrossRef] [PubMed]
42. Stamatakis, A. RAxML version 8: A tool for phylogenetic analysis and post-analysis of large phylogenies. *Bioinformatics* **2014**, *30*, 1312–1313. [CrossRef]
43. Zhai, L.; Dietrich, A.; Skurat, A.V.; Roach, P.J. Structure-function analysis of GNIP, the glycogenin-interacting protein. *Arch. Biochem. Biophys.* **2004**, *421*, 236–242. [CrossRef] [PubMed]
44. Fletcher, A.J.; Vaysburd, M.; Maslen, S.; Zeng, J.; Skehel, J.M.; Towers, G.J.; James, L.C. Trivalent RING Assembly on retroviral capsids activates TRIM5 ubiquitination and innate immune signaling. *Cell Host Microbe* **2018**, *24*, 761–775.e6. [CrossRef]
45. Reymond, A.; Meroni, G.; Fantozzi, A.; Merla, G.; Cairo, S.; Luzi, L.; Riganelli, D.; Zanaria, E.; Messali, S.; Cainarca, S.; et al. The tripartite motif family identifies cell compartments. *EMBO J.* **2001**, *20*, 2140–2151. [CrossRef] [PubMed]
46. Zeng, J.; Santos, A.F.; Mukadam, A.S.; Osswald, M.; Jacques, D.A.; Dickson, C.F.; McLaughlin, S.H.; Johnson, C.M.; Kiss, L.; Luptak, J.; et al. Target-induced clustering activates trim-away of pathogens and proteins. *Nat. Struct. Mol. Biol.* **2021**, *28*, 278–289. [CrossRef] [PubMed]
47. Meng, B.; Ferreira, I.A.T.M.; Abdullahi, A.; Goonawardane, N.; Saito, A.; Kimura, I.; Yamasoba, D.; Gerba, P.P.; Fatihi, S.; Rathore, S.; et al. SARS-CoV-2 Omicron spike mediated immune escape and tropism shift. *bioRxiv* **2022**. [CrossRef]
48. Liang, X.; Xiao, J.; Li, X.; Wen, Y.; Che, X.; Ma, Y.; Zhang, X.; Zhang, Y.; Jian, D.; Wang, P.; et al. Structural insights into the viral proteins binding by TRIM7 reveal a general C-terminal glutamine recognition mechanism. *bioRxiv* **2022**. [CrossRef]
49. Bachmair, A.; Finley, D.; Varshavsky, A. In vivo half-life of a protein is a function of its amino-terminal residue. *Science* **1986**, *234*, 179–186. [CrossRef]
50. Sriram, S.M.; Kwon, Y.T. The molecular principles of N-end rule recognition. *Nat. Struct. Mol. Biol.* **2010**, *17*, 1164–1165. [CrossRef]
51. Ozato, K.; Shin, D.M.; Chang, T.H.; Morse, H.C., 3rd. TRIM family proteins and their emerging roles in innate immunity. *Nat. Rev. Immunol.* **2008**, *8*, 849–860. [CrossRef] [PubMed]
52. Sardiello, M.; Cairo, S.; Fontanella, B.; Ballabio, A.; Meroni, G. Genomic analysis of the TRIM family reveals two groups of genes with distinct evolutionary properties. *BMC Evol. Biol.* **2008**, *8*, 225. [CrossRef] [PubMed]
53. Malfavon-Borja, R.; Sawyer, S.L.; Wu, L.I.; Emerman, M.; Malik, H.S. An evolutionary screen highlights canonical and non-canonical candidate antiviral genes within the primate TRIM gene family. *Genome Biol. Evol.* **2013**, *5*, 2141–2154. [CrossRef]
54. Kumar, S.; Stecher, G.; Suleski, M.; Hedges, S.B. TimeTree: A resource for timelines, timetrees, and divergence times. *Mol. Biol. Evol.* **2017**, *34*, 1812–1819. [CrossRef]
55. Carthagena, L.; Bergamaschi, A.; Luna, J.M.; David, A.; Uchil, P.D.; Margottin-Goguet, F.; Mothes, W.; Hazan, U.; Transy, C.; Pancino, G.; et al. Human TRIM gene expression in response to interferons. *PLoS ONE* **2009**, *4*, e4894. [CrossRef] [PubMed]
56. Aguet, F.; Barbeira, A.N.; Bonazzola, R.; Brown, A.; Castel, S.E.; Jo, B.; Kasela, S.; Kim-Hellmuth, S.; Liang, Y.; Oliva, M.; et al. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *bioRxiv* **2019**. [CrossRef]

（参照文献２：Garcia-Diaz et al., Interferon Receptor Signaling Pathways Regulating PD-L1 and PD-L2 Expression, Cell Rep. 19(6):1189-1201 (2017)）

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

HHS Public Access
Author manuscript
Cell Rep. Author manuscript; available in PMC 2019 March 16.

Published in final edited form as:
Cell Rep. 2017 May 09; 19(6): 1189–1201. doi:10.1016/j.celrep.2017.04.031.

Interferon Receptor Signaling Pathways Regulating PD-L1 and PD-L2 Expression

Angel Garcia-Diaz[1,*], **Daniel Sanghoon Shin**[1], **Blanca Homet Moreno**[1,2], **Justin Saco**[1], **Helena Escuin-Ordinas**[1], **Gabriel Abril Rodriguez**[1], **Jesse M. Zaretsky**[1], **Lu Sun**[3], **Willy Hugo**[3], **Xiaoyan Wang**[4], **Giulia Parisi**[1], **Cristina Puig Saus**[1], **Davis Y. Torrejon**[1], **Thomas G. Graeber**[5,6,7], **Begonya Comin-Anduix**[7,8], **Siwen Hu-Lieskovan**[1], **Robert Damoiseaux**[5,9], **Roger S. Lo**[3,5,7], and **Antoni Ribas**[1,5,7,8,10,*]

[1]Division of Hematology-Oncology, Department of Medicine, David Geffen School of Medicine, University of California, Los Angeles (UCLA), Los Angeles, CA 90095, USA

[2]Division of Translational Oncology, Carlos III Health Institute, 28029 Madrid, Spain

[3]Division of Dermatology, Department of Medicine, UCLA, Los Angeles, CA 90095, USA

[4]Statistics Core, Department of Medicine, UCLA, Los Angeles, CA 90095, USA

[5]Department of Molecular and Medical Pharmacology, UCLA, Los Angeles, CA 90095, USA

[6]Crump Institute for Molecular Imaging, UCLA, Los Angeles, CA 90095, USA

[7]Jonsson Comprehensive Cancer Center, UCLA, Los Angeles, CA 90095, USA

[8]Division of Surgical Oncology, Department of Surgery, UCLA, Los Angeles, CA 90095, USA

[9]California NanoSystems Institute, UCLA, Los Angeles, CA 90095, USA

SUMMARY

PD-L1 and PD-L2 are ligands for the PD-1 immune inhibiting checkpoint that can be induced in tumors by interferon exposure, leading to immune evasion. This process is important for immunotherapy based on PD-1 blockade. We examined the specific molecules involved in

This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

*Correspondence: agdiaz@ipb.csic.es (A.G.-D.), aribas@mednet.ucla.edu (A.R.) http://dx.doi.org/10.1016/j.celrep.2017.04.031.
10Lead Contact

ACCESSION NUMBERS

The accession number for the RNA-seq data reported in this paper is GEO: GSE96619. The nCounter transcriptome is listed in Table S2.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, seven figures, and four tables and can be found with this article online at http://dx.doi.org/10.1016/j.celrep.2017.04.031.

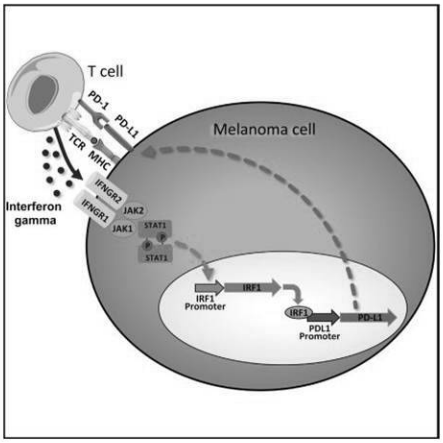
AUTHOR CONTRIBUTIONS

A.R. and A.G.-D. supervised the project and developed the concepts. A.G.-D., D.S.S., B.H.M., J.M.Z., G.P., C.P.S., R.D., B.C.-A., T.G.G., and A.R. designed the experiments. A.G.-D. performed vector cloning, promoter truncation analysis, qPCR assays, transcriptional profile analysis, and reporter cell line generation. A.G.-D. and D.S.S. performed ChIP analysis. J.S. assisted in the construct cloning and performed truncation studies. D.S.S. performed flow cytometry analysis. A.G.-D., R.D., and J.M.Z. carried out shRNA screen analysis. D.S.S. and H.E.O. performed western blot analysis. L.S., W.H., and R.S.L. performed RNA-seq analyses. S.H.-L. performed pathological analyses in biopsies. X.W. carried out statistical analysis. A.G.-D. and A.R. wrote the manuscript. All authors contributed to the manuscript and approved the final version.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

interferon-induced signaling that regulates PD-L1 and PD-L2 expression in melanoma cells. These studies revealed that the interferon-gamma-JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 axis primarily regulates PD-L1 expression, with IRF1 binding to its promoter. PD-L2 responded equally to interferon beta and gamma and is regulated through both IRF1 and STAT3, which bind to the PD-L2 promoter. Analysis of biopsy specimens from patients with melanoma confirmed interferon signature enrichment and upregulation of gene targets for STAT1/STAT2/STAT3 and IRF1 in anti-PD-1-responding tumors. Therefore, these studies map the signaling pathway of interferon-gamma-inducible PD-1 ligand expression.

In Brief



Garcia-Diaz et al. performed a small hairpin RNA screen and genetic and functional studies to map the signaling pathways that result in reactive PD-L1 and PD-L2 on melanoma cells upon interferon gamma exposure. The authors highlight the importance of the JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 axis for clinical responses to PD-1 blockade therapy.

INTRODUCTION

The signaling pathway resulting in adaptive expression of PD-L1 and PD-L2 upon exposure to interferons is of high importance for the clinical development of PD-1 blockade therapies for cancer. Upon tumor antigen recognition by T cells, the released interferons trigger the inducible expression of PD-L1 by cancer cells or other tumor microenvironment cells, thereby inhibiting the antitumor immune response in a process known as adaptive immune resistance. Adaptive immune resistance allows the specific inhibition of T cell recognition of cancer while it spares the rest of the immune responses to other antigens, avoiding a systemic immune-suppressive state (Pardoll, 2012; Ribas, 2015). Interferons were first described in the 1950s as agents that interfere with viral replication (Isaacs and Lindenmann, 1957), and signaling from the interferon receptors has been well characterized (Domanski and Colamonici, 1996; Novick et al., 1994; Velazquez et al., 1992). Janus kinase (JAK) and

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

signal transducer and activators of transcription (STAT) are the main signaling pathways mediating interferon-induced gene expression (Darnell et al., 1994; Velazquez et al., 1992) and resulting in the activation of interferon-stimulated response elements (ISREs) (Darnell et al., 1994; Kessler et al., 1988) and gamma interferon activation sites (GASs) (Decker et al., 1991; Lew et al., 1991). There is a renewed interest in interferon signaling given its key role in regulating PD-1 ligand expression and emerging evidence of its role in primary and acquired resistance to immune checkpoint blockade therapy for cancer (Shin et al., 2017; Gao et al., 2016; Zaretsky et al., 2016).

Type I interferons (alpha, beta, and omega) bind to interferon receptor type 1, which is composed of two subunits, IFNAR1 and IFNAR2, and they signal through JAK1 and TYK2, which phosphorylate STAT1, STAT2, and STAT3, as well as other STAT family members, depending on the cellular context. Activated phosphorylated STAT1 (pSTAT1) typically dimerizes with pSTAT2 to form the ISGF3 complex together with the interferon regulatory factor 9 (IRF9) (Smith et al., 2005). This complex binds at the genomic level to the ISRE sequences to control a long list of interferon-induced genes (Friedman and Stark, 1985). Type I interferons can also trigger phosphorylation and subsequent activation of homo- or hetero-dimers of STAT1, STAT3, STAT4, STAT5, and STAT6.

Type II interferon gamma binds to the interferon gamma receptor, leading to phosphorylation of JAK1 and JAK2, with receptor phosphorylation followed by receptor attachment and phosphorylation of STAT1 in most cells and STAT3 in some cells. The activated dimers then accumulate in the nucleus to act as transcription factors (Schroder et al., 2004; Aaronson and Horvath, 2002). There, they bind to the GAS elements present in most interferon gamma inducible genes, such as the IRF1 gene (Platanias, 2005). Negative regulators of interferon signaling, such as the suppressor of cytokine signaling protein family (SOCS; mostly SOCS1 and SOCS3) are involved in negative feedback regulation of cytokines that signal mainly through JAK2 binding, thereby modulating the activity of both STAT1 and STAT3 (Qing and Stark, 2004).

PD-1 has two known ligands, PD-L1 (CD274 or B7-H1) and PD-L2 (CD273 or B7-DC), and both have been reported to be expressed on cell surfaces upon exposure to interferons, in particular interferon gamma (Kim et al., 2005; Dong et al., 2002; Tseng et al., 2001). Evidence has been generated for the role of STAT1 and STAT3, as well as the downstream transcription factor IRF1, in regulating the surface expression of PD-L1 upon interferon gamma exposure (Lee et al., 2006; Loke and Allison, 2003). However, there has not been a systematic analysis of the molecules involvedin this signal transduction pathway. Given the importance of this process, we undertook a detailed analysis of molecules responsible for interferon receptor signaling and mapping of the PD-L1 and PD-L2 promoters to define the specific signaling that regulates their expression. Our studies demonstrate the key roles of signaling through the interferon-gamma-JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 axis, resulting in binding of the IRF1 transcription factor tothePD-L1 promoter and weaker binding to thePD-L2 promoter, which is also regulated by STAT3 in melanoma cells.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

RESULTS

Interferon Receptor Signal Transduction Pathway Regulating PD-L1 Expression

We analyzed the interferon-inducible surface expression of PD-L1 in three human melanoma cell lines, M244, M263, and M381, using flow cytometry (Figures 1A–1C). In all three cell lines, upregulation of PD-L1 was strongest with interferon gamma. Western blot analysis in the three cell lines (Figures 1D–1F) revealed the induction of proteins and phosphorylated proteins involved in the JAK-STAT signaling pathway, including STAT1/pSTAT1, STAT2/pSTAT2, and STAT3/pSTAT3 and increased expression of IRF1 and IRF9. Induction of pSTAT1 and IRF1 was consistently stronger upon interferon gamma exposure compared to interferon alpha or beta exposure. pSTAT1, pSTAT3, and IRF9 induction was also observed through interferon alpha and beta, consistent with the canonical type I interferon signaling pathway (Ivashkiv and Donlin, 2014). We also analyzed interferon receptor mRNA expression in these three lines. Expression of interferon receptor type II was higher than that of interferon receptor type I (Figure S1), which could explain in part the increased sensitivity of these cells lines to interferon gamma. Interestingly, IFNGR1 expression correlated with the PD-L1 induction under interferon gamma stimulation.

Interferon Receptor Pathway shRNA Screen to Define Signaling Molecules Involved in PD-L1 Regulation

To analyze PD-L1 regulation upon interferon gamma exposure, we generated reporter cell lines with luciferase expression downstream of the PD-L1 promoter to be used in a small hairpin RNA (shRNA) screen. The human melanoma cell lines M244, M263, and M381 were infected with a lentiviral vector carrying the PD-L1 promoter driving a polycistronic reporter cassette expressing both a DsRed-expressDR protein and firefly luciferase linked by a 2A picornavirus sequence (Figure 2A). This vector also included an elongation factor alpha promoter (EF1a)-BSD cassette as selectable marker. Cells were also transduced with a second vector used for assay signal normalization containing a constitutively expressed EF1-alpha promoter driving Renilla luciferase and RSV-Neo as selectable marker. After transduction and antibiotic selection, double stably transduced melanoma cell lines were validated for interferon gamma response and reporter expression (Figure S2).

In order to screen for individual targets known or anticipated to be involved in interferon receptor signaling and related pathways, we selected 180 shRNA hairpins targeting 33 genes to carry out the shRNA screen of known interferon receptor pathway signaling molecules. The doubly transduced reporter cell lines were then additionally transduced with the 180 shRNA lentiviral vectors. A non-hairpin-containing vector was used as a negative control. Transduced cells were induced for 8 hr with 100 U/mL interferon gamma, and firefly luciferase expression was normalized to the Renilla luciferase signal and the percentage of transduction of each virus. A redundant small interfering RNA (siRNA) activity (RSA) statistical analysis was performed to compensate for the effects of nonfunctional hairpins.

Inhibition of several genes involved in both type I and type II interferon signaling strongly affected PD-L1 reporter expression (Figures 2B–2D). We calculated the percentage of inhibition for each hairpin, and we assigned a score depending on the effect of each silenced

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

gene on the reporter expression in the three different cell lines. We then rank listed the silenced genes that had the strongest global effect in inhibiting PD-L1 expression. Each gene was then represented in a color heatmap in a schematic representation of the interferon receptor signaling pathways (Figure 2E). The data suggest that there are two converging bottlenecks in the signaling pathway: at the upstream tyrosine kinases JAK1, JAK2, and TYK2 and at the downstream transcription factors IRF1 and IRF9. In between, silencing of STAT1, STAT2, or STAT3 had a moderate effect on the PD-L1 reporter expression, suggesting redundancy in the signaling at this level. There was a strong effect in PD-L1 expression when silencing MAK14 (p38), CRKL, and phosphatidylinositol 3-kinase (PI3K), which have been previously reported to be involved in modulating interferon signaling pathways (Platanias et al., 1999). However, when we knocked out these three genes using CRISPR/Cas9, we could not detect a detrimental effect on PDL1 expression upon interferon gamma exposure (Figure S3); therefore, we believe these were off-target effects of the shRNA screen. We finally confirmed the role of JAK1, JAK2, STAT1, and IRF1 silencing in inhibiting PD-L1 expression using M381 cells stably transduced with shRNAs and analyzed by qPCR (Figure S4), and for JAK1 and JAK2, we have previously reported that their CRISPR/Cas9 knockout results in loss of PD-L1 upregulation upon interferon gamma exposure (Zaretsky et al., 2016).

IRF1 Is the Key Factor for PD-L1 Promoter Function

Analysis of the PD-L1 promoter sequence using the MotEvo algorithm (Pachkov et al., 2007) revealed putative binding sites for STAT1/STAT3, STAT2/STAT5, and IRF1 (Figure 3A). We performed site-directed mutagenesis to delete the STAT1/STAT3, STAT2/STAT5, and IRF1 putative binding sites in a PD-L1 promoter firefly luciferase reporter plasmid. Transiently transfected M381 cells were exposed to interferon gamma, and luciferase activity was quantitated (Figure 3B). Deletion of the IRF1 site dramatically decreased the expression of the PD-L1 reporter construct upon interferon gamma induction. Deletion of the putative STAT2/STAT5 site also affected interferon-gamma-induced luciferase expression, but at a lower level than IRF1. It should be noted that the proximity of these two sequences might be affecting the same activity. On the other hand, deletion of the STAT1/STAT3 putative binding site resulted in strong activation of the PD-L1 reporter upon interferon gamma induction. These data suggest the binding of a putative repressor factor at this level or the presence of genomic elements as silencers or insulator/boundary elements that could block the action of distal enhancers.

In order to confirm the binding of the predicted factors to the specific sequence sites, we carried out chromatin immunoprecipitation (ChIP) assays at the PD-L1 promoter. We confirmed IRF1 binding to the PD-L1 promoter in an interferon-gamma-inducible manner at a level that was much stronger than the positive control HLA-B promoter (Figure 3C). We also detected IRF1 binding to the PD-L1 promoter upon interferon beta exposure but at lower rates than under interferon gamma treatment (Figure S5A). ChIP analysis using STAT3 antibodies did not reveal direct binding of this factor at the PD-L1 promoter in M381 melanoma cells (Figure S5B), which is different from the reported binding of STAT3 to the PD-L1 promoter in a chimeric nucleophosmin (NPM)/anaplastic lymphoma kinase (ALK) T cell lymphoma (ALK-TCL) (Marzec et al., 2008).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

The JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 Axis Controls PD-1 Ligand and Antigen-Presenting Machinery upon Interferon Gamma Exposure

In order to analyze the relevant genes activated at transcriptional level by interferon gamma, we analyzed transcripts of 750 immune-related genes that capture the great majority of known interferon response genes (Table S2). Upon interferon gamma exposure of the three melanoma cell lines, there was strong activation of PD-L1 transcripts compared with PD-L2 (Figures 4A– 4C). We documented a very repetitive pattern of activation of JAK2, STAT1, STAT2, STAT3, IRF1, IRF9, and SOCS1 transcription upon interferon gamma stimulation. All of these molecules are known to be involved in the JAK-STAT signaling pathway downstream of the interferon gamma receptor (Aaronson and Horvath, 2002).

We also observed upregulation of a second group of genes (light-blue labels) related to immunoregulation and the antigen processing-presentation machinery through major histocompatibility complex (MHC) class I, including the transporters associated with antigen processing 1 and 2 (TAP1/TAP2) and the proteasome subunit beta types 8, 9, and 10 (PSMB9/LMP2, PSMB8/LMP7, and PSMB10/LMP10). Most of these genes are also known to be regulated by interferon gamma. TAP1 and PSMB9 share a bidirectional promoter activated through the interferon gamma-STAT1-IRF1 axis (Saha et al., 2010), and they work at the antigen-processing and antigen-presentation level through the immunoproteasome.

In order to confirm the importance of the JAK1/JAK2-STAT1/STAT3-IRF1 axis, we analyzed the expression profile induced by interferon gamma in three additional human melanoma cell lines with altered interferon receptor signaling (Figures 4D–4F). M368 has a *JAK2* loss-of-function mutation and M395 has a *JAK1* loss-of-function mutation, and we tested the effect of pharmacological JAK2 inhibition in M233, which is a good interferon gamma responder cell line (Shin et al., 2015, 2017). Interferon-related gene expression was tested in each cell line with or without interferon gamma exposure, and M233 was treated with or without co-incubation with the JAK2 inhibitor CEP33779 (Stump et al., 2011).

The *JAK1* mutated M395 cell line was strongly affected and dramatically failed to upregulate most of the previously seen interferon-inducible genes, such as JAK2, STAT1, STAT3, IRF1, PD-L1, and PD-L2, although it still conserved some degree of interferon gamma activation for some genes. The *JAK2* mutated M368 cell line presented a complete loss of the interferon gamma induction of the JAK-STAT genes, IRF1, and both PD-1 ligands. The antigen-presentation-related group of genes had equally flat responses to interferon gamma. To confirm the critical role of JAK2 in regulating interferon response, we treated the good responder M233 with the JAK2 inhibitor CEP-33779, which led to a downregulation of STAT3, IRF1, and the two PD-1 ligands, as well as some of the antigen-presentation-related group of genes, such as TAP1 and TAP2. Other interesting genes that were not induced are the chemokine CXCL10 and the interferon-inducible metabolic immune suppressor indoleamine 2,3-dioxygenase-1 (IDO1). These results confirm the importance of the JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 axis in the PD-L1 induction upon interferon gamma exposure.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

PD-L1 and PD-L2 Differential Regulation

Given the observations of different regulation of PD-L2 compared to PD-L1, we extended our studies to the PD-L2 promoter function and expression pattern. Flow cytometry analysis of PD-L2 surface expression (Figures 5A–5C) revealed similar or even stronger effect of interferon beta, compared to interferon gamma, in two of the three analyzed cell lines, suggesting a shared role of type I and II interferon signaling and differential regulation for the PD-1 ligands, as previously reported in other cell types (Loke and Allison, 2003). Because the expression of interferon type I receptors is lower than that of type II receptors in all the analyzed cell lines, there must be contribution of downstream elements to explain the similar or higher expression of PD-L2 in M244 and M263 under interferon beta exposure compared to interferon gamma exposure.

PD-L1 and PD-L2 are paralog genes, and their promoters share a similar architecture in terms of putative binding sites. We generated PD-L2 luciferase reporter constructs with specific deletions in STAT1/STAT3 and two IRF1 putative binding sites (α and β) (Figure 5D). First, we checked that PD-L2 promoter behaved very similar under interferon beta or gamma exposure (no statistically significant differences) in M381 and M244 cells (Figures 5E and S6), and then we performed reporter truncation assays comparing the PD-L2 reporter expression under interferon beta and gamma exposure or in untreated cells.

All mutations had a very similar effect on the PD-L2 promoter function independently of the type of interferon stimulation. Deletion of the IRF1α putative binding site decreased PD-L2 reporter expression upon interferon gamma induction, but at lower rates than on the PD-L1 promoter compared to the intact promoter (compare Figures 3B and 5E). In contrast, IRF1β site deletion had no effect on PD-L2 expression. Disruption of the STAT1/STAT3 putative binding site dramatically decreased PD-L2 promoter activation. These data suggest opposite effects of these sites at the PD-L1 and PD-L2 promoters, which is consistent with a differential regulation of the two PD-1 ligands. Interestingly, the double mutation of these sites had the lower reporter expression of the PD-L2 promoter, suggesting the cooperation of both sites and their binding factors IRF1 and STAT3 in the regulation of the PD-L2 expression.

In order to confirm the IRF1 and STAT3 participation in PD-L2 regulation, ChIP analysis was performed upon interferon gamma or beta exposure for both PD-L1 and PD-L2 promoters in M381 cells (Figures 5F and 5G). Upon interferon gamma exposure, binding of IRF1 to the PD-L2 promoter was strong, yet weaker than PD-L1 promoter binding. We also demonstrated a moderated, interferon-beta-inducible binding of STAT3 to the PD-L2 promoter, but not to the PD-L1 promoter or under interferon gamma exposure (Figures 5G and S5B). These results support the key role of IRF1 with STAT3 contribution to the differential regulation of PD-L1 and PD-L2 expression in melanoma cells, respectively.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

STAT1 and IRF1 Upregulation Correlates with PD-1 Ligand Expression and Interferon Signatures Enrichment in Biopsy Specimens of Patients Responding to Anti-PD-1 Blockade Therapy

To address whether our observations were recapitulated in human melanoma tumors, we analyzed RNA-sequencing (RNAseq) data from The Cancer Genome Atlas (TCGA) skin cutaneous melanoma RNA-seq database (Figures 6A and 6B). We found a strong correlation between IRF1 and PD-L1 or PD-L2 expression (Pearson correlation: R = 0.73 and R = 0.83, respectively), as well as between STAT1 and PD-L1 or PD-L2 expression (R = 0.78 and R = 0.74, respectively). These data support our observations in the cell line analyses that the correlation between STAT1/IRF1 and PD-L1 expression is strong.

We analyzed by RNA-seq the differential expression of interferon gamma and interferon gamma responsive gene signatures (Figure 6E) and of immune cell marker genes (Figure 6D) in biopsy specimens from five patients responding (red, n = 2) or not responding (black, n = 3) to anti-PD-1 therapy. These biopsy specimens represented a range of baseline CD8 and PD-L1 expression both in the tumor center and the invasive margin that were felt to be representative of most cases with melanoma (Table S3). Biopsy specimens from patients responding to anti-PD-1 had enriched expression of interferon gamma responsive genes. The non-responding biopsy specimens did not show interferon gamma upregulation or increased interferon signatures, which is supportive of our in vitro data.

On the other hand, we observed at least 1.5 (log2) fold changes in the expression of immune cell marker genes in on-treatment tumors compared to their respective baselines (Figure 6D). For the on-treatment tumors of the two responders, there was a general increase in the mRNA expression levels of multiple immune lineage markers, especially T cell and natural killer (NK) cell. On the contrary, for the three non-responders, T cell and NK cell markers were either not changed or downregulated.

Finally, we analyzed the expression of target genes of the IRF1, STAT1, STAT2, and STAT3 transcription factors in anti-PD-1 on-treatment biopsy specimens (Figure 6C). The small nodes represent the up-expressed genes in each sample, with binding associations indicted by the gray lines connected to the transcription factors. The absolute numbers of upregulated genes with the motifs of IRF1, STAT1, STAT1/STAT2 dimer, and STAT3 were higher in the responder biopsy specimens than in those from biopsy specimens of patients who did not respond to therapy (n indicates the number of up-expressed genes with binding motifs of transcription factors, while indicates N the total number of up-expressed genes in each sample).

DISCUSSION

Blocking the inducible PD-L1 expression upon tumor-antigen-specific T cell infiltration is the key event leading to response to anti-PD-1 or anti-PD-L1 antibody therapy in patients with cancer (Herbst et al., 2014; Tumeh et al., 2014). The clinical significance of the current work mapping the pathways that allow interferons to regulate the expression of PD-L1 and PD-L2 is highlighted by recent evidence that biopsy specimens of patients with metastatic melanoma who do not respond to anti-CTLA-4 or anti-PD-1 therapy are enriched for

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

mutations in the interferon receptor pathway (Gao et al., 2016; Shin et al., 2017) and the evidence that the selective pressure induced by a longstanding T cell response to melanoma with anti-PD-1 therapy can result in acquired resistance with loss-of-function mutations, with concomitant loss of heterozygosity, of *JAK1* or *JAK2* (Zaretsky et al., 2016). Therefore, a detailed understanding of the signaling pathways regulating the induction of PD-L1 and PD-L2 may help in defining additional mechanisms of primary or acquired resistance and test further improvements in this mode of therapy.

Our studies show that PD-L1 is mainly regulated by the type II interferon receptor singling pathway through JAK1 and JAK2, several STATs, and other modulators of the pathway and converged on the binding of IRF1 to the PD-L1 promoter. On the contrary, PD-L2 is regulated by both interferon beta and interferon gamma, with STAT3 and IRF1 being the transcription factors binding to its promoter in melanoma cells. This detailed knowledge may allow defining patients who cannot respond to PD-1 blockade therapy due to the cancer's inability to engage downstream interferon signaling, resulting in adaptive immune resistance when mutating or epigenetically silencing key molecules in this signaling, as demonstrated in preclinical models in vitro and in vivo (Dunn et al., 2005; Kaplan et al., 1998). Furthermore, this knowledge would allow defining new druggable molecules that could specifically modulate PD-L1 or PD-L2 signaling pathways, which may have advantages over current antitumor approaches given the lower cost of small molecules compared to antibodies. These agents may increase or decrease this signaling pathway, thereby having opposing potential uses in autoimmunity and transplantation tolerance or for cancer treatment.

Our studies document a differential regulation between the two PD-1 ligands depending on the cytokine signals, as well as cross talk between the interferon signaling pathways. Interferon gamma treatment resulted in a clear and repetitive upregulation pattern of the JAK2/STAT1/IRF1 axis and PD-L1, which is typical of the type II interferon canonical pathway. However, there was also upregulation of STAT2, STAT3 and IRF9 genes, more typically related to the type I interferon-signaling pathway, further highlighting the overlap between the pathways. We also found a very consistent IL-6 activation between the cell lines that could be contributing to the STAT3 upregulation via autocrine/paracrine action as previously described for other cancer cells (Sriuranpong et al., 2003).

To find the link between the interferon signaling pathways and the regulation of the expression of the PD-1 ligands, we dissected the promoters of both genes. It had been previously reported using electrophoretic mobility shift assays (EMSAs) that IRF1 binds to the PD-L1 promoter in vitro in human lung cancer cells (Lee et al., 2006). We carried out ChIP assays in M381melanoma cells, and we confirmed a strong binding of IRF1 to the PD-L1 promoter in an interferon-gamma-inducible manner. The PD-L1 promoter truncation assay indicated the likely presence of a repressor at the STAT1/STAT3 in-silico-predicted putative binding site, although a ChIP assay failed to detect STAT3 binding to this region, indicating possible participation of additional factors. For the PD-L2 promoter, both a transient truncation reporter assay and a ChIP assay revealed STAT3 participation in the regulation of this promoter. Interestingly, STAT3 binding was only detected upon interferon

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

beta induction, pointing out again the fact of the differential regulation of these two promoters.

The importance of the JAK1/JAK2-STAT1/STAT2/STAT3-IRF1 axis for the activation of the PD-1 ligands and many other interferon response genes is highlighted by the transcriptional profile of the two mutant melanoma cell lines that had loss of function of *JAK1* or *JAK2* genes and studies of the pharmacological inhibition of JAK2. Furthermore, we confirmed the relevance of our laboratory findings in patient-derived biopsy specimens. Tumor biopsy specimens from patients treated with PD-1 blocking antibody therapy upregulated interferon gamma signatures and target genes for IRF1, STAT1, STAT2, and STAT3 in anti-PD-1 on-treatment tumors. These data provide evidence of the importance of the related signaling pathway for an effective response to anti-PD-1 therapy. Whether this response is related exclusively to PD-1 ligand expression or is a consequence of antigen- presentation machinery regulation should be further investigated.

In conclusion, understanding the signaling pathway used by melanomas to respond to interferon exposure leading to the expression of PD-L1 and PD-L2 is of high importance to develop prognostic molecular markers and PD-1 blockade cancer immunotherapy. Genetic or epigenetic alterations affecting molecules in this pathway that result in lack of adaptive PD-1 ligand expression could be used for better patient selection for PD-1 blockade therapy. Modulation of the expression of factors involved in these pathways could define pharmaceutical targets to develop inhibitors that may block specific signaling resulting in adaptive immune resistance.

EXPERIMENTAL PROCEDURES

Cell Lines

Human melanoma cell lines of the M series were established from patient's biopsies under University of California, Los Angeles (UCLA) institutional review board (IRB) approval (11-003254), as previously described (Atefi et al., 2014). The double stable (DS) reporter cell lines DS244, DS263, and DS381 were generated by transducing M244, M263, and M381 cell lines with two different lentiviral particles (Figure 2A): EF1AProm-Renilla luciferase/RSV-BSD (GenTarget) and PD-L1Prom-DSRed-FireflyLuciferase/Neo lentivirus. Cells were infected at MOI 5–10 with 5 μg/mL polybrene (Santa Cruz Biotechnology) and then selected using 500–1,000 μg/mL G418 and 3–60 μg/mL of BSD during 3 weeks in RPMI media (Invitrogen, Thermo Fisher Scientific) with 10% fetal calf serum (FCS; Omega Scientific), 2 mM L-glutamine, and 1% penicillin, streptomycin (Invitrogen). Expression of both luciferases was validated after interferon gamma treatment for each generated reporter cell line as described in the reporter assay section (Figure S2).

Plasmids and Vectors

For the transient reporter analyses, PD-L1 and PD-L2 promoters were amplified by PCR from genomic DNA obtained from Jurkat cells using the primers listed in Table S1 and cloned into the multicloning site (MCS) of the PGL3 Basic Vector (Promega Corporation) as previously described (Sambrook et al., 1989; full methods can be found in Supplemental

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Experimental Procedures). Specific deletions of the putative binding sites were carried out using the Q5 Site-Directed Mutagenesis Kit (New England Biolabs), with primers and templates listed in Table S1. Normalization pRL-SV40P plasmid (Addgene plasmid 27163) was a gift from Ron Prywes (Chen and Prywes, 1999), who deposited the plasmid at the Addgene public repository.

For the stable reporter cell lines, the PD-L1Prom-DSRed-FireflyLuciferase/Neo lentiviral construct was cloned as described in Supplemental Experimental Procedures. shRNA lentiviral particles based on the pGIPZ lentiviral vector (Dharmacon) carrying hairpins for the specific genes were generated at UCLA's Molecular Screening Shared Resource (MSSR) from the shRNA Hannon collection. Plasmids were prepared using the PureLink HiPure Filter Plasmid Maxiprep Kit (Invitrogen) or NucleoSpin 96-well Miniprep kits (Macherey-Nagel), and lentiviral particles were produced as previously described (Kappes and Wu, 2001; Silva et al., 2005).

Surface Flow Cytometry Analysis of PD-L1 and PD-L2

Melanoma cell lines were seeded into six-well plates on day 1, targeting 70%–80% of confluence on the day of surface staining. On day 2, cells were exposed to 5,000 IU/mL interferon alpha (Merck Millipore), 500 IU/mL interferon beta (Merck Millipore), or 100 IU/mL interferon gamma (Becton Dickinson) for 18 hr. Interferon concentrations were defined after dose-response curve (PD-L1 mean fluorescence intensity as a function of interferon concentration) optimization processes for all three interferons (Shin et al., 2017).

On day 3, cells were trypsinized and incubated at 37°C for 2 hr with media containing the same concentrations of interferon alpha, beta, or gamma. After 2 hr of incubation, cells were stained with allophycocyanin (APC) anti-PD-L1 and phycoerythrin (PE) anti-PD-L2 antibodies on ice for 20 min and analyzed by flow cytometry using an LSRII (Becton, Dickinson and Company). Data were analyzed using FlowJo software (Tree Star). Experiments were performed at least twice for each cell line. Specificity of the PD-L1 antibody was previously reported (Atefi et al., 2014), while PDL2 antibody specificity was checked as described below using a PD-L2 siRNA approach (Figure S7).

siRNA Transfection

PD-L2 antibody specificity checking was performed on the two cell lines (M244 and M381), which were seeded on a six-well plate (target confluency of ~80% on the day of flow cytometry analysis) on day 1. Cells were transfected with 25 nM of a PD-L2 siRNA (GE Dharmacon, SMARTpool: siGENOME PDCD1LG2 siRNA and non-targeting control siRNA) as per the manufacturer's protocol on day 2. On day 3, the selected groups were exposed to 100 IU/mL interferon gamma, and flow cytometry analyses were performed on day 4 as described above.

Western Blot

Western blotting was performed as previously described (Escuin-Ordinas et al., 2014). Primary antibodies were purchased from Cell Signaling Technology (CST). Immunoreactivity was revealed with an ECL-Plus kit (Amersham Biosciences) using the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

ChemiDoc MP system (Bio-Rad). Selected melanoma cells were maintained in 10-cm cell culture dishes and treated with 5,000 IU/mL interferon alpha, 500 IU/mL interferon beta, or 100 IU/mL interferon gamma for 30 min and 18 hr.

Transient Luciferase Reporter Assays

M381 cells were seeded in 24-well plates for 18 hr and then transfected in triplicate using the TransIT-X2 (Mirus Bio) according to the manufacturer's manual with 0.5 μg of each experimental plasmid and 0.5 μg of the Renilla pRL-SV40P plasmid used for normalization per well. After 24 hr in culture, relative luciferase units (RLUs) were measured in non-treated and interferon-treated cells (3 hr, 100 IU/mL interferon gamma; 500, IU/ml interferon beta) using the Dual-Glo Luciferase Assay System and a GloMax 96 Microplate Luminometer (Promega) according to the manufacturer's instructions. RLUs from firefly luciferase signal were normalized by RLUs from Renilla signal.

shRNA Lentiviral Screen

1,000–1,500 cells of each reporter cell line were seeded in 384-well plates and transduced with 15 μL of each virus containing a shRNA hairpin or with the same empty vector without any active hairpin as a control. After 72 hr of culture, transduced cells were treated with 100 U/mL interferon gamma for 8 hr and then stained with 35 μM propidium iodide (PI) and 5 μg/mL Hoechst (Invitrogen). Cell images were acquired using an Image press XL (Molecular Devices) with a 10× objective (0.6 numerical aperture [NA]) in the DAPI, GFP (virus encoded), and CY3 channel. Images were analyzed using the MetaXpress multi-wavelength cell-scoring algorithm with standard settings in order to score cells and calculate percent viability and efficiency of transduction. RLUs for firefly and Renilla were analyzed using the Dual Glo Luciferase Assay System (Promega) according to the manufacturer instructions in a Victor3V luminometer (Perkin Elmer) with 0.1 s integration time. Firefly RLUs were normalized to the Renilla signal and the percentage of transduction to take into account any possible difference in cell viability and virus titer. A probability-based RSA analysis was performed in order to minimize the impact of the shRNA off-target activities as described in the statistical analysis section. Relative downregulation compared to the control were calculated and plotted for each gene and cell line (Figures 3B–3D). A global score based ranking was generated taking into consideration the position of each gene in the 3 cell lines and then was mapped onto the interferon pathways using PathVisio (v 3.2.1) (Kutmon et al., 2015). Pathway representation was based on interferon type I signaling (*Homo sapiens*) from Wikipathways (WP585) and edited for size and clarity and to include type II interferon pathway components (Kutmon et al., 2016).

ChIP

Formaldehyde-cross-linked chromatin was prepared from 2×10^7 M381 melanoma cells, and ChIP was performed using the SimpleChip Plus Enzymatic Chromatin IP Kit (Magnetic Beads 9005) from CST according to the manufacturer's instructions. Antibodies were purchased from CST (normal rabbit immunoglobulin G [IgG] 2729, STAT3 12640) and Abcam (IRF1 ab26109). To calculate DNA enrichment in the ChIP assays, Real-time qPCR was performed in an ABI 7500 (Applied Biosystems, Thermo Fisher Scientific) or a CFX96 real-time PCR system (Bio-Rad) using the iQ SYBR Green supermix (Bio-Rad) and the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

primers for the PD-L1, PD-L2, and HLA-B promoters listed in Table S1. SimpleChip C-FOS and human tRNA-Leu anti-codon (TAG) primers were purchased from CST, and HLA-B primers were previously reported (Stefan et al., 2011).

Gene Expression qPCR Assays

Downregulation of the genes in the generated cells lines was measured by qPCR using TaqMan gene expression assays and the TaqMan RNA-to-CT 1-Step Kit (Applied Biosystems, Thermo Fisher Scientific) following the vendor's specifications. Total RNA was extracted from cell lines in the absence or presence of interferon-gamma (100 IU/mL) at 3 hr using the QIAGEN AllPrep DNA/RNA Mini Kit RNeasy kit according to the manufacturer's protocol (QIAGEN).

nCounter Transcriptional Profiling Analysis

Total RNA was extracted from human melanoma cell lines in the absence or presence of interferon gamma (100 IU/mL) at 3 hr as described above. nCounter (NanoString Technologies) analysis was performed at the Center for Systems Biomedicine, a part of the Integrated Molecular Technologies Core (IMTC) at UCLA analyzing the genes detailed in Table S2. mRNA transcripts of specific cellular genes, including housekeeping genes for normalization, were quantified in untreated M233, M244, M263, M368, M381, and M395 cells and after 3 hr of interferon gamma exposure. M233 cells were also coincubated for 3 hr with 1 μM JAK2 pharmacological inhibitor CEP-33779 (Apex Bio). Results were analyzed using the 2.5 nSolver Software (NanoString Technologies).

Tumor Biopsies and Immunohistochemical Staining—Tumor biopsy specimens were obtained from patients receiving anti-PD-1 therapy under UCLA IRB 11-001918. Samples were immediately fixed in formalin followed by paraffin embedding and processed for snap-freezing in liquid nitrogen when an additional sterile piece of the tumor was present (Tumeh et al., 2014). For CD8 and PD-L1 analyses of tumor biopsy specimens, slides were stained with H&E, S100, CD8, and PD-L1 at UCLA's anatomic pathology immunohistochemistry laboratory. Immunostaining was performed on Leica Bond III autostainers using Leica Bond ancillary reagents and REFINE polymer 3,3′-diaminobenzidine (DAB) detection system. Percent positivity in the invasive margin or intratumoral area was calculated using the Indica Labs Halo platform as previously described (Tumeh et al., 2014).

CRISPR/Cas9-Mediated Knockout

The human melanoma cell line M233 was subjected to CRISPR/Cas9-mediated knockout of MAK14 (p38), CRKL, and PI3K by lentiviral transduction using particles encoding guide RNAs, a fully functional CAS9 cassette, GFP, and puromycin as selectable markers (Sigma-Aldrich), as previously described (Zaretsky et al., 2016). Two guide sequences were used per gene. GFP-positive single-cell clones were isolated using a FACSARIA sorter (Becton Dickinson). Disruption was confirmed by Sanger sequencing with tracking of indels by decomposition (TIDE) analysis (Netherlands Cancer Institute (NKI); https://tide.nki.nl) and finally by western blot.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

RNA-Seq Gene Set Enrichment and Transcription Factor Enrichment Analysis

RNA-seq data were generated using 2 × 100 bp paired-end sequencing using the Illumina HiSeq2000 platform. Paired-end reads were mapped to the UCSC hg19 reference genome using Tophat2 (Kim et al., 2013); full methods are included in Supplemental Experimental Procedures.

Statistical Analysis

Descriptive statistics such as mean and SD were calculated and are presented in the figures. Unpaired t tests were performed for reporter assay analysis as well as comparing percent input in ChIP analysis. For the shRNA screening, a probability-based RSA analysis was performed in order to minimize the impact of off-target activities of the different shRNA hairpins used in the study (König et al., 2007). The activities of the top hit shRNAs of each gene identified by RSA algorithm were then summarized and presented. nCounter expression profile analysis was carried out using the 2.5 nSolver Software (NanoString Technologies). The normalization module of this software used the popular geNorm algorithm (Vandesompele et al., 2002) to identify the most stable subset of housekeeping genes. From this, a gene expression normalization factor was calculated for each sample based on the geometric mean of the selected housekeeping genes. The normalized expression values of each sample were visualized and presented in scatterplots. Graph Pad Prism Version 6 was also used to generate plots and additional statistical analysis.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

This work was funded by NIH grants R35 CA197633 and P01 CA168585, the Parker Institute for Cancer Immunotherapy, the Dr. Robert Vigen Memorial Fund, the Ressler Family Foundation, the Samuels Family Fund, and the Garcia-Corsini Family Fund (to A.R.). A.R. was supported by a Stand Up To Cancer – Cancer Research Institute Cancer Immunology Dream Team Translational Research grant SU2C-AACR-DT1012). Stand Up To Cancer is a program of the Entertainment Industry Foundation administered by the American Association for Cancer Research. D.S.S. was supported by oncology (5T32CA009297-30), dermatology (5T32AR058921-05), and tumor immunology (5T32CA009120-39) training grants and a Tower Cancer Research Foundation grant. B.H.M. was supported by Juan Rodes Scholarship G83727016 from the Hospital 12 de Octubre, Madrid, Spain. Flow cytometry was performed in the UCLA Jonsson Comprehensive Cancer Center (JCCC) and Center for AIDS Research Flow Cytometry Core Facility, which is supported by NIH awards CA-16042 and AI-28697, and the JCCC, the UCLA AIDS Institute, and the David Geffen School of Medicine at UCLA. The NanoString experiments were run at the Center for Systems Biomedicine, a part of the Integrated Molecular Technologies Core (IMTC) supported by CURE/P30DK41301-26. We are grateful to Ron Prywes for the pRL-SV40P plasmid (Addgene plasmid 27163). Thanks to Bryan France and Bobby Tofig from the CNSI Molecular Screening Share Facility at UCLA for the help with the shRNA screen, Drs. Bin Liu and Kei Shuai from UCLA for guidance on interferon signaling and ChIP analyses, Shauna Gerold and Emmanuelle Faure-Kumar for assistance in NanoString data analysis, and Jeff Calimlim and the rest of the staff from the flow cytometry core laboratory at UCLA for the flow cytometry support.

References

Aaronson DS, Horvath CM. 2002; A road map for those who don't know JAK-STAT. Science. 296:1653–1655. [PubMed: 12040185]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Atefi M, Avramis E, Lassen A, Wong DJ, Robert L, Foulad D, Cerniglia M, Titz B, Chodon T, Graeber TG, et al. 2014; Effects of MAPK and PI3K pathways on PD-L1 expression in melanoma. Clin. Cancer Res. 20:3446–3457. [PubMed: 24812408]

Chen X, Prywes R. 1999; Serum-induced expression of the cdc25A gene by relief of E2F-mediated repression. Mol. Cell. Biol. 19:4695–4702. [PubMed: 10373518]

Darnell JE Jr, Kerr IM, Stark GR. 1994; Jak-STAT pathways and transcriptional activation in response to IFNs and other extracellular signaling proteins. Science. 264:1415–1421. [PubMed: 8197455]

Decker T, Lew DJ, Mirkovitch J, Darnell JE Jr. 1991; Cytoplasmic activation of GAF, an IFN-gamma-regulated DNA-binding factor. EMBO J. 10:927–932. [PubMed: 1901265]

Domanski P, Colamonici OR. 1996; The type-I interferon receptor. The long and short of it. Cytokine Growth Factor Rev. 7:143–151. [PubMed: 8899292]

Dong H, Strome SE, Salomao DR, Tamura H, Hirano F, Flies DB, Roche PC, Lu J, Zhu G, Tamada K, et al. 2002; Tumor-associated B7-H1 promotes T-cell apoptosis: a potential mechanism of immune evasion. Nat. Med. 8:793–800. [PubMed: 12091876]

Dunn GP, Sheehan KCF, Old LJ, Schreiber RD. 2005; IFN unresponsiveness in LNCaP cells due to the lack of JAK1 gene expression. Cancer Res. 65:3447–3453. [PubMed: 15833880]

Escuin-Ordinas H, Atefi M, Fu Y, Cass A, Ng C, Huang RR, Yashar S, Comin-Anduix B, Avramis E, Cochran AJ, et al. 2014; COX-2 inhibition prevents the appearance of cutaneous squamous cell carcinomas accelerated by BRAF inhibitors. Mol. Oncol. 8:250–260. [PubMed: 24345644]

Friedman RL, Stark GR. 1985; α-Interferon-induced transcription of HLA and metallothionein genes containing homologous upstream sequences. Nature. 314:637–639. [PubMed: 3990797]

Gao J, Shi LZ, Zhao H, Chen J, Xiong L, He Q, Chen T, Roszik J, Bernatchez C, Woodman SE. 2016; Loss of IFN-γ pathway genes in tumor cells as a mechanism of resistance to anti-CTLA-4 therapy. Cell. 167:397–404. [PubMed: 27667683]

Herbst RS, Soria J-C, Kowanetz M, Fine GD, Hamid O, Gordon MS, Sosman JA, McDermott DF, Powderly JD, Gettinger SN, et al. 2014; Predictive correlates of response to the anti-PD-L1 antibody MPDL3280A in cancer patients. Nature. 515:563–567. [PubMed: 25428504]

Isaacs A, Lindenmann J. 1957; Virus interference. I. The interferon. Proc. R. Soc. Lond. B Biol. Sci. 147:258–267. [PubMed: 13465720]

Ivashkiv LB, Donlin LT. 2014; Regulation of type I interferon responses. Nat. Rev. Immunol. 14:36–49. [PubMed: 24362405]

Kaplan DH, Shankaran V, Dighe AS, Stockert E, Aguet M, Old LJ, Schreiber RD. 1998; Demonstration of an interferon γ-dependent tumor surveillance system in immunocompetent mice. Proc. Natl. Acad. Sci. USA. 95:7556–7561. [PubMed: 9636188]

Kappes JC, Wu X. 2001; Safety considerations in vector development. Somat. Cell Mol. Genet. 26:147–158. [PubMed: 12465466]

Kessler DS, Levy DE, Darnell JE Jr. 1988; Two interferon-induced nuclear factors bind a single promoter element in interferon-stimulated genes. Proc. Natl. Acad. Sci. USA. 85:8521–8525. [PubMed: 2460869]

Kim J, Myers AC, Chen L, Pardoll DM, Truong-Tran Q-A, Lane AP, McDyer JF, Fortuno L, Schleimer RP. 2005; Constitutive and inducible expression of b7 family of ligands by human airway epithelial cells. Am. J. Respir. Cell Mol. Biol. 33:280–289. [PubMed: 15961727]

Kim D, Pertea G, Trapnell C, Pimentel H, Kelley R, Salzberg SL. 2013; TopHat2: accurate alignment of transcriptomes in the presence of insertions, deletions and gene fusions. Genome Biol. 14:R36. [PubMed: 23618408]

König R, Chiang CY, Tu BP, Yan SF, DeJesus PD, Romero A, Bergauer T, Orth A, Krueger U, Zhou Y, Chanda SK. 2007; A probability-based approach for the analysis of large-scale RNAi screens. Nat. Methods. 4:847–849. [PubMed: 17828270]

Kutmon M, van Iersel MP, Bohler A, Kelder T, Nunes N, Pico AR, Evelo CT. 2015; PathVisio 3: an extendable pathway analysis toolbox. PLoS Comput. Biol. 11:e1004085. [PubMed: 25706687]

Kutmon M, Riutta A, Nunes N, Hanspers K, Willighagen EL, Bohler A, Mélius J, Waagmeester A, Sinha SR, Miller R, et al. 2016; WikiPathways: capturing the full diversity of pathway knowledge. Nucleic Acids Res. 44(D1):D488–D494. [PubMed: 26481357]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Lee S-J, Jang B-C, Lee S-W, Yang Y-I, Suh S-I, Park Y-M, Oh S, Shin J-G, Yao S, Chen L, Choi IH. 2006; Interferon regulatory factor-1 is prerequisite to the constitutive expression and IFN-γ-induced upregulation of B7-H1 (CD274). FEBS Lett. 580:755–762. [PubMed: 16413538]

Lew DJ, Decker T, Strehlow I, Darnell JE. 1991; Overlapping elements in the guanylate-binding protein gene promoter mediate transcriptional induction by alpha and gamma interferons. Mol. Cell. Biol. 11:182–191. [PubMed: 1898761]

Loke P, Allison JP. 2003; PD-L1 and PD-L2 are differentially regulated by Th1 and Th2 cells. Proc. Natl. Acad. Sci. USA. 100:5336–5341. [PubMed: 12697896]

Marzec M, Zhang Q, Goradia A, Raghunath PN, Liu X, Paessler M, Wang HY, Wysocka M, Cheng M, Ruggeri BA, Wasik MA. 2008; Oncogenic kinase NPM/ALK induces through STAT3 expression of immunosuppressive protein CD274 (PD-L1, B7-H1). Proc. Natl. Acad. Sci. USA. 105:20852–20857. [PubMed: 19088198]

Novick D, Cohen B, Rubinstein M. 1994; The human interferon α/β receptor: characterization and molecular cloning. Cell. 77:391–400. [PubMed: 8181059]

Pachkov M, Erb I, Molina N, van Nimwegen E. 2007; SwissRegulon: a database of genome-wide annotations of regulatory sites. Nucleic Acids Res. 35:D127–D131. [PubMed: 17130146]

Pardoll DM. 2012; The blockade of immune checkpoints in cancer immunotherapy. Nat. Rev. Cancer. 12:252–264. [PubMed: 22437870]

Platanias LC. 2005; Mechanisms of type-I- and type-II-interferon-mediated signalling. Nat. Rev. Immunol. 5:375–386. [PubMed: 15864272]

Platanias LC, Uddin S, Bruno E, Korkmaz M, Ahmad S, Alsayed Y, Van Den Berg D, Druker BJ, Wickrema A, Hoffman R. 1999; CrkL and CrkII participate in the generation of the growth inhibitory effects of interferons on primary hematopoietic progenitors. Exp. Hematol. 27:1315–1321. [PubMed: 10428508]

Qing Y, Stark GR. 2004; Alternative activation of STAT1 and STAT3 in response to interferon-γ. J. Biol. Chem. 279:41679–41685. [PubMed: 15284232]

Ribas A. 2015; Adaptive immune resistance: how cancer protects from immune attack. Cancer Discov. 5:915–919. [PubMed: 26272491]

Saha B, Jyothi Prasanna S, Chandrasekar B, Nandi D. 2010; Gene modulation and immunoregulatory roles of interferon γ. Cytokine. 50:1–14. [PubMed: 20036577]

Sambrook, J, Fritsch, EF, Maniatis, T. Molecular cloning: a laboratory manual. Cold spring harbor laboratory press; 1989.

Schroder K, Hertzog PJ, Ravasi T, Hume DA. 2004; Interferon-γ: an overview of signals, mechanisms and functions. J. Leukoc. Biol. 75:163–189. [PubMed: 14525967]

Shin D, Garcia-Diaz A, Zaretsky J, Escuin-Ordinas H, Hu-Lieskovan S, Palaskas NJ, Hugo W, Komenan MS, Chmielowski B, Cherry G, et al. 2015; Innate resistance of PD-1 blockade through loss of function mutations in JAK resulting in inability to express PD-L1 upon interferon exposure. J. Immunother. Cancer. 3:311.

Shin DS, Zaretsky JM, Escuin-Ordinas H, Garcia-Diaz A, Hu-Lieskovan S, Kalbasi A, Grasso CS, Hugo W, Sandoval S, Torrejon DY, et al. 2017; Primary resistance to PD-1 blockade mediated by JAK1/2 mutations. Cancer Discov. 7:188–201. [PubMed: 27903500]

Silva JM, Li MZ, Chang K, Ge W, Golding MC, Rickles RJ, Siolas D, Hu G, Paddison PJ, Schlabach MR, et al. 2005; Second-generation shRNA libraries covering the mouse and human genomes. Nat. Genet. 37:1281–1288. [PubMed: 16200065]

Smith PL, Lombardi G, Foster GR. 2005; Type I interferons and the innate immune response–more than just antiviral cytokines. Mol. Immunol. 42:869–877. [PubMed: 15829276]

Sriuranpong V, Park JI, Amornphimoltham P, Patel V, Nelkin BD, Gutkind JS. 2003; Epidermal growth factor receptor-independent constitutive activation of STAT3 in head and neck squamous cell carcinoma is mediated by the autocrine/paracrine stimulation of the interleukin 6/gp130 cytokine system. Cancer Res. 63:2948–2956. [PubMed: 12782602]

Stefan M, Jacobson EM, Huber AK, Greenberg DA, Li CW, Skrabanek L, Conception E, Fadlalla M, Ho K, Tomer Y. 2011; Novel variant of thyroglobulin promoter triggers thyroid autoimmunity through an epigenetic interferon a-modulated mechanism. J. Biol. Chem. 286:31168–31179. [PubMed: 21757724]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Stump KL, Lu LD, Dobrzanski P, Serdikoff C, Gingrich DE, Dugan BJ, Angeles TS, Albom MS, Ator MA, Dorsey BD, et al. 2011; A highly selective, orally active inhibitor of Janus kinase 2, CEP-33779, ablates disease in two mouse models of rheumatoid arthritis. Arthritis Res. Ther. 13:R68. [PubMed: 21510883]

Tseng S-Y, Otsuji M, Gorski K, Huang X, Slansky JE, Pai SI, Shalabi A, Shin T, Pardoll DM, Tsuchiya H. 2001; B7-DC, a new dendritic cell molecule with potent costimulatory properties for T cells. J. Exp. Med. 193:839–846. [PubMed: 11283156]

Tumeh PC, Harview CL, Yearley JH, Shintaku IP, Taylor EJ, Robert L, Chmielowski B, Spasic M, Henry G, Ciobanu V, et al. 2014; PD-1 blockade induces responses by inhibiting adaptive immune resistance. Nature. 515:568–571. [PubMed: 25428505]

Vandesompele J, De Preter K, Pattyn F, Poppe B, Van Roy N, De Paepe A, Speleman F. 2002; Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. Genome Biol. 3:H0034.

Velazquez L, Fellous M, Stark GR, Pellegrini S. 1992; A protein tyrosine kinase in the interferon α/β signaling pathway. Cell. 70:313–322. [PubMed: 1386289]

Zaretsky JM, Garcia-Diaz A, Shin DS, Escuin-Ordinas H, Hugo W, Hu-Lieskovan S, Torrejon DY, Abril-Rodriguez G, Sandoval S, Barthly L, et al. 2016; Mutations associated with acquired resistance to PD-1 blockade in melanoma. N. Engl. J. Med. 375:819–829. [PubMed: 27433843]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Highlights

- PD-L1 is primarily regulated by interferon gamma signaling in melanoma cells
- PD-L2 is regulated by both interferon beta and gamma signaling
- Regulation of PD-1 ligands works mainly through the JAK1/2-STAT1/3-IRF1 axis

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

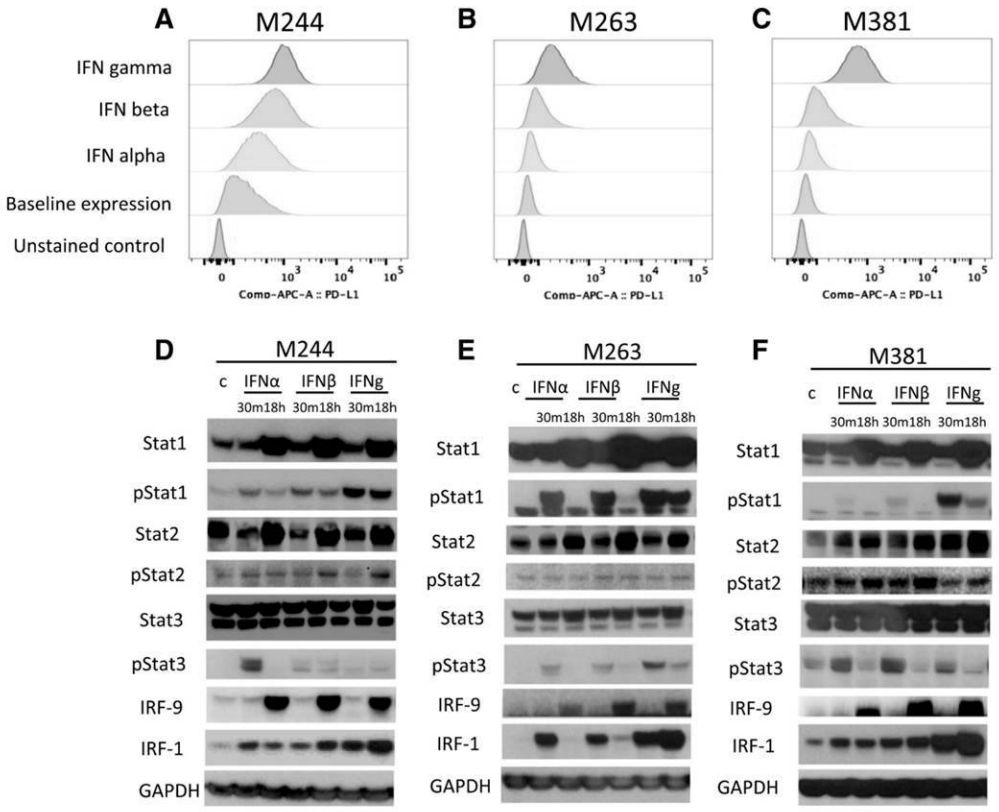


Figure 1. Induction of PD-L1 by Interferon Alpha, Beta, and Gamma

(A–C) Flow cytometry analysis of PD-L1 surface expression upon interferon treatment in the human melanoma cell lines M244 (A), M263 (B), and M381 (C) exposed to interferon alpha, beta, or gamma for 18 hr. Histograms represent changes in mean fluorescence intensity by flow cytometry compared to baseline. (D–F) Western blot analysis of interferon receptor signaling proteins in M244 (D), M263 (E), and M381 (F), including a set of proteins involved in interferon signaling pathways. Basal and activated (phosphorylated) states of the proteins are included to compare the induction trough these different mediators. The first lane of each panel represents untreated control cells. Each cell line was exposed 30 min or 18 hr with interferon alpha, beta, or gamma, respectively.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

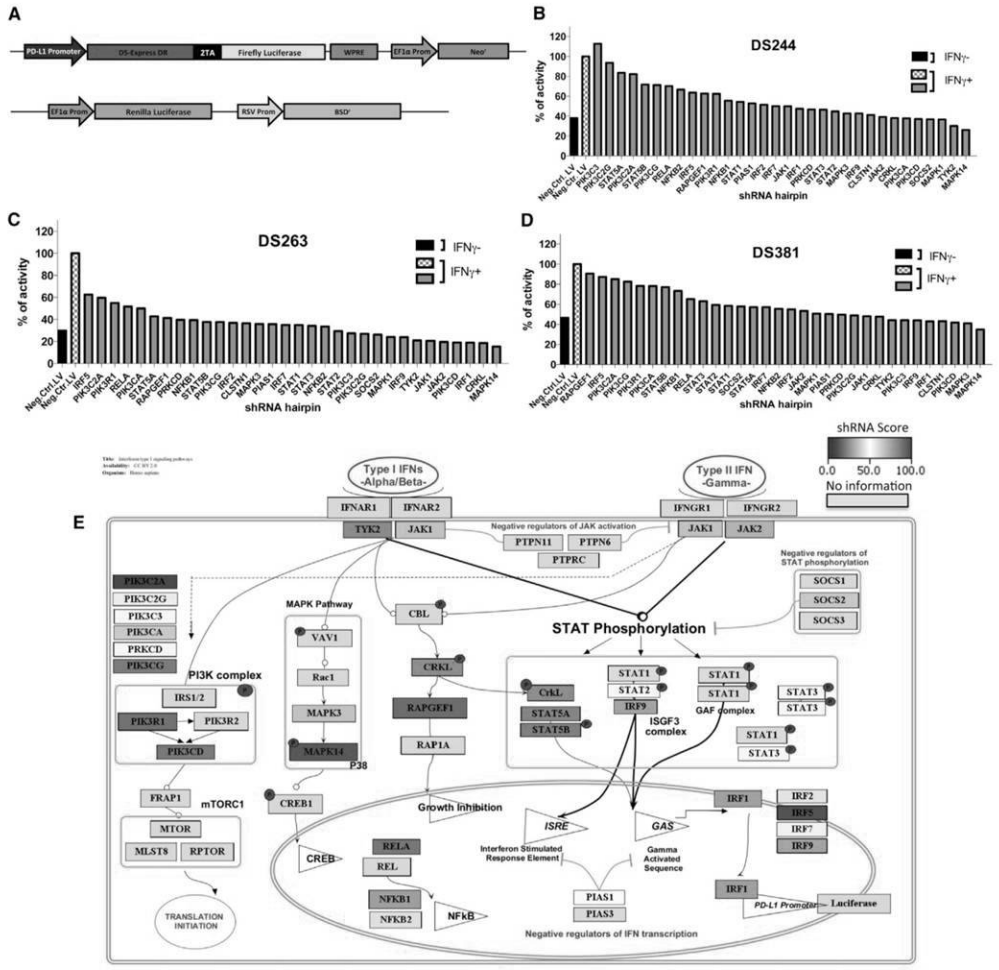


Figure 2. Effects on PD-L1 Reporter Expression upon shRNA Silencing of 33 Genes Involved in the Interferon Signaling Pathway

(A) Schematic representation of the PD-L1Prom-DSRed-FireflyLuciferase/Neo and EF1AProm-Renilla luciferase/RSV-BSD constructs used to generate the reporter melanoma cell lines. Reporter cells contain the PD-L1 promoter driving the expression of a DSRedDR-T2A-Firefly luciferase cassette and also an EF1-alpha promoter driving the Renilla luciferase gene used for normalization.

(B–D) Normalized luciferase reporter expression of each cell line, M244 (B), M263 (C), and M381 (D), transduced with different sets of lentiviral shRNA hairpins. Black and checkered white bars represent the cells transduced with a lentiviral control containing no shRNA, with

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

(black) or without (checkered) interferon gamma treatment. Gray bars represent the expression level of the cells transduced with a set of lentiviral shRNA hairpins with interferon gamma treatment to compare changes upon the interferon gamma induction of PD-L1 expression. Results are represented as a percentage of luciferase expression compared with the interferon-gamma-treated negative control.

(E) Schematic representation of the interferon receptor signaling pathway depicting the hits taking into account the redundant siRNA/shRNA activity (RSA) score obtained for each factor in the three reporter cell lines. Color heatmap represents the summed rank score, with red indicating the greatest impact on interferon-gamma-induced PD-L1 reporter activity and dark blue the least impact. Spatial orientation places each gene in the context of its signaling pathway.

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

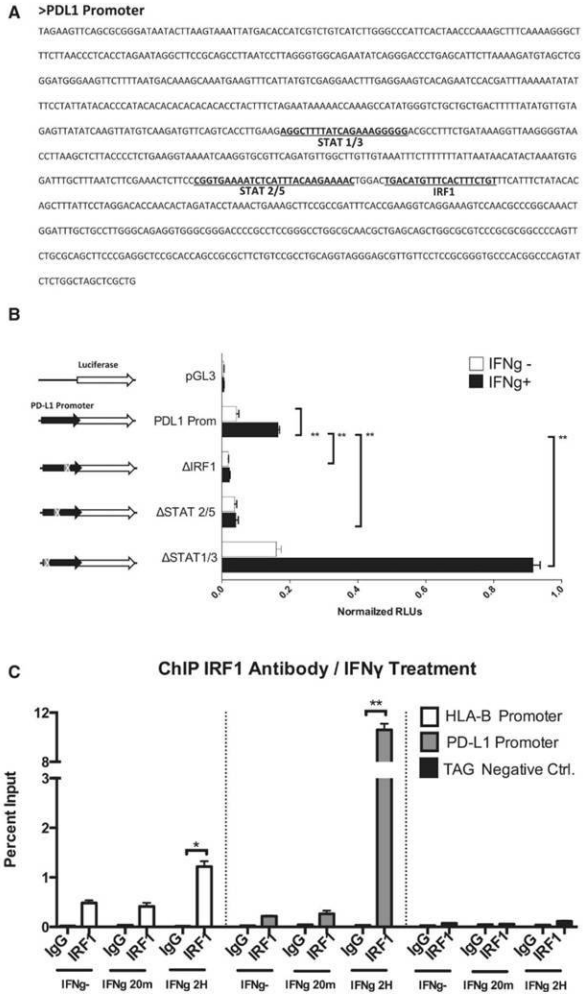


Figure 3. Transient Luciferase Reporter Assays and ChIP Analysis for the PD-L1 Promoter in M381 Melanoma Cells

(A) Sequence of the PD-L1 promoter showing the position of the most representative putative binding sites of the promoter, STAT1/STAT3, STAT2/STAT5, and IRF1.

(B) PD-L1 promoter transient reporter assay including deletions of the putative binding sites. Results are represented as normalized relative luciferase units (RLUs).

(C) ChIP assay in M381 cells at the PD-L1 promoter (gray), the HLA-B promoter as a positive control (white), and the human tRNA-Leu anti-codon (TAG) as irrelevant sequence for IRF1 binding (negative control). Results are represented as percent enrichment relative to

input. Asterisks denote significance in an unpaired t test (*$p < 0.005$, **$p < 0.001$), and error bars denote SD.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

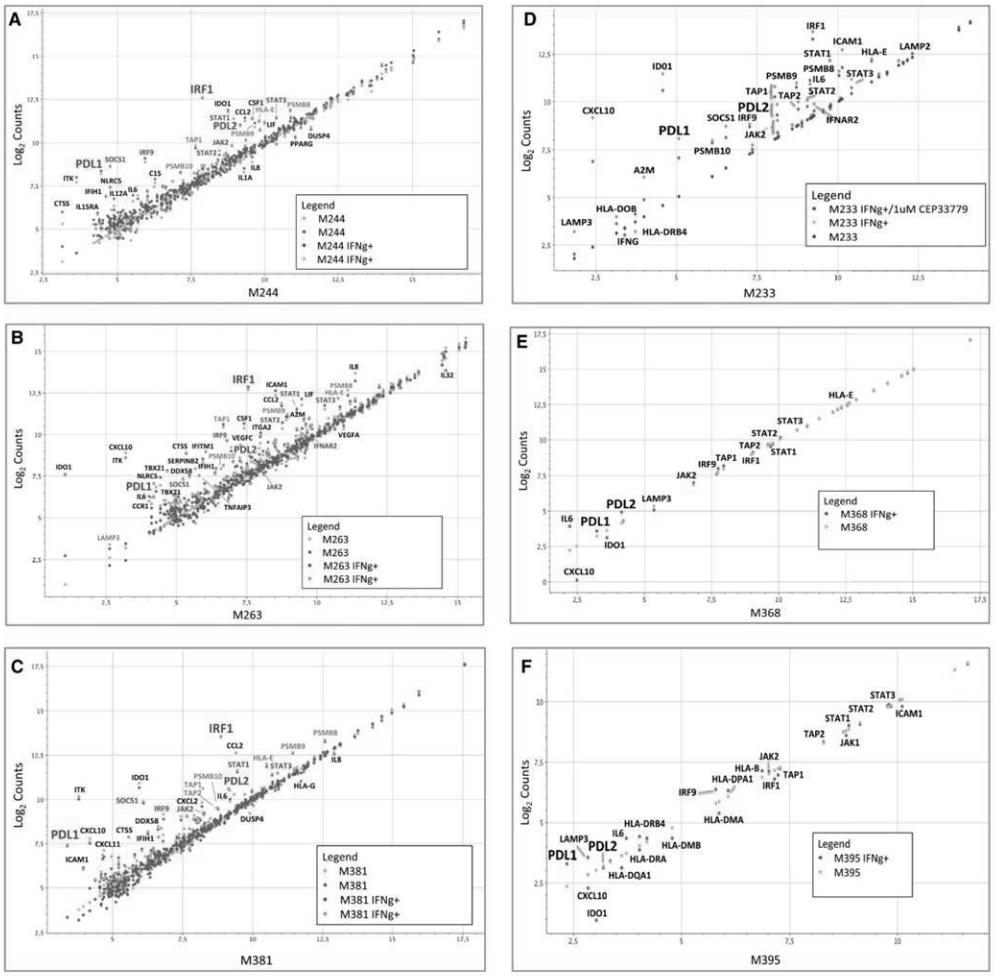


Figure 4. Selected mRNA Expression Profiling of Interferon-Gamma-Induced Genes

(A–C) Changes in selected gene expression upon interferon gamma exposure in three melanoma cell lines: M244 (A), M263 (B), and M381 (C). Purple and blue dots represent the expression level of the transcripts at basal level in two different biological duplicates (untreated cells), and red and green dots represent the expression level of the transcripts after 3 hr of interferon gamma treatment.

(D) Expression profile analysis of the good interferon gamma responder M233 cell line using the JAK2-specific inhibitor CEP33779. Dark blue dots represent untreated cells, green dots represent interferon-gamma-treated cells, and red dots represent cells treated with the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

JAK2 inhibitor CEP-33779 and interferon gamma. Data are presented as normalized counts (log2) of each transcript.
(E) Same analysis of the *JAK2* mutant cell line M368 with (red) or without (light blue) interferon gamma treatment.
(F) Expression profile of the *JAK1* mutant cell line M395 with (red) or without (light blue) interferon gamma treatment.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

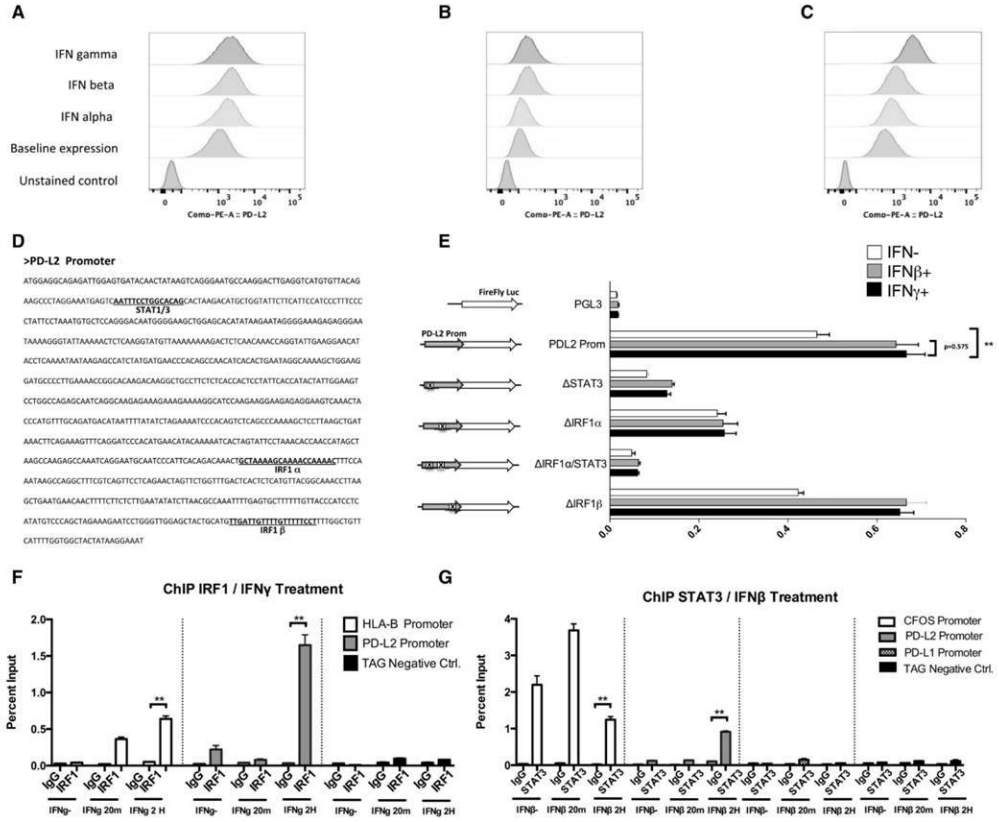


Figure 5. PD-L2 Expression and Promoter Function Analysis

(A–C) Flow cytometry analysis of PD-L2 surface expression upon interferon treatment in M244 (A), M263 (B), and M381 (C).

(D) Sequence of the PD-L2 promoter and position of the putative transcription factor binding sites.

(E) Transient reporter assay including deletions of STAT1/STAT3, IRF1, or double mutations for the putative binding sites. Analysis was performed in untreated cells (white bars) and under interferon beta (gray bars) and interferon gamma exposure (black bars).

(F) ChIP assay using IRF1 antibody in interferon-gamma-treated cells, including primers for the PD-L2 promoter (gray), HLA-B promoter as positive control (white), and TAG gene (black) as an irrelevant sequence for IRF1 binding.

(G) ChIP assay using STAT3 antibody in interferon-beta-treated cells, including primers for the C-FOS promoter as a positive control (white), the PD-L2 promoter (gray), the PD-L1 promoter (checkered), and TAG gene (black) as an irrelevant sequence for STAT3 binding.

Asterisks denote significance in an unpaired t test (*$p < 0.05$, **$p < 0.005$, ***$p < 0.001$), and error bars denote SD.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

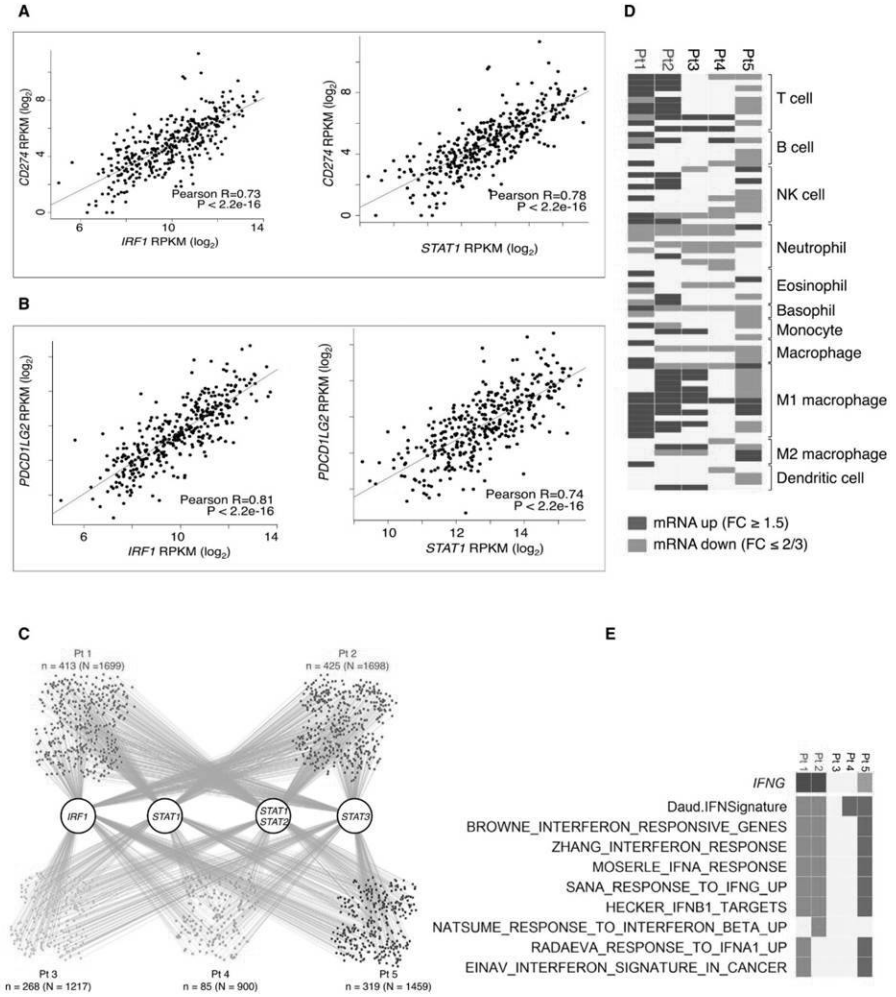


Figure 6. RNA-Seq Analysis of TCGA Tumors and Anti-PD-1-Treated Biopsies

(A) Pearson correlations between log2 normalized mRNA expression levels (RPKM) of IRF1 and *STAT1* versus *PD-L1* in the TCGA skin cutaneous melanoma RNA-seq database.

(B) Pearson correlations between log2 normalized mRNA expression levels (RPKM) of IRF1 and *STAT1* versus PD-L2 in the TCGA skin cutaneous melanoma RNA-seq database.

(C) Potential target genes of IRF1, STAT1, STAT3, and STAT1:STAT2 heterodimer among all up-expressed genes in anti-PD1 on-treatment tumor samples (upexpression is defined by fold change ≥1.5). Gray lines indicate the occurrence of the respective transcription factor binding motifs in the target genes.

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

(D) Tilling of differential immune cell marker genes in on-treatment anti-PD1 tumors compared to their respective baselines (red and green, up- and down-regulation of at least 1.5 (log2) fold changes.
(E) Tiling of differential *IFNG* expression (red and green, up- and down-expression) and interferon signature enrichment (orange and blue, positive and negative enrichments) in on-treatment tumors derived from patients receiving anti-PD1 treatment (red, patients who responded to treatment; black, patients who did not respond to treatment).

（参照文献３：Nowicki et al., Mechanisms of Resistance to PD-1 and PD-L1 blockade, Cancer J. (1):47-53 (2018)）

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript

HHS Public Access
Author manuscript
Cancer J. Author manuscript; available in PMC 2019 January 01.

Published in final edited form as:
Cancer J. 2018 ; 24(1): 47–53. doi:10.1097/PPO.0000000000000303.

Mechanisms of Resistance to PD-1 and PD-L1 blockade

Theodore S. Nowicki, M.D., Ph.D.[1], **Siwen Hu-Lieskovan, M.D., Ph.D.**[2], and **Antoni Ribas, M.D., Ph.D.**[2,3,4,5]

[1]Division of Pediatric Hematology-Oncology, Department of Pediatrics, University of California Los Angeles, Los Angeles, California

[2]Division of Hematology-Oncology, Department of Medicine, University of California Los Angeles, Los Angeles, California

[3]Jonsson Comprehensive Cancer Center, Los Angeles, California

[4]Department of Molecular and Medical Pharmacology, University of California Los Angeles, Los Angeles, California

[5]Division of Surgical-Oncology, Department of Surgery, University of California Los Angeles, Los Angeles, California

Abstract

Cancer immunotherapy utilizing blockade of the PD-1/PD-L1 checkpoint has revolutionized the treatment of a wide variety of malignancies, leading to durable therapeutic responses not typically seen with traditional cytotoxic anti-cancer agents. However, these therapies are ineffective in a significant percentage of patients, and some initial responders eventually develop resistance to these therapies with relapsed disease. The mechanisms leading to both primary and acquired resistance to PD-1/PD-L1 inhibition are varied, and can be both multifactorial and overlapping in an individual patient. As the mechanisms of resistance to PD-1/PD-L1 blockade continue to be further characterized, new strategies are being developed to prevent or reverse resistance to therapy, leading to improved patient outcomes.

Keywords

Immunotherapy; PD-1; PD-L1; resistance

CORRESPONDING AUTHOR: Antoni Ribas, M.D., Ph.D. Professor of Medicine, Hematology-Oncology, Professor of Surgery, Surgical Oncology, Professor of Molecular and Medical Pharmacology, Director, Tumor Immunology Program, Jonsson Comprehensive Cancer Center (JCCC), David Geffen School of Medicine, University of California Los Angeles (UCLA), Jonsson Comprehensive Cancer Center, Phone: 310-206-3928, aribas@mednet.ucla.edu.

DISCLOSURE OF POTENTIAL CONFLICTS OF INTEREST: The authors declare no potential conflicts of interest.

DISCLAIMER: The contents of this article are solely the responsibility of the authors and do not necessarily represent the official view of the NIH, the NCI, the NICHD, the Parker Institute for Cancer Immunotherapy, the Grimaldi Family Fund, the Samuels Family Fund, the Garcia-Corsini Family Fund, or the Ressler Family Foundation.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

INTRODUCTION

Immunotherapy represents a paradigm shift in the treatment of cancer, particularly the inhibition of various checkpoints that control host T cell activity. Programmed cell death 1 (PD-1) interacting with its corresponding ligand PD- L1 has emerged as a common means by which cancer cells evade the host immune response (1). PD-1 is expressed on activated $CD8^+$ T cells (as well as B cells and natural killer cells) in the setting of chronic antigen exposure. The interaction between PD-1 and PD-L1 or PD-L2 on host tissues leads to the inhibition of T cell receptor (TCR) signaling and CD28 co-stimulation (2, 3), limits T cell interactions with target cells, and ultimately leads to their inactivation and loss of proliferative capacity (4–7). PD-L1 expression is induced by localized inflammatory stimuli, such as interferons (IFNs) released by the infiltrating T cells (5). The PD-L1 induction process in cancer has been termed "adaptive immune resistance" (1), and represents a mechanism by which cancer cells protect themselves from T cell mediated destruction.

This new approach to treating cancer has led to the clinical development of therapeutic monoclonal antibodies blocking PD-1 or PD-L1. Pembrolizumab (Keytruda®) and nivolumab (Opdivo®) are two such agents targeting PD-1, while atezolizumab (Tecentriq®), avelumab (Bavencio®), and durvalumab (Imfinzi®) block PD-L1 instead (8). Collectively, these agents are approved for the treatment of a wide variety of malignancies, including metastatic melanoma, non-small-cell lung cancer (NSCLC), head and neck squamous cell cancer, Hodgkin's lymphoma, renal cell carcinoma (RCC), urothelial carcinoma, Merkel cell carcinoma, gastric carcinoma, and hepatocellular carcinomas (8–22). These agents inhibit the negative regulatory effects of PD-L1 on patient T cells via PD-1, resulting in the enhancement of a pre-existent antitumor immune activity. This unleashes a focused T cell response against a patient's tumor, as it increases proliferation of tumor infiltration lymphocytes (TILs), and leads to a more clonal TCR repertoire within the T cell population directed against the tumor (23). These effects ultimately provide patients with significant and durable immune responses against their malignancy, with complete responses lasting for years in some cases.

Although PD-1/PD-L1 checkpoint blockade can result in dramatic therapeutic responses, this therapy is only effective in a subset of patients, and many patients are only partial responders to therapy (9, 12, 24). Patients who do not respond to initial therapy with PD-1/PD-L1 blockade are referred to as having "primary resistance" to therapy (25). Furthermore, there is a growing subset of patients who, despite showing a robust initial response to therapy, go on to develop progressive disease. This phenomenon, in which disease is either refractory to resumption of therapy, or develops despite continuation of therapy, is known as "acquired resistance" to PD-1/PD-L1 blockade immunotherapy (25, 26). Both phenomena are highly complex, as the mechanisms for both types of resistance can be overlapping and/or multifactorial. Furthermore, each patient's individual environmental and genetic factors, as well as prior treatments, can create an evolving therapeutic landscape unique to a given patient. As the mechanisms responsible for both primary and acquired resistance to PD-1/PD-L1 blockade are further elucidated, improved treatments with superior therapeutic efficacy can be developed. This review will explore the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

mechanisms responsible for primary and acquired resistance to immunotherapy via PD-1/PD-L1 blockade, as well as techniques being studied to overcome such resistance.

PRIMARY RESISTANCE TO PD-1/PD-L1 BLOCKADE

Patients treated with PD-1/PD-L1 blockade who never demonstrate a clinical response or stabilized disease are referred to as having "primary resistance" to therapy. Early data from clinical trials demonstrated that the presence of pre-existing $CD8^+$ T cells within the tumor and its periphery, along with co-localized PD-1 and PD-L1 expression on the T cells and tumor cells, respectively, predicted therapeutic response to anti-PD-1 therapy in patients with malignant melanoma (23). Therefore, many studies have explored primary resistance as a phenomenon in which a patient's $CD8^+$ T cells are either unable to recognize and localize to the tumor, or are rendered ineffective despite seemingly adequate localization. The latter mechanism can occur due to other cell types that exert local immunosuppressive effects within the tumor microenvironment, or by inherent resistance of tumor cells to the effects of the T cells themselves.

Inadequate T cell infiltration due to lack of tumor immunogenicity

The inability of host $CD8^+$ T cells to localize to a tumor can be most simply attributed to an absence of sufficiently immunogenic tumor antigens for T cell recognition (27). This may be the case in tumors that are either not significantly dedifferentiated from their tissue of origin, or possess insufficient mutational burden to express tumor antigens which are able to produce a focused $CD8^+$ T cell response. The resulting absence of T cells that can differentially recognize unique tumor antigens renders such tumors non-responsive to PD-1/PD-L1 blockade therapy. Indeed, tumors with high mutational burden and increased tumor neoantigen expression, such as melanoma, head and neck, NSCLC, bladder, and microsatellite unstable cancers are generally more responsive to anti-PD-1/PD-L1 therapy (12, 28–30). RCC, which has a relatively low mutational burden, appears to be a notable exception to this trend (31), demonstrating the importance of other factors in its responsiveness. However, tumors with few somatic mutations, such as pancreatic and prostate cancers, are generally more resistant to PD-1/PD-L1 blockade (28, 32). One approach to overcome this obstacle is to supply with the patient with T cells targeting antigens that might not inherently display strong immunogenicity. Adoptive cell therapy (ACT) involves the generation of large quantities of $CD8^+$ T cells directed at specific tumor antigens, such as MART-1 in melanoma and NY-ESO-1 in sarcoma (33–35). T cells specific for the tumor antigen can be generated by *ex vivo* expansion of endogenous, low-frequency $CD8^+$ T cells specific for a characterized tumor antigen, which are then re-infused into the patient (36, 37). Alternatively, such cells can be generated by *ex vivo* genetic modification of peripheral mononuclear blood cells, which are transduced with a TCR directed against a given tumor antigen, expanded, and re-infused into the patient (34). Tumor vaccines are also an effective means of stimulating patients' dendritic cells to generate T cell activity against tumor antigens (38), including with the use of mutational neoantigens as vaccines (39, 40). Combining ACT or dendritic cell vaccines with PD-1 inhibition can be effective in generating a therapeutic response in these cases by providing a focused T cell response against the tumor (41–44). More novel ways of inducing T cell infiltration within tumors to

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

sensitize them to PD-1 inhibition have also been studied. Ribas et al recently reported the novel combination of intralesional injection of a modified human herpes simplex virus with systemic anti-PD-1 therapy for the treatment of melanoma in a phase 1b clinical trial (45). This lead to a 62% objective response rate, as well as 33% complete response rate in patients, and was accompanied by enhanced T cell infiltration in virus-injected lesions. Additionally, responses to this combination appeared to be independent of baseline $CD8^+$ T cell infiltration. As new methods of stimulating T cell infiltration within tumors are characterized, their combination with anti-PD-1/PD-L1 therapies may continue to lead to improved response rates.

Intracellular antigens require cell surface presentation via the major histocompatability complex (MHC) and a lack of antigen presentation can lead to tumor resistance to T cells. Various mechanisms to disrupt the machinery of antigen presentation have been described including mutations that interfere with the proteasome, transporters associated with antigen processing (TAP), and the structural components of the MHC itself (46). Several groups had demonstrated long ago that mutations resulting in loss of beta-2-microglobulin (β2M) could render melanoma tumors resistant to T cell infiltration, due to the role β2M plays in MHC transport and stable expression on the cell surface (47–49). The resultant inability to present antigens via MHC class I renders the tumor cells effectively undetectable by $CD8^+$ T cells. Antigen processing and surface presentation via the MHC can be disrupted by epigenetic changes as well, such as hypermethylation and histone acetylation (50, 51). Therefore, agents reversing these processes may augment T-cell dependent immunotherapy with PD-1 (52). Demethylating agents and histone deacetylation agents alike have been shown to increase the MHC presentation of tumor-specific antigens in models of lymphoma and melanoma, leading to increased infiltration by $CD8^+$ T cells and subsequent tumor responses (53, 54). As the ability to classify epigenetic modulation status of individual tumors becomes better refined and more specific epigenetic drugs are being developed, combinatorial strategies involving epigenetic modification therapy with PD-1/PD-L1 blockade may be effective in overcoming these cases of primary resistance.

T cell exclusion

Primary mutagenesis events within a tumor can also lead to a phenomenon known as T cell exclusion, in which T cell tracking to the tumor microenvironment is inhibited without impacting antigen expression or presentation. One mutation reported to contribute to T cell exclusion is activation of β-catenin/Wnt signaling. In a mouse model of melanoma, Spranger et al demonstrated an inverse correlation between β-catenin/Wnt activation in tumors and their degree of $CD8^+$ T cell infiltration (55). Increased β-catenin/Wnt activity was also associated with decreased $CD103^+$ dendritic cell infiltration due to decreased CCL4 expression, a chemokine responsible for their attraction. These melanoma tumors with β-catenin/Wnt activation responded poorly to PD-1 blockade therapy, whereas tumors without β-catenin/Wnt mutations responded well to treatment with PD-1 blockade. β-catenin/Wnt expression has also been correlated with T cell exclusion in urothelial bladder cancer, another tumor type which can be treated by PD-1/PD-L1 inhibition (56). Further study regarding this mechanism of T cell exclusion can lead to better classification of the likelihood of a tumor's response to checkpoint blockade. Several drugs inhibiting

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

constitutive β-catenin/Wnt pathway signaling are on clinical development, and can be combined with anti-PD-1/L1 therapy to overcome this mode of primary resistance.

Other mutations associated with T cell exclusion and subsequent resistance to PD-1/PD-L1 blockade are those within the mitogen activated protein kinase (MAPK) signaling cascade. Constitutive oncogenic signaling through this pathway results in the production of the immunosuppressive cytokines vascular endothelial growth factor (VEGF) and interleukin 8 (IL-8), which inhibit T cell recruitment to the tumor, as well as their functionality (57, 58). Activating mutations within the MAPK cascade are common in melanoma, and inhibition of this cascade has been shown to both improve $CD8^+$ T cell infiltration within tumors, as well as sensitize them to PD-1 blockade therapy (59). These data have provided strong rationale for combination therapy of multi-kinase inhibition with PD-1 blockade in tumors with such mutations. Similarly, loss of the tumor suppressor phosphatase and tensin homolog (PTEN), which leads to activation of phosphatidylinositol 3-kinase (PI3K) signaling, has also been associated with increased VEGF production and reduced $CD8^+$ T cell infiltration of tumors and subsequent resistance to PD-1 blockade therapy. In clinical samples of melanoma with loss of functional PTEN, Peng et al showed a decrease in T cell infiltration proportional to the frequency of PTEN deletions and mutations. Furthermore, in a murine model, addition of a PI3K inhibitor to PD-1 blockade produced superior regression of PTEN-deficient tumors (60). These studies demonstrate the need for continued study of targetable mutations leading to primary resistance to PD-1/PD-L1 blockade, which can lead to clinical therapeutic regimens combining kinase inhibitor therapy with immune checkpoint blockade to improve response rates.

Tumor cell resistance to interferon

Tumors can display inherent primary resistance to PD-1/PD-L1 blockade despite effective CD8 T cell recognition. This can occur due to lack of interferon-gamma (IFNγ) responsiveness in a tumor, which can occur in the setting of both primary and acquired resistance to checkpoint inhibition therapy. IFNγ is produced by $CD8^+$ T cells which have recognized and engaged an appropriate tumor antigen, and represents a primary means of increasing MHC expression/antigen presentation, recruiting additional T cells to tumors, and inducing direct anti-proliferative effects and apoptosis in cancer cells (61). Furthermore, this IFNγ stimulation is a primary cause of the induction of PD-L1 on the tumor cells themselves, as previously discussed (62, 63). IFNγ binds to the interferon-gamma receptor (IFNGR), which leads to JAK1 and JAK2 activation and subsequent STAT1 and STAT3 recruitment and phosphorylation (64). This complex then translocates to the nucleus, where it activates interferon regulatory factor 1 (IRF1), the transcriptional activity of which ultimately gives rise to the anti-tumor effects of IFNγ, as well as PD-L1 expression (62, 63, 65). Primary mutations resulting in loss of function of JAK1 and JAK2 have been shown to impair the IFNGR signaling pathway, resulting in loss of the anti-tumor effects of IFNγ (66–68). This phenomenon also results in decreased or absent PD-L1 expression by the tumor cells. However, rather than augmenting the $CD8^+$ T cell response against the tumor by removing the inhibitory effect of PD-L1 on the T cells via PD-1, such cancer cells with JAK1/2 mutations display primary resistance to anti-PD-1 therapy by virtue of their overall resistance to the anti-tumor effects of IFNγ itself (68). Additionally, in a preclinical model

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

of CRISPR screens, mutations in the apelin receptor, which interacts with JAK1 to modulate IFNγ responses, have also been associated with primary resistance to immunotherapy with PD-1 blockade via IFNγ resistance (69). Furthermore, in another CRISPR screen, activating mutations in tyrosine-protein phosphatase non-receptor type 2 (Ptpn2), which negatively regulates JAK1 and STAT1 signaling, have also been associated with primary resistance to PD-1 blockade via resistance to IFNγ, and deletion of Ptpn2 via CRISPR-Cas9 genome editing was able to restore IFNγ sensitivity in a melanoma model (70). Further classification of tumors at the genetic level can provide evidence of functional IFNγ response elements, which would be a prerequisite for an effective response to PD-1/PD-L1 blockade therapy.

In addition to reflecting dynamic IFNγ responsiveness, PD-L1 expression can also be constitutively expressed in some cases. Although the prognostic implications of constitutive expression of PD-L1 are not always clear, there are some that are associated with poor response to therapeutic inhibition of the PD-1/PD-L1 checkpoint. In patients with NSCLC, patients with EGFR mutations and ALK rearrangements have poor response rates to PD-1/PD-L1 blockade. Patients with such mutations display high PD-L1 expression rates despite low or absent $CD8^+$ T cell infiltration, implying that the PD-L1 expression was constitutive rather than induced by local inflammatory stimuli; these tumors correspondingly demonstrated a lack of responsiveness to PD-1/PD-L1 blockade (71). Additionally, preclinical models of NSCLC harboring EGFR mutations and ALK rearrangements have demonstrated that these genetic aberrations are directly responsible for the constitutive PD-L1 expression seen in these tissues (72, 73). Although the mechanism for the apparent immune exclusion caused by these mutations remains unclear, they nevertheless underscore the danger in predicting a tumor as being responsive to PD-1/PD-L1 blockade therapy based solely on the expression of a single marker.

Local immunosuppressive factors within the tumor microenvironment

The wide variety of other cell types within the tumor microenvironment (TME) represents another diverse pool of potential modulators of immune activity against a tumor. Certain non-tumor cells within the TME can impair a response to PD-1/PD-L1 blockade despite seemingly adequate infiltration by $CD8^+$ T cells and interferon-responsive tumor cells. One of these is the myeloid-derived suppressor cell (MDSC), a heterogeneous population of immature myeloid cells recruited by tumors with the ability to regulate local immune function. MDSCs dampen T cell responses via multiple mechanisms, including depletion of local nutrients, production of reactive oxygen species, and nitrosylation of local chemokines (74–77). Several studies have demonstrated that depletion of intra-tumor MDSCs restores the effectiveness of PD-1 blockade (78, 79). Another immunosuppressive cell type within the TME is the regulatory T (Treg) cell. Treg cells are a subtype of $CD4^+$ T cells that suppress the activity and proliferation of local effector CD8 T cells; when present within the TME, this results in a poor immunologic response against the tumor (80, 81). Strategies which deplete or impair Tregs have led to improved responsiveness to PD-1/PD-L1 checkpoint blockade (82, 83). Further studies characterizing intra-tumor Tregs and MDSCs and their inhibition will aid in overcoming this mechanism of primary resistance to checkpoint blockade.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Tumor cells themselves and their surrounding stroma can modulate immunosuppressive states within the TME, resulting in tumors with primary resistance to PD-1/PD-L1 blockade therapy. Enrichment of a collection of genes termed innate anti-PD-1 resistance (IRPES) has been characterized in patients with melanoma who were non-responsive to checkpoint blockade (84). These include locally immunosuppressive genes (VEGF, IL-10), as well as genes responsible for mesenchymal transition, monocyte/macrophage/MDSC chemotaxis, wound healing, and angiogenesis. IPRES signatures, interestingly, represent a transcriptomic program that exists across multiple cancer subtypes, including melanoma, kidney clear cell carcinoma, colon adenocarcinoma, lung adenocarcinoma, and pancreatic adenocarcinoma (84). Correlation of such patterns across other malignancies may provide a distinct means of predicting a tumor's response to immunotherapy with PD-1/PD-L1 blockade.

ACQUIRED RESISTANCE TO PD-1/PD-L1 BLOCKADE

Immunotherapy with PD-1/PD-L1 blockade has been characterized by a subset of patients who exhibit long-lasting responses to treatment. However, over time, cohorts of patients have emerged who either eventually progress while on therapy despite an initially robust response, or who are unresponsive to re-initiation of checkpoint blockade. The mechanisms behind acquired resistance to therapy are diverse, and can occur through similar mechanisms responsible for primary resistance. Broadly, the pathogenesis of acquired resistance to anti-PD-1/PD-L1 therapy can include eventual loss of T cell function (due to epigenetic dysfunction or the acquisition of other immunosuppressive signals), disruption of antigen presentation (leading to decreased T cell recognition of the tumor), and development of resistance to the effects of interferon generated by the T cells.

Loss of T cell function

PD-1 expression occurs on $CD8^+$ T cells in the setting of persistent antigen stimulation, and its stimulation leads to impaired effector functionality within these cells. PD-1 blockade can re-invigorate these hypofunctional "exhausted" $CD8^+$ T cells (T_{EX}), restoring their anti-tumor effector functionality (85). However, recent data have elucidated the degree to which this phenomenon is sustainable *in vivo*, which has significant implications for relapse to PD-1 blockade therapy. Tumor-infiltrating T_{EX} cells can be characterized by increasing dysfunction driven by progression through discrete epigenetic states. While initially plastic and able to be modified to restore an effector functionality, they eventually reach a state of fixed epigenetic dysfunction, in which their chromatin is rendered inaccessible and resistant to further remodeling and reinvigoration (86). The overall ratio of reinvigorated T_{EX} to pretreatment tumor burden has also been correlated with degree of clinical response to PD-1 blockade (87). If the tumor burden remains high and reinvigorated T_{EX} fail to clear the tumor, they eventually become re-exhausted, acquiring a fixed epigenetic state that renders them resistant to reinvigoration via PD-1 blockade (88). Additionally, PD-1 blockade in T_{EX} activates transcriptional changes associated with effector T cells (T_{EFF}), but not with memory T cells (T_{MEM}) (88). T_{MEM} have greater proliferative capacity and longevity than T_{EFF}, and are associated with greater long-term potentiation of immune responses, including those directed against tumors (89–91); a lack of T_{MEM} cells combined with re-exhausted T_{EX} leads to tumor progression after an initial response. This phenomenon underscores the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

need for development of epigenomic modification strategies which can perpetuate reinvigoration of T cells in response to PD-1/PD-L1 blockade, resulting in a more sustainable anti-tumor activity.

In addition to PD-1, there are other immunosuppressive signaling receptors that can impair anti-tumor T cell functionality. These include cytotoxic T-lymphocyte antigen 4 (CTLA-4), T-cell immunoglobulin and mucin domain-containing molecule-3 (TIM3), lymphocyte activation gene-3 (LAG-3), and V-domain Ig suppressor of T cell activation (VISTA), among others (92, 93). While their expression can occur at any time, TIM3 has been shown to be up-regulated in patients with NSCLC who developed acquired resistance to anti-PD-1 therapy (94, 95). The resistance of these tumors to anti-PD-1 therapy was able to be overcome in a murine model with the addition of a TIM-3 blocking antibody. Up-regulation of VISTA on TILs over time has also been demonstrated in cases of acquired resistance to PD-1 blockade (96). While the expression patterns of these markers and their correlation with response to immunotherapy are still being characterized, blocking multiple immune inhibitory checkpoints is a logical extension of these data. Several trials are currently underway testing the effectiveness of combining PD-1/PD-L1 blockade with inhibitors of other co-existing immune regulatory checkpoints as a means to overcome resistance to therapy (97, 98). Indeed, the clinical combination of the anti-CTLA-4 antibody ipilimumab with nivolumab (an anti-PD-1 antibody) is FDA-approved for the treatment of metastatic melanoma, and has demonstrated superior outcomes compared to nivolumab alone (99, 100). As the expression patterns of other immunosuppressive factors are further characterized, future immunotherapy regimens can be designed combining multiple levels of checkpoint blockade to improve clinical outcomes and overcome resistance to therapy.

Disruption of antigen presentation

As previously discussed, antigen/neoantigen expression and presentation are vital to T cell recognition of a tumor and engagement of the TCR. While primary mutagenesis events can interfere with this process in a variety of ways and contribute to resistance, this can also occur following initiation of immunotherapy, resulting in acquired resistance to PD-1/PD-L1 blockade. Inflammatory stress has been shown to induce melanoma cell dedifferentiation along the neural crest lineage, resulting in increased expression of the nerve growth factor receptor (NGFR) and decreased expression of melanosomal antigens such as MART-1 (101–103). This is also true of mutational neoantigens, which are especially important in T cell recognition of tumors in the setting of checkpoint blockade (28, 104). This has recently been demonstrated in NSCLC, where patients who developed acquired resistance to PD-1/PD-L1 blockade were found to have lost multiple mutation-associated neoantigens while on treatment (105). These neoantigens, which had been present in the tumors prior to initiation of checkpoint blockade, were found to have superior MHC binding affinity than the remaining neoantigens present at disease progression. Furthermore, the lost neoantigens were found to induce clonal T cell expansion *in vitro*, demonstrating their ability to elicit a T cell response. These findings demonstrate the need for better understanding of how the neoantigen landscape changes in patients who develop acquired resistance to PD-1/PD-L1 blockade. Classification of putative neoantigens in real time could provide early evidence of patients more likely to develop resistance to therapy.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Antigen/neoantigen presentation can also be disrupted via mutations in the antigen presenting machinery. While this can occur at baseline in the setting of primary resistance to PD-1/PD-L1 blockade, it has been shown to occur to patients while on treatment as well. Loss of β2M, which is required for surface expression of class I MHC, has been demonstrated to occur following initiation of adoptive cell therapy or IL-2 treatment, leading to loss of T cell recognition of the tumor (48). This same phenomenon has been demonstrated in clinical examples of acquired resistance to PD-1 blockade in melanoma (106), in which a homozygous truncating mutation in β2M led to lack of surface expression of MHC class I. This mutation was not present in the baseline tumor, and ultimately led to acquired resistance to therapy in the patient. This demonstrates the ability of immunotherapy with PD-1/PD-L1 blockade to induce mutations that allow a tumor cell to escape detection by the T cells. Such tumor cells are then clonally expanded via selective pressure, resulting in disease relapse in patients who had initially responded to therapy.

Evolution of interferon resistance

Interferon responsiveness of a tumor is vital to the success of any T cell-based immunotherapy, including PD-1/PD-L1 blockade. While mutations within interferon signaling elements have been described in the setting of primary resistance to therapy, these mutations can also develop following initiation of therapy. Zaretsky et al demonstrated that patients with melanoma who developed resistance to anti-PD-1 therapy had acquired loss-of-function mutations in JAK1 and JAK2 (106), similar to the reported mechanisms responsible for primary resistance to therapy involving mutations in this signaling cascade (68). Although the tumor cells could still be recognized and engaged by T cells, their JAK1/2 mutations rendered them resistant to the effects of interferon stimulation. Tumor cells from these patients displayed absent activity in the IFNγ signaling pathway, insensitivity to the anti-proliferative effects of IFNγ, and lack of IFNγ-induced surface expression of both PD-L1 and MHC class I. As with disruption of antigen presentation machinery, this phenomenon underscores the ability of PD-1/PD-L1 blockade to clonally select for cells that can develop new mutations capable of withstanding inflammatory stresses brought on by the anti-tumor T cells.

CONCLUSIONS

Although immunotherapy with PD-1/PD-L1 blockade has shown great promise for the treatment of a variety of advanced cancers, significant durable responses only occur in a minority of patients, and patients who initially respond can ultimately relapse despite continued treatment. The reasons for response or resistance to therapy are multifaceted, highly individualized, and can evolve over time during treatment. There remains a critical need for the development of comprehensive monitoring of patients being treated with these agents. As the mechanisms responsible for resistance continue to be characterized, therapies can be personalized to optimize a patient's chance of responding, and real-time changes to therapeutic regimens can be made to overcome relapse. These highly individualized approaches in response to therapeutic monitoring will ultimately allow many more patients to benefit from these agents.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Acknowledgments

GRANT SUPPORT: This research was supported by the Parker Institute for Cancer Immunotherapy, NIH/NCI grants R35 CA197633 and P01 CA168585 (A.R.), the Ressler Family Fund, the Garcia-Corsini Family Fund, the Samuels Family Fund, and the Grimaldi Family Fund (A.R.). S.H.-L. is supported by a Career Development Award from the American Society of Clinical Oncology (ASCO), a SU2C/AACR Phil Sharp Award, a Melanoma Foundation Alliance Young Investigator Award and a UCLA KL2 Award. T.S.N. is supported by the NIH/NICHD grant K12-HD000850 (Pediatric Scientist Development Program).

References

1. Pardoll DM. The blockade of immune checkpoints in cancer immunotherapy. Nat Rev Cancer. 2012; 12:252–264. [PubMed: 22437870]
2. Kamphorst AO, Wieland A, Nasti T, et al. Rescue of exhausted CD8 T cells by PD-1-targeted therapies is CD28-dependent. Science. 2017; 355:1423–1427. [PubMed: 28280249]
3. Hui E, Cheung J, Zhu J, et al. T cell costimulatory receptor CD28 is a primary target for PD-1-mediated inhibition. Science. 2017; 355:1428–1433. [PubMed: 28280247]
4. Ribas A. Tumor immunotherapy directed at PD-1. N Engl J Med. 2012; 366:2517–2519. [PubMed: 22658126]
5. Keir ME, Liang SC, Guleria I, et al. Tissue expression of PD-L1 mediates peripheral T cell tolerance. J Exp Med. 2006; 203:883–895. [PubMed: 16606670]
6. Francisco LM, Sage PT, Sharpe AH. The PD-1 pathway in tolerance and autoimmunity. Immunol Rev. 2010; 236:219–242. [PubMed: 20636820]
7. Fife BT, Pauken KE, Eagar TN, et al. Interactions between PD-1 and PD-L1 promote tolerance by blocking the TCR-induced stop signal. Nat Immunol. 2009; 10:1185–1192. [PubMed: 19783989]
8. Abril-Rodriguez G, Ribas A. SnapShot: Immune Checkpoint Inhibitors. Cancer Cell. 2017; 31:848–848 e841. [PubMed: 28609660]
9. Hamid O, Robert C, Daud A, et al. Safety and tumor responses with lambrolizumab (anti-PD-1) in melanoma. N Engl J Med. 2013; 369:134–144. [PubMed: 23724846]
10. Robert C, Ribas A, Wolchok JD, et al. Anti-programmed-death-receptor-1 treatment with pembrolizumab in ipilimumab-refractory advanced melanoma: a randomised dose-comparison cohort of a phase 1 trial. Lancet. 2014; 384:1109–1117. [PubMed: 25034862]
11. Sundar R, Cho BC, Brahmer JR, et al. Nivolumab in NSCLC: latest evidence and clinical potential. Ther Adv Med Oncol. 2015; 7:85–96. [PubMed: 25755681]
12. Topalian SL, Hodi FS, Brahmer JR, et al. Safety, activity, and immune correlates of anti-PD-1 antibody in cancer. N Engl J Med. 2012; 366:2443–2454. [PubMed: 22658127]
13. Brahmer JR, Tykodi SS, Chow LQ, et al. Safety and activity of anti-PD-L1 antibody in patients with advanced cancer. N Engl J Med. 2012; 366:2455–2465. [PubMed: 22658128]
14. Armand P, Nagler A, Weller EA, et al. Disabling immune tolerance by programmed death-1 blockade with pidilizumab after autologous hematopoietic stem-cell transplantation for diffuse large B-cell lymphoma: results of an international phase II trial. J Clin Oncol. 2013; 31:4199–4206. [PubMed: 24127452]
15. Herbst RS, Soria JC, Kowanetz M, et al. Predictive correlates of response to the anti-PD-L1 antibody MPDL3280A in cancer patients. Nature. 2014; 515:563–567. [PubMed: 25428504]
16. Powles T, Eder JP, Fine GD, et al. MPDL3280A (anti-PD-L1) treatment leads to clinical activity in metastatic bladder cancer. Nature. 2014; 515:558–562. [PubMed: 25428503]
17. Robert C, Long GV, Brady B, et al. Nivolumab in previously untreated melanoma without BRAF mutation. N Engl J Med. 2015; 372:320–330. [PubMed: 25399552]
18. Ansell SM, Lesokhin AM, Borrello I, et al. PD-1 blockade with nivolumab in relapsed or refractory Hodgkin's lymphoma. N Engl J Med. 2015; 372:311–319. [PubMed: 25482239]
19. Robert C, Schachter J, Long GV, et al. Pembrolizumab versus Ipilimumab in Advanced Melanoma. N Engl J Med. 2015; 372:2521–2532. [PubMed: 25891173]
20. Le DT, Uram JN, Wang H, et al. PD-1 Blockade in Tumors with Mismatch-Repair Deficiency. N Engl J Med. 2015; 372:2509–2520. [PubMed: 26028255]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

21. Garon EB, Rizvi NA, Hui R, et al. Pembrolizumab for the treatment of non-small-cell lung cancer. N Engl J Med. 2015; 372:2018–2028. [PubMed: 25891174]
22. Nghiem PT, Bhatia S, Lipson EJ, et al. PD-1 Blockade with Pembrolizumab in Advanced Merkel-Cell Carcinoma. N Engl J Med. 2016; 374:2542–2552. [PubMed: 27093365]
23. Tumeh PC, Harview CL, Yearley JH, et al. PD-1 blockade induces responses by inhibiting adaptive immune resistance. Nature. 2014; 515:568–571. [PubMed: 25428505]
24. Zou W, Wolchok JD, Chen L. PD-L1 (B7-H1) and PD-1 pathway blockade for cancer therapy: Mechanisms, response biomarkers, and combinations. Science Translational Medicine. 2016; 8:328rv324–328rv324.
25. Sharma P, Hu-Lieskovan S, Wargo JA, et al. Primary, Adaptive, and Acquired Resistance to Cancer Immunotherapy. Cell. 2017; 168:707–723. [PubMed: 28187290]
26. O'Donnell JS, Smyth MJ, Teng MW. Acquired resistance to anti-PD1 therapy: checkmate to checkpoint blockade? Genome Med. 2016; 8:111. [PubMed: 27782862]
27. Gubin MM, Zhang X, Schuster H, et al. Checkpoint blockade cancer immunotherapy targets tumour-specific mutant antigens. Nature. 2014; 515:577–581. [PubMed: 25428507]
28. Schumacher TN, Schreiber RD. Neoantigens in cancer immunotherapy. Science. 2015; 348:69–74. [PubMed: 25838375]
29. Rizvi NA, Hellmann MD, Snyder A, et al. Cancer immunology. Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. Science. 2015; 348:124–128. [PubMed: 25765070]
30. Le DT, Durham JN, Smith KN, et al. Mismatch repair deficiency predicts response of solid tumors to PD-1 blockade. Science. 2017; 357:409–413. [PubMed: 28596308]
31. de Velasco G, Miao D, Voss MH, et al. Tumor Mutational Load and Immune Parameters across Metastatic Renal Cell Carcinoma Risk Groups. Cancer Immunol Res. 2016; 4:820–822. [PubMed: 27538576]
32. Martin AM, Nirschl TR, Nirschl CJ, et al. Paucity of PD-L1 expression in prostate cancer: innate and adaptive immune resistance. Prostate Cancer Prostatic Dis. 2015; 18:325–332. [PubMed: 26260996]
33. Chodon T, Comin-Anduix B, Chmielowski B, et al. Adoptive transfer of MART-1 T-cell receptor transgenic lymphocytes and dendritic cell vaccination in patients with metastatic melanoma. Clin Cancer Res. 2014; 20:2457–2465. [PubMed: 24634374]
34. Robbins PF, Kassim SH, Tran TL, et al. A pilot trial using lymphocytes genetically engineered with an NY-ESO-1-reactive T-cell receptor: long-term follow-up and correlates with response. Clin Cancer Res. 2015; 21:1019–1027. [PubMed: 25538264]
35. Robbins PF, Morgan RA, Feldman SA, et al. Tumor regression in patients with metastatic synovial cell sarcoma and melanoma using genetically engineered lymphocytes reactive with NY-ESO-1. J Clin Oncol. 2011; 29:917–924. [PubMed: 21282551]
36. Pollack SM, Jones RL, Farrar EA, et al. Tetramer guided, cell sorter assisted production of clinical grade autologous NY-ESO-1 specific CD8(+) T cells. J Immunother Cancer. 2014; 2:36. [PubMed: 25317334]
37. Hunder NN, Wallen H, Cao J, et al. Treatment of metastatic melanoma with autologous CD4+ T cells against NY-ESO-1. N Engl J Med. 2008; 358:2698–2703. [PubMed: 18565862]
38. Somaiah N, B MS, Kim JW, Shapiro G, Hwu P, Eder JP, Jones RL, Gnjatic S, Lu, Frank H, Hsu J, Pollack S. Phase 1 First-in-Human Trial Of LV305 In Patients With Advanced Or Metastatic Cancer Expressing NY-ESO-1. J Clin Oncol. 2015; 33 abstr 3021.
39. Ott PA, Hu Z, Keskin DB, et al. An immunogenic personal neoantigen vaccine for patients with melanoma. Nature. 2017; 547:217–221. [PubMed: 28678778]
40. Sahin U, Derhovanessian E, Miller M, et al. Personalized RNA mutanome vaccines mobilize poly-specific therapeutic immunity against cancer. Nature. 2017; 547:222–226. [PubMed: 28678784]
41. Moon EK, Ranganathan R, Eruslanov E, et al. Blockade of Programmed Death 1 Augments the Ability of Human T Cells Engineered to Target NY-ESO-1 to Control Tumor Growth after Adoptive Transfer. Clin Cancer Res. 2016; 22:436–447. [PubMed: 26324743]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

42. Rosenblatt J, Glotzbecker B, Mills H, et al. PD-1 blockade by CT-011, anti-PD-1 antibody, enhances ex vivo T-cell responses to autologous dendritic cell/myeloma fusion vaccine. J Immunother. 2011; 34:409–418. [PubMed: 21577144]
43. Stecher C, Battin C, Leitner J, et al. PD-1 Blockade Promotes Emerging Checkpoint Inhibitors in Enhancing T Cell Responses to Allogeneic Dendritic Cells. Front Immunol. 2017; 8:572. [PubMed: 28588576]
44. John LB, Devaud C, Duong CP, et al. Anti-PD-1 antibody therapy potently enhances the eradication of established tumors by gene-modified T cells. Clin Cancer Res. 2013; 19:5636–5646. [PubMed: 23873688]
45. Ribas A, Dummer R, Puzanov I, et al. Oncolytic Virotherapy Promotes Intratumoral T Cell Infiltration and Improves Anti-PD-1 Immunotherapy. Cell. 2017; 170:1109–1119 e1110. [PubMed: 28886381]
46. Marincola FM, Jaffee EM, Hicklin DJ, et al. Escape of human solid tumors from T-cell recognition: molecular mechanisms and functional significance. Advances in immunology. 2000; 74:181–273. [PubMed: 10605607]
47. Sucker A, Zhao F, Real B, et al. Genetic evolution of T-cell resistance in the course of melanoma progression. Clin Cancer Res. 2014; 20:6593–6604. [PubMed: 25294904]
48. Restifo NP, Marincola FM, Kawakami Y, et al. Loss of functional beta 2-microglobulin in metastatic melanomas from five patients receiving immunotherapy. J Natl Cancer Inst. 1996; 88:100–108. [PubMed: 8537970]
49. D'Urso CM, Wang ZG, Cao Y, et al. Lack of HLA class I antigen expression by cultured melanoma cells FO-1 due to a defect in B2m gene expression. J Clin Invest. 1991; 87:284–292. [PubMed: 1898655]
50. Kim HJ, Bae SC. Histone deacetylase inhibitors: molecular mechanisms of action and clinical trials as anti-cancer drugs. American journal of translational research. 2011; 3:166–179. [PubMed: 21416059]
51. Karpf AR, Jones DA. Reactivating the expression of methylation silenced genes in human cancer. Oncogene. 2002; 21:5496–5503. [PubMed: 12154410]
52. Héninger E, Krueger TE, Lang JM. Augmenting antitumor immune responses with epigenetic modifying agents. Frontiers in immunology. 2015; 6:29. [PubMed: 25699047]
53. Vo DD, Prins RM, Begley JL, et al. Enhanced antitumor activity induced by adoptive T-cell transfer and adjunctive use of the histone deacetylase inhibitor LAQ824. Cancer Res. 2009; 69:8693–8699. [PubMed: 19861533]
54. Wang L-X, Mei Z-Y, Zhou J-H, et al. Low dose decitabine treatment induces CD80 expression in cancer cells and stimulates tumor specific cytotoxic T lymphocyte responses. PloS one. 2013; 8:e62924. [PubMed: 23671644]
55. Spranger S, Bao R, Gajewski TF. Melanoma-intrinsic beta-catenin signalling prevents anti-tumour immunity. Nature. 2015; 523:231–235. [PubMed: 25970248]
56. Sweis RF, Spranger S, Bao R, et al. Molecular Drivers of the Non-T-cell-Inflamed Tumor Microenvironment in Urothelial Bladder Cancer. Cancer Immunol Res. 2016; 4:563–568. [PubMed: 27197067]
57. Liu C, Peng W, Xu C, et al. BRAF inhibition increases tumor infiltration by T cells and enhances the antitumor activity of adoptive immunotherapy in mice. Clin Cancer Res. 2013; 19:393–403. [PubMed: 23204132]
58. Loi S, Dushyanthen S, Beavis PA, et al. RAS/MAPK Activation Is Associated with Reduced Tumor-Infiltrating Lymphocytes in Triple-Negative Breast Cancer: Therapeutic Cooperation Between MEK and PD-1/PD-L1 Immune Checkpoint Inhibitors. Clin Cancer Res. 2016; 22:1499–1509. [PubMed: 26515496]
59. Hu-Lieskovan S, Mok S, Homet Moreno B, et al. Improved antitumor activity of immunotherapy with BRAF and MEK inhibitors in BRAF(V600E) melanoma. Sci Transl Med. 2015; 7:279ra241.
60. Peng W, Chen JQ, Liu C, et al. Loss of PTEN Promotes Resistance to T Cell-Mediated Immunotherapy. Cancer Discov. 2016; 6:202–216. [PubMed: 26645196]
61. Platanias LC. Mechanisms of type-I- and type-II-interferon-mediated signalling. Nat Rev Immunol. 2005; 5:375–386. [PubMed: 15864272]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

62. Ribas A. Adaptive Immune Resistance: How Cancer Protects from Immune Attack. Cancer Discov. 2015; 5:915–919. [PubMed: 26272491]
63. Benci JL, Xu B, Qiu Y, et al. Tumor Interferon Signaling Regulates a Multigenic Resistance Program to Immune Checkpoint Blockade. Cell. 2016; 167:1540–1554.e1512. [PubMed: 27912061]
64. Darnell JE Jr, Kerr IM, Stark GR. Jak-STAT pathways and transcriptional activation in response to IFNs and other extracellular signaling proteins. Science. 1994; 264:1415–1421. [PubMed: 8197455]
65. Garcia-Diaz A, Shin DS, Moreno BH, et al. Interferon Receptor Signaling Pathways Regulating PD-L1 and PD-L2 Expression. Cell reports. 2017; 19:1189–1201. [PubMed: 28494868]
66. Kaplan MH, Wurster AL, Grusby MJ. A signal transducer and activator of transcription (Stat)4-independent pathway for the development of T helper type 1 cells. J Exp Med. 1998; 188:1191–1196. [PubMed: 9743537]
67. Dunn GP, Bruce AT, Sheehan KC, et al. A critical function for type I interferons in cancer immunoediting. Nat Immunol. 2005; 6:722–729. [PubMed: 15951814]
68. Shin DS, Zaretsky JM, Escuin-Ordinas H, et al. Primary Resistance to PD-1 Blockade Mediated by JAK1/2 Mutations. Cancer Discov. 2017; 7:188–201. [PubMed: 27903500]
69. Patel SJ, Sanjana NE, Kishton RJ, et al. Identification of essential genes for cancer immunotherapy. Nature. 2017; 548:537–542. [PubMed: 28783722]
70. Manguso RT, Pope HW, Zimmer MD, et al. In vivo CRISPR screening identifies Ptpn2 as a cancer immunotherapy target. Nature. 2017; 547:413–418. [PubMed: 28723893]
71. Gainor JF, Shaw AT, Sequist LV, et al. EGFR Mutations and ALK Rearrangements Are Associated with Low Response Rates to PD-1 Pathway Blockade in Non-Small Cell Lung Cancer: A Retrospective Analysis. Clin Cancer Res. 2016; 22:4585–4593. [PubMed: 27225694]
72. Akbay EA, Koyama S, Carretero J, et al. Activation of the PD-1 pathway contributes to immune escape in EGFR-driven lung tumors. Cancer Discov. 2013; 3:1355–1363. [PubMed: 24078774]
73. Ota K, Azuma K, Kawahara A, et al. Induction of PD-L1 Expression by the EML4-ALK Oncoprotein and Downstream Signaling Pathways in Non-Small Cell Lung Cancer. Clin Cancer Res. 2015; 21:4014–4021. [PubMed: 26019170]
74. Gabrilovich DI, Ostrand-Rosenberg S, Bronte V. Coordinated regulation of myeloid cells by tumours. Nat Rev Immunol. 2012; 12:253–268. [PubMed: 22437938]
75. Gabrilovich DI, Nagaraj S. Myeloid-derived suppressor cells as regulators of the immune system. Nat Rev Immunol. 2009; 9:162–174. [PubMed: 19197294]
76. Arina A, Bronte V. Myeloid-derived suppressor cell impact on endogenous and adoptively transferred T cells. Curr Opin Immunol. 2015; 33:120–125. [PubMed: 25728992]
77. Ostrand-Rosenberg S. Myeloid-derived suppressor cells: more mechanisms for inhibiting antitumor immunity. Cancer Immunol Immunother. 2010; 59:1593–1600. [PubMed: 20414655]
78. Highfill SL, Cui Y, Giles AJ, et al. Disruption of CXCR2-mediated MDSC tumor trafficking enhances anti-PD1 efficacy. Sci Transl Med. 2014; 6:237ra267.
79. Steinberg SM, Shabaneh TB, Zhang P, et al. Myeloid Cells That Impair Immunotherapy Are Restored in Melanomas with Acquired Resistance to BRAF Inhibitors. Cancer Res. 2017; 77:1599–1610. [PubMed: 28202513]
80. Bettelli E, Carrier Y, Gao W, et al. Reciprocal developmental pathways for the generation of pathogenic effector TH17 and regulatory T cells. Nature. 2006; 441:235–238. [PubMed: 16648838]
81. Elpek KG, Lacelle C, Singh NP, et al. CD4+CD25+ T regulatory cells dominate multiple immune evasion mechanisms in early but not late phases of tumor development in a B cell lymphoma model. J Immunol. 2007; 178:6840–6848. [PubMed: 17513732]
82. Arce Vargas F, Furness AJS, Solomon I, et al. Fc-Optimized Anti-CD25 Depletes Tumor-Infiltrating Regulatory T Cells and Synergizes with PD-1 Blockade to Eradicate Established Tumors. Immunity. 2017; 46:577–586. [PubMed: 28410988]
83. Taylor NA, Vick SC, Iglesia MD, et al. Treg depletion potentiates checkpoint inhibition in claudin-low breast cancer. J Clin Invest. 2017; 127:3472–3483. [PubMed: 28825599]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

84. Hugo W, Zaretsky JM, Sun L, et al. Genomic and Transcriptomic Features of Response to Anti-PD-1 Therapy in Metastatic Melanoma. Cell. 2016; 165:35–44. [PubMed: 26997480]
85. Wherry EJ, Kurachi M. Molecular and cellular insights into T cell exhaustion. Nat Rev Immunol. 2015; 15:486–499. [PubMed: 26205583]
86. Philip M, Fairchild L, Sun L, et al. Chromatin states define tumour-specific T cell dysfunction and reprogramming. Nature. 2017; 545:452–456. [PubMed: 28514453]
87. Huang AC, Postow MA, Orlowski RJ, et al. T-cell invigoration to tumour burden ratio associated with anti-PD-1 response. Nature. 2017; 545:60–65. [PubMed: 28397821]
88. Pauken KE, Sammons MA, Odorizzi PM, et al. Epigenetic stability of exhausted T cells limits durability of reinvigoration by PD-1 blockade. Science. 2016; 354:1160–1165. [PubMed: 27789795]
89. Klebanoff CA, Gattinoni L, Torabi-Parizi P, et al. Central memory self/tumor-reactive CD8+ T cells confer superior antitumor immunity compared with effector memory T cells. Proc Natl Acad Sci U S A. 2005; 102:9571–9576. [PubMed: 15980149]
90. Klebanoff CA, Gattinoni L, Restifo NP. CD8+ T-cell memory in tumor immunology and immunotherapy. Immunol Rev. 2006; 211:214–224. [PubMed: 16824130]
91. Klebanoff CA, Gattinoni L, Palmer DC, et al. Determinants of successful CD8+ T-cell adoptive immunotherapy for large established tumors in mice. Clin Cancer Res. 2011; 17:5343–5352. [PubMed: 21737507]
92. Topalian SL, Drake CG, Pardoll DM. Immune checkpoint blockade: a common denominator approach to cancer therapy. Cancer Cell. 2015; 27:450–461. [PubMed: 25858804]
93. Shin DS, Ribas A. The evolution of checkpoint blockade as a cancer therapy: what's here, what's next? Curr Opin Immunol. 2015; 33:23–35. [PubMed: 25621841]
94. Koyama S, Akbay EA, Li YY, et al. Adaptive resistance to therapeutic PD-1 blockade is associated with upregulation of alternative immune checkpoints. Nature communications. 2016; 7:10501.
95. Lines JL, Pantazi E, Mak J, et al. VISTA is an immune checkpoint molecule for human T cells. Cancer Res. 2014; 74:1924–1932. [PubMed: 24691993]
96. Kakavand H, Jackett LA, Menzies AM, et al. Negative immune checkpoint regulation by VISTA: a mechanism of acquired resistance to anti-PD-1 therapy in metastatic melanoma patients. Mod Pathol. 2017
97. Sharma P, Allison JP. Immune checkpoint targeting in cancer therapy: toward combination strategies with curative potential. Cell. 2015; 161:205–214. [PubMed: 25860605]
98. Anderson AC, Joller N, Kuchroo VK. Lag-3, Tim-3, and TIGIT: Co-inhibitory Receptors with Specialized Functions in Immune Regulation. Immunity. 2016; 44:989–1004. [PubMed: 27192565]
99. Larkin J, Chiarion-Sileni V, Gonzalez R, et al. Combined Nivolumab and Ipilimumab or Monotherapy in Untreated Melanoma. N Engl J Med. 2015; 373:23–34. [PubMed: 26027431]
100. Wolchok JD, Chiarion-Sileni V, Gonzalez R, et al. Overall Survival with Combined Nivolumab and Ipilimumab in Advanced Melanoma. N Engl J Med. 2017
101. Landsberg J, Kohlmeyer J, Renn M, et al. Melanomas resist T-cell therapy through inflammation-induced reversible dedifferentiation. Nature. 2012; 490:412–416. [PubMed: 23051752]
102. Ribas A, Tumeh PC. Cancer therapy: Tumours switch to resist. Nature. 2012; 490:347–348. [PubMed: 23051745]
103. Reinhardt J, Landsberg J, Schmid-Burgk JL, et al. MAPK Signaling and Inflammation Link Melanoma Phenotype Switching to Induction of CD73 during Immunotherapy. Cancer Res. 2017; 77:4697–4709. [PubMed: 28652246]
104. van Rooij N, van Buuren MM, Philips D, et al. Tumor exome analysis reveals neoantigen-specific T-cell reactivity in an ipilimumab-responsive melanoma. J Clin Oncol. 2013; 31:e439–442. [PubMed: 24043743]
105. Anagnostou V, Smith KN, Forde PM, et al. Evolution of Neoantigen Landscape during Immune Checkpoint Blockade in Non-Small Cell Lung Cancer. Cancer Discov. 2017; 7:264–276. [PubMed: 28031159]

106. Zaretsky JM, Garcia-Diaz A, Shin DS, et al. Mutations Associated with Acquired Resistance to PD-1 Blockade in Melanoma. N Engl J Med. 2016; 375:819–829. [PubMed: 27433843]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript